

# SOFTWARE REQUIREMENTS SPECIFICATION

## Inventory Management System (IMS)

Version 1.0

Date: July 2026

## Table of Contents

|                                                            |          |
|------------------------------------------------------------|----------|
| <b>1. INTRODUCTION .....</b>                               | <b>4</b> |
| 1.1 Purpose .....                                          | 4        |
| 1.2 Scope .....                                            | 4        |
| 1.3 Definitions, Acronyms, and Abbreviations .....         | 4        |
| 1.4 References .....                                       | 5        |
| 1.5 Overview .....                                         | 5        |
| <b>2. OVERALL DESCRIPTION .....</b>                        | <b>6</b> |
| 2.1 Product Perspective .....                              | 6        |
| 2.2 Product Functions .....                                | 6        |
| 2.3 User Classes and Characteristics .....                 | 7        |
| 2.4 Operating Environment .....                            | 7        |
| 2.5 Design and Implementation Constraints .....            | 7        |
| 2.6 Assumptions and Dependencies .....                     | 9        |
| 3.1 PROCUREMENT & PURCHASE ORDER MANAGEMENT .....          | 9        |
| 3.3 INVENTORY & STOCK CONTROL .....                        | 47       |
| 3.4 GRN & INVOICE MANAGEMENT .....                         | 75       |
| 3.5 INDENT & ISSUE MANAGEMENT .....                        | 89       |
| Figure: 3.5 Indent & Issue Management – Process Flow ..... | 90       |
| 3.6 ASSET MANAGEMENT .....                                 | 100      |
| 3.7 BUDGET & FINANCE INTEGRATION .....                     | 111      |
| 3.8 CRM & TICKETING .....                                  | 120      |
| Figure: 3.8 Crm & Ticketing – Process Flow .....           | 120      |
| 3.9 REPORTS & ANALYTICS .....                              | 128      |
| 3.10 PRODUCT MASTER MANAGEMENT .....                       | 140      |
| 3.11 PRINTING CONSUMABLES MANAGEMENT .....                 | 147      |
| 3.12 UNIFORMS & LINEN TRACKING .....                       | 152      |

|                                                       |                                                  |            |
|-------------------------------------------------------|--------------------------------------------------|------------|
| 3.13 WORKFLOW & APPROVALS                             | Figure: 3.13 Workflow & Approvals – Process Flow | 157        |
| 3.14 TRACKING & MONITORING.....                       |                                                  | 167        |
| 3.15 GATE PASS MANAGEMENT.....                        |                                                  | 172        |
| 3.16 CONSIGNMENT STOCK MANAGEMENT .....               |                                                  | 179        |
| <b>4. EXTERNAL INTERFACE REQUIREMENTS .....</b>       |                                                  | <b>186</b> |
| 4.1 User Interfaces.....                              |                                                  | 186        |
| 4.2 Hardware Interfaces.....                          |                                                  | 186        |
| 4.3 Software Interfaces.....                          |                                                  | 186        |
| 4.4 Communications Interfaces.....                    |                                                  | 187        |
| <b>5. NON-FUNCTIONAL REQUIREMENTS.....</b>            |                                                  | <b>188</b> |
| 5.1 Performance Requirements .....                    |                                                  | 188        |
| 5.2 Security Requirements .....                       |                                                  | 188        |
| 5.3 Availability & Reliability.....                   |                                                  | 189        |
| 5.4 Scalability .....                                 |                                                  | 189        |
| 5.5 Data Integrity & Backup.....                      |                                                  | 189        |
| <b>6. DATA REQUIREMENTS .....</b>                     |                                                  | <b>190</b> |
| 6.1 Data Models.....                                  |                                                  | 190        |
| 6.2 Data Retention .....                              |                                                  | 192        |
| Appendix A: Implementation Phases.....                |                                                  | 193        |
| Appendix B: Glossary of Hospital-Specific Terms ..... |                                                  | 194        |
| Appendix C: Compliance & Regulatory Requirements..... |                                                  | 195        |
| Appendix D: Risk Mitigation .....                     |                                                  | 196        |

## 1. INTRODUCTION

### 1.1 Purpose

This SRS defines functional and non-functional requirements for the Inventory Management System (IMS). The system streamlines procurement, inventory control, asset management, vendor relations, and financial integration for a multi-department hospital environment.

### 1.2 Scope

- Procurement and Purchase Order lifecycle management
- Vendor portal with self-service capabilities
- Inventory tracking, stock transfers, and reorder management
- Goods Receipt Note (GRN) and Invoice processing
- Department indenting and stock issuance
- Asset lifecycle management with depreciation
- Budget preparation and financial reporting
- CRM/Ticketing for service and material requests
- Integration with HIS and Tally
- Comprehensive reporting and MIS dashboards

### 1.3 Definitions, Acronyms, and Abbreviations

| Term                               | Abbreviation | Definition                                |
|------------------------------------|--------------|-------------------------------------------|
| Annual Maintenance Contract        | AMC          | Yearly service contract for equipment     |
| Comprehensive Maintenance Contract | CMC          | Full coverage including spare parts       |
| Goods Receipt Note                 | GRN          | Document acknowledging receipt of goods   |
| Hospital Information System        | HIS          | Existing hospital patient/ward management |
| Head of Department                 | HOD          | Department head responsible for approvals |
| Inventory Management System        | IHMS         | The software system being specified       |
| Management Information System      | MIS          | Reports for management decisions          |
| Purchase Order                     | PO           | Formal document issued to vendor          |
| Permanent Account Number           | PAN          | Indian tax identification                 |
| Goods and Services Tax             | GST          | Indian indirect tax system                |
| Tax Deducted at Source             | TDS          | Indian tax deduction mechanism            |
| Delivery Challan                   | DC           | Document with goods delivery              |
| Rate Contract                      | RC           | Fixed-price agreement with vendor         |

|         |    |                                    |
|---------|----|------------------------------------|
| Red Tag | RT | Status marking products out of use |
|---------|----|------------------------------------|

## 1.4 References

- Hospital procurement policy documents
- Indian GST Act, 2017
- Tally ERP integration specs
- Existing HIS API documentation
- Internal hospital SOPs

## 1.5 Overview

This document is organized into seven sections: Introduction, Overall Description, System Features, External Interfaces, Non-Functional Requirements, Data Requirements, and Appendices.

## 2. OVERALL DESCRIPTION

### 2.1 Product Perspective

IHMS is a web-based system integrating with existing HIS and Tally. It operates on hospital LAN with external vendor portal access via secure internet.

### 2.2 Product Functions

| Functional Area       | Key Capabilities                                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------|
| Procurement           | PO creation, approval workflows, rate contracts, addendums, cash purchases                                       |
| Vendor Management     | Portal registration, document upload, reconciliation, GST validation                                             |
| Inventory Control     | Stock tracking, transfers, reorder levels, expiry alerts, barcode                                                |
| GRN & Invoice         | Goods receipt, invoice validation, batch handling, payment confirmation                                          |
| Indent & Issue        | Ward requests, auto-indent, HIS integration, stock issuance                                                      |
| Asset Management      | Lifecycle tracking, depreciation, maintenance, warranty reminders                                                |
| Financial Integration | Tally sync, budget prep, outstanding reconciliation, GST cross-validation                                        |
| CRM & Ticketing       | Service requests, sample tracking, quality ratings, communication                                                |
| Reporting             | Daily/weekly/monthly reports, MIS dashboards, price comparisons, audit trails                                    |
| Product Master        | Categorization, alias management, image upload, duplicate prevention                                             |
| Gate Pass Management  | Outward/inward returnable and non-returnable passes, Security gate verification, overdue tracking                |
| Consignment Stock     | Vendor-owned on-site stock, consumption-triggered ownership transfer, periodic vendor billing and reconciliation |

## 2.3 User Classes and Characteristics

| User Class       | Description             | Privileges                                          | Priority |
|------------------|-------------------------|-----------------------------------------------------|----------|
| System Admin     | IT staff                | Full system access, user management                 | High     |
| Store Manager    | Head of stores          | All procurement, inventory, GRN, vendor functions   | High     |
| Store Keeper     | Stores staff            | PO creation, GRN, stock issuance, indent processing | High     |
| Department HOD   | Department heads        | Approve indents, item requests, quality feedback    | High     |
| Accounts/Finance | Finance staff           | Budget, payment, Tally integration, reconciliation  | High     |
| Management       | Hospital administration | All reports, high-value PO approval, MIS            | High     |
| Ward Staff       | Nurses/ward admins      | Create indents, receive stock, return items         | Medium   |
| Doctors          | Medical practitioners   | Request products/services, approve samples          | Medium   |
| Vendor/Supplier  | External suppliers      | Portal access, upload docs, view payments           | Medium   |
| Security         | Gate staff              | Record material entry, verify deliveries            | Low      |

## 2.4 Operating Environment

- Server: Cloud or on-premise Windows/Linux
- Database: PostgreSQL/MySQL/MS SQL Server
- Web Server: IIS/Apache/Nginx
- Client Browsers: Chrome, Firefox, Edge (latest 2 versions)
- Mobile: Responsive web for tablet/phone
- Network: Hospital LAN + VPN/SSL for vendor portal
- Backup: Daily automated with 30-day retention

## 2.5 Design and Implementation Constraints

- Must integrate with existing Tally ERP

- Must support HIS integration for ward indents
- Must comply with Indian GST regulations
- Must handle multiple currencies for imports
- Must support online and offline barcode printing
- Vendor portal accessible via standard browsers
- Must maintain audit trail for all financial transactions
- Data export must support Excel format
- Maximum response time: 3 seconds
- Minimum 50 concurrent users



## 2.6 Assumptions and Dependencies

- Hospital network supports HTTPS
- Vendors have basic internet access
- HIS provides API or database access
- Tally supports external data import
- Barcode printers/scanners available
- Email server configured for notifications
- GST portal data exportable to Excel
- HODs have computer/mobile access for approvals

## 3. SYSTEM FEATURES

This section details functional requirements by module. Each requirement has a unique ID (REQ-XXX), priority (High/Medium/Low), and detailed description.

### 3.1 PROCUREMENT & PURCHASE ORDER MANAGEMENT

## 3.1 Procurement & Purchase Order Management Process Flow

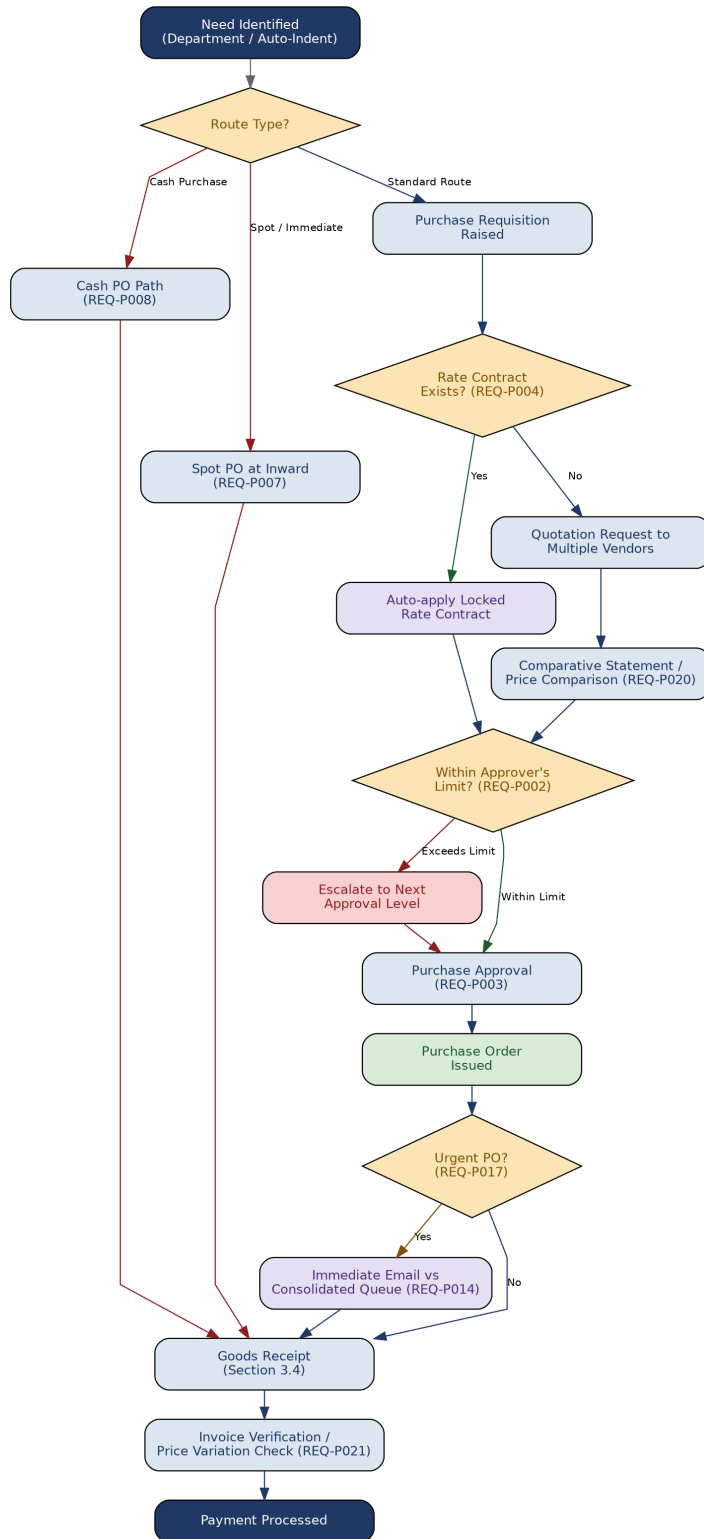


Figure: 3.1 Procurement & Purchase Order Management – Process Flow

## REQ-P001 – Vendor Terms & Conditions Management

Priority: High | Status: Proposed

Description: Configure vendor T&C based on total PO value, advance percentage, credit period, penalty clauses. Auto-populate on PO and require vendor acknowledgment through portal.

Preconditions: Vendor is registered and approved in the vendor master. PO value slabs and T&C templates are configured by Finance.

Inputs: Vendor ID, PO value band, advance percentage, credit period (days), penalty clause text, warranty clause, applicable T&C template code.

Processing Logic: 1. Finance configures T&C templates against PO value bands (e.g., <2L, 2L-10L, >10L) with default advance %, credit period, and penalty terms. 2. On PO creation, system auto-selects the T&C template matching the PO's computed value. 3. Purchase Executive may override specific clauses with a mandatory reason, subject to Finance re-approval. 4. Selected T&C text is embedded in the PO printout/PDF sent to the vendor. 5. Vendor portal presents the PO with an 'Accept T&C' action; acknowledgment is time-stamped. 6. PO cannot proceed to dispatch confirmation by the vendor until T&C is acknowledged. 7. Any change to standard templates is version-controlled and audit-logged.

Outputs: PO document with embedded T&C clause, vendor acknowledgment record, T&C version history.

Postconditions: 1. PO carries a legally referenceable, vendor-acknowledged T&C. 2. Acknowledgment status is visible to Purchase and Finance.

Business Rules: 1. If vendor does not acknowledge within a configurable SLA (e.g., 3 working days), system escalates to Purchase Executive. 2. Manual override of a clause requires Finance sign-off before the PO is released. 3. Penalty clause invocation on a delayed delivery must reference the exact clause version applied at PO issue.

Exception Handling: 1. If no T&C template matches the PO value band, system blocks PO release and prompts Finance to configure one. 2. If the vendor disputes a clause after acceptance, the dispute is logged against the PO and routed to Purchase Manager for resolution. 3. If the vendor portal is unavailable, acknowledgment can be captured manually (signed hard copy scan) with the same audit weight.

## REQ-P002 – User Approval Limits

Priority: High | Status: Proposed

Description: Hierarchical approval by role and value. Multi-level workflows: Store Keeper → Store Manager → HOD → Finance → Management. Emergency override with audit trail.

Preconditions: User roles and their monetary approval limits are defined in the User Master. The approval matrix (Store Keeper → Store Manager → HOD → Finance → Management) is configured and active.

Inputs: User ID, role, department, configured approval limit (₹), PO/PR value, transaction type, emergency-override flag.

Processing Logic: 1. On PR/PO submission, system computes the total transaction value. 2. System determines the minimum chain of approvers whose combined limits cover the value, per the configured hierarchy. 3. Each approver in sequence receives a notification; the transaction advances only after the current tier approves. 4. If a user attempts to approve above their configured limit, the action is rejected and auto-escalated to the next tier. 5. For emergency-override use (Section 3.1, REQ-P006/P007), the system logs the override reason and notifies the next two tiers regardless of value. 6. Approval limits can be temporarily delegated (see Business Rules) during leave/absence.

Outputs: Routed approval chain per transaction, approval/rejection log with timestamps, escalation notifications, override audit entries.

Postconditions: 1. Every approved transaction carries a complete, sequential approval trail matching the configured hierarchy. 2. No transaction can be approved by a single user beyond their delegated limit.

Business Rules: 1. Approval limits are role- and department-specific and can differ between departments for the same role title. 2. A user cannot approve a transaction they themselves raised (segregation of duties). 3. Delegated approval authority is time-bound (start/end date) and is itself audit-logged. 4. Limit changes require Management approval and are versioned.

Exception Handling: 1. If the designated approver is on leave without a delegate configured, the system escalates automatically to that approver's reporting manager after a configurable wait period. 2. If a value exceeds every configured limit in the hierarchy (i.e., requires Board approval), the system routes it to a Board-approval queue rather than blocking it. 3. If two approvers are required at the same tier and one rejects, the transaction returns to the originator with both comments visible.

## REQ-P003 – Routine Purchase Approval

Priority: High | Status: Proposed

Description: Routine consumables/assets approved by department leads within predefined limits. Auto-approval for routine items. Flag non-routine for additional approval.

Preconditions: Item is classified as 'Routine' in the product master, and the requesting department has an approved, unfrozen budget for the period.

Inputs: Item code, item classification (Routine/Non-Routine), requested quantity, department budget balance, department lead's auto-approval limit.

Processing Logic: 1. System checks the item classification flag at PR creation. 2. For Routine items within the department lead's configured auto-approval limit and within budget, the PR is auto-approved without manual intervention. 3. For Routine items exceeding the auto-approval limit, or for any Non-Routine item, the PR is flagged and routed for manual HOD/Finance approval per the standard matrix. 4. All auto-approved PRs are still logged in the audit trail and appear on a daily auto-approval summary sent to the department head for visibility.

Outputs: Auto-approved PR (routine, in-limit), flagged PR for manual approval (non-routine or over-limit), daily auto-approval summary report.

Postconditions: 1. Routine, budget-compliant, in-limit purchases move to PO generation without manual delay. 2. All non-routine or over-limit requests are captured for management visibility before approval.

Business Rules: 1. Item classification (Routine/Non-Routine) can only be changed by the Product Master Administrator with Store Manager sign-off. 2. Auto-approval is disabled automatically if the department's budget utilization crosses a configurable threshold (e.g., 90%). 3. A department head may opt out of auto-approval for their department at any time.

Exception Handling: 1. If the department budget is frozen or exhausted, the PR is blocked regardless of routine classification and routed to Finance for a budget exception decision. 2. If the same item is flagged as routine but requested in unusually high quantity (e.g., >3x rolling average), the system flags it for manual review instead of auto-approving. 3. If item classification is missing/unset, the system defaults to manual approval routing.

## REQ-P004 – Rate Contract Management

Priority: High | Status: Proposed

Description: Lock item rates for 1yr/2yr contract duration. Alert 30 days before expiry. Track contract value consumption. Allow PO generation from active contracts.

Preconditions: Vendor is approved and a rate contract negotiation/agreement has concluded with agreed items, rates, and validity dates.

Inputs: Vendor ID, contract items and rates, contract start/end date, contract ceiling value (if any), renewal terms, approving authority.

Processing Logic: 1. Purchase/Finance creates a Rate Contract record referencing the vendor, item list, locked rates, and validity window. 2. Contract is routed for Management approval before activation. 3. On activation, the contracted rates become the default, non-negotiable rate for those items when the specific vendor is selected on a PO within the validity period. 4. System tracks cumulative PO value raised against the contract and displays remaining ceiling (if a value cap applies). 5. System sends an automated alert 30 days before contract expiry to Purchase and the assigned relationship owner. 6. On expiry without renewal, the system reverts the item(s) to the standard quotation-based procurement path.

Outputs: Active rate contract record, contract-linked PO generation, contract consumption tracker, expiry alert notifications, renewal workflow trigger.

Postconditions: 1. All POs raised against an active contract automatically inherit the locked rate, eliminating rate negotiation at the transaction level. 2. Contract consumption and remaining validity are visible to Purchase at all times.

Business Rules: 1. A PO cannot be raised at a rate different from the active contract rate for a contracted item/vendor combination without explicit Finance override and reason capture. 2. If the contract ceiling value is reached before the validity period ends, the system blocks further POs against that contract and requires either a ceiling revision or fallback to the quotation process. 3. Renewal of an expiring contract follows the same approval workflow as a new contract.

Exception Handling: 1. If a contracted item's market price has fallen significantly below the locked rate, Purchase may raise a rate-revision request, which requires Management re-approval before the new rate applies. 2. If the vendor is unable to fulfil an order at the contracted rate/quantity, the shortfall is documented and the PR reverts to the open-market quotation path for that shortfall quantity. 3. If two active contracts exist for the same item from different vendors, the system prompts the user to select and logs the selection rationale.

**REQ-P005 – Department HOD Item Approval**

Priority: High | Status: Proposed

Description: HOD approves indents/requests for their department. Approve/Reject/Clarify options. Approval required before PO generation. Delegation during absence.

Preconditions: Indent/item request has been raised by a department user or auto-generated by the reorder engine, and is pending at the HOD's approval queue.

Inputs: Indent/request ID, department, requested items and quantities, requester, justification/reason, HOD ID, delegate ID (if applicable).

Processing Logic: 1. Indent/request appears in the HOD's approval queue (portal and email/SMS notification). 2. HOD reviews item, quantity, and justification against departmental need. 3. HOD selects Approve, Reject, or Request Clarification. 4. On Approve, the request proceeds to Purchase for PO generation (or to Stores for direct issue if stock is available). 5. On Reject, a mandatory reason is captured and the request returns to the originator. 6. On Request Clarification, the request is held with a comment thread until the originator responds, then re-enters the HOD queue. 7. If the HOD will be unavailable, a delegate is configured in advance with a defined validity window.

Outputs: Approved/rejected/clarification-pending indent status, comment history, delegate approval log, downstream PO or stock-issue trigger.

Postconditions: 1. No item leaves stores or is procured for a department without a traceable HOD decision. 2. Rejected or clarification-held requests do not silently disappear; they remain visible in the requester's dashboard.

Business Rules: 1. HOD approval is required in addition to (not instead of) the value-based approval matrix for items requiring PO. 2. A delegate can approve on the HOD's behalf only within the configured delegation window; the system labels such approvals as 'Approved by Delegate' in the audit trail. 3. Requests pending HOD action beyond a configurable SLA are escalated with a reminder, then to the HOD's superior.

Exception Handling: 1. If no delegate is configured and the HOD is unreachable beyond the SLA, the system escalates to the next department in the reporting hierarchy for interim approval, flagged for the HOD's retrospective review. 2. If the requester disputes a rejection, they may resubmit with additional justification, which is tracked as a revised version of the same indent, not a new one. 3. If the item requested is not in the department's typical consumption pattern, the system adds a soft warning to the HOD's review screen.

**REQ-P006 – Immediate Purchase (Without PO)**

Priority: High | Status: Proposed

Description: Emergency purchases without prior PO. Mandatory: reason, department, estimated value, vendor. Post-facto PO generation. Management notification. Monthly justification report.

Preconditions: A genuine urgent/emergency operational or clinical need exists that cannot wait for the standard PR-to-PO cycle.

Inputs: Department, item(s), estimated value, vendor (if known), urgency reason, requesting user, department budget status.

Processing Logic: 1. Requesting user or Store Keeper initiates an 'Immediate Purchase' transaction, bypassing the standard multi-step PR workflow. 2. Mandatory fields are captured: reason for immediacy, department, estimated value, and vendor. 3. Purchase is made (cash, card, or direct vendor billing) and goods/services received. 4. System requires a Purchase Order to be generated post-facto within a configurable window (e.g., 24-48 hours) referencing the immediate purchase transaction. 5. Management receives an automatic notification at the time of the immediate purchase, not only after the fact. 6. All immediate purchases for the month are compiled into a justification report reviewed by Management, analyzing frequency, department, and recurring causes.

Outputs: Immediate purchase transaction record, post-facto PO, management notification, monthly justification/exception report.

Postconditions: 1. Every emergency purchase is fully documented and eventually reconciled into the normal PO/GRN/payment cycle. 2. Recurring emergency-purchase patterns are visible to Management for root-cause action (e.g., reorder-level tuning).

Business Rules: 1. Immediate purchases above a configured value threshold require a higher-tier approver's post-facto sign-off, not just notification. 2. If the post-facto PO is not raised within the configured window, the transaction is auto-flagged as a compliance exception. 3. Repeated immediate purchases for the same item by the same department beyond a configurable frequency triggers a reorder-level review recommendation.

Exception Handling: 1. If the vendor used for the emergency purchase is not in the approved vendor master, the transaction is still recorded but flagged for vendor-onboarding follow-up. 2. If Finance later determines the purchase was not genuinely urgent, it is reclassified and routed for the disciplinary/process-improvement workflow defined by hospital policy. 3. If budget is unavailable at the time of post-facto PO creation, the transaction is still honored (goods already received) but the budget variance is separately reported to Finance.



## REQ-P007 – Spot PO Approval at Inward

Priority: High | Status: Proposed

Description: Generate spot PO at inward time for emergencies. Auto-notification for immediate approval. Mobile/email approval within 30 minutes. Auto-convert to regular PO.

Preconditions: Goods have arrived at the inward gate without a prior Purchase Order, and the receiving department confirms genuine urgency.

Inputs: Vendor, item(s) and quantity received, estimated value, department, security/gate entry reference, approver mobile/email.

Processing Logic: 1. Security/Store Keeper logs the inward material and initiates a Spot PO at the point of receipt. 2. System auto-notifies all eligible approvers (per the value-based matrix) simultaneously via mobile and email. 3. Approver reviews and approves/rejects from mobile or web within a target SLA (e.g., 30 minutes). 4. On approval, the Spot PO is auto-converted into a regular PO in the sequential PO numbering series, and a GRN can proceed. 5. If no response within the SLA, the system escalates to the next configured approver or triggers a reminder call/SMS. 6. Goods are held at the inward dock (not moved to stores) until Spot PO approval, unless the item is perishable/critical, in which case a supervised conditional release is logged.

Outputs: Spot PO record, converted regular PO, approval SLA log, GRN eligibility flag, escalation/reminder log.

Postconditions: 1. No goods are formally received into stock without a traceable Spot PO approval. 2. Approval turnaround time is measured and reportable for process-improvement review.

Business Rules: 1. Spot PO approvals still respect the value-based approval matrix; high-value spot receipts require multiple approvers even under the expedited flow. 2. A conditional release of perishable/critical goods before formal approval must be co-signed by the receiving department head and is separately flagged in the audit trail. 3. Spot PO frequency by vendor/department is monitored to detect vendors habitually bypassing the standard PO process.

Exception Handling: 1. If the Spot PO is ultimately rejected after goods have already been used/consumed, the transaction is escalated to Management for a post-facto regularization decision. 2. If quantity/value at inward differs materially from what is eventually invoiced, the variance is logged against REQ-P021 (Price Variation Report). 3. If the vendor is unknown/unregistered, the Spot PO triggers a parallel fast-track vendor registration request.

## REQ-P008 – Cash PO and Auto Cash GRN

Priority: High | Status: Proposed

Description: Cash PO auto-generates Cash GRN. Payment confirmation workflow. Image upload for Cash GRN. Separate numbering series. Petty cash integration.

Preconditions: Petty cash/imprest fund is available to the requesting department or store, and the purchase qualifies under the organization's cash-purchase value ceiling.

Inputs: Item(s), amount, vendor (may be unregistered/local), department, petty-cash ledger reference, cash memo/bill image.

Processing Logic: 1. Store Keeper raises a Cash PO for a low-value, immediate purchase within the configured cash-purchase ceiling. 2. On confirmation of purchase, system auto-generates a corresponding Cash GRN in a distinct numbering series (separate from regular GRNs). 3. User uploads an image/scan of the cash memo/bill against the Cash GRN. 4. Petty-cash ledger is debited automatically and linked to the transaction for reconciliation. 5. Cash GRN routes for a lightweight payment-confirmation workflow (Store Manager/Finance sign-off) rather than the full three-way match used for credit purchases. 6. Monthly petty-cash utilization and replenishment request are generated from cumulative Cash GRN value.

Outputs: Cash PO, Cash GRN (distinct series), uploaded bill image, petty-cash ledger entry, payment confirmation, monthly petty-cash utilization report.

Postconditions: 1. Every cash purchase is traceable to a specific petty-cash debit and an uploaded proof of purchase. 2. Petty-cash balance is always reconcilable to the sum of Cash GRNs raised against it.

Business Rules: 1. Cash purchases above the configured ceiling are blocked and must follow the standard PO route instead. 2. A Cash GRN without an uploaded bill image cannot be marked payment-confirmed. 3. Petty-cash ledger cannot go negative; replenishment must be processed before further cash purchases are permitted.

Exception Handling: 1. If the cash-purchase ceiling is breached inadvertently (e.g., price change at point of purchase), the transaction is flagged for retrospective Finance approval rather than silently accepted. 2. If the bill image is illegible or missing, the Store Keeper is required to obtain a duplicate/attested copy before the Cash GRN can be closed. 3. If the same vendor is repeatedly used for cash purchases beyond a configurable frequency, the system recommends onboarding them as a regular registered vendor with rate-contract discussion.

## REQ-P009 – Quotation vs Invoice Number Validation

Priority: High | Status: Proposed

Description: When PO raised after invoice: capture both Quotation Ref and Invoice No. Indicator compares and validates match. Alert on mismatch. Prevent GRN if unresolved.

Preconditions: A PO is being raised after goods/services have already been quoted and, in some cases, invoiced (e.g., in a Spot PO or Immediate Purchase scenario).

Inputs: Quotation reference number, vendor invoice number, PO value, item-level rate from quotation vs. invoice.

Processing Logic: 1. On PO creation in this scenario, the system requires both the Quotation Reference and the Invoice Number to be captured. 2. System compares quoted rate/terms against invoiced rate/terms, item by item. 3. If values match within a configurable tolerance, the PO is auto-flagged 'Validated – Match'. 4. If a mismatch is found, the system displays the discrepancy and requires the Purchase Executive to select a reason (price revision, quantity change, tax change, error). 5. GRN creation is blocked until the mismatch is either resolved (corrected data) or formally accepted with a documented reason and, where required, Finance sign-off.

Outputs: Quotation-vs-invoice validation flag, discrepancy report, GRN-block/release status, discrepancy reason log.

Postconditions: 1. No GRN is processed against an unresolved quotation-invoice mismatch. 2. All accepted mismatches carry a documented, auditable reason.

Business Rules: 1. Mismatches beyond a configurable percentage threshold require Finance approval even if a reason is provided. 2. Repeated mismatches from the same vendor are surfaced in the vendor performance score. 3. If the quotation reference cannot be located in the system (e.g., informal/telephonic quote), the Purchase Executive must attach supporting evidence before proceeding.

Exception Handling: 1. If the invoice is received before any quotation was formally recorded, the system allows the quotation reference to be entered retrospectively but flags the transaction as 'documented after the fact' for audit visibility. 2. If the vendor disputes the discrepancy, the item is held pending vendor clarification via the vendor portal query/ticket workflow. 3. If the mismatch is due to a GST rate change effective between quotation and invoice dates, the system auto-suggests this as the likely reason based on the applicable GST notification date.

## REQ-P010 – Multi-Currency PO for Imports

Priority: Medium | Status: Proposed

Description: Support USD, EUR, GBP. Store exchange rate at PO creation. Convert to INR for accounting. Track currency fluctuations. Support LC and TT payments.

Preconditions: Vendor is a foreign/import supplier and the transaction is designated as a multi-currency PO at creation.

Inputs: Currency (USD/EUR/GBP/etc.), PO value in foreign currency, exchange rate at PO date, payment mode (LC/TT), Incoterms.

Processing Logic: 1. Purchase Executive selects the transaction currency at PO creation; the system fetches or allows manual entry of the prevailing exchange rate. 2. The exchange rate used is locked and stored against the PO (not recalculated later) to preserve an auditable basis. 3. System computes and displays the INR-equivalent value alongside the foreign-currency value throughout the PO, GRN, and invoice lifecycle. 4. At invoice booking, if the settlement exchange rate differs from the PO-date rate, the system computes and posts the resulting exchange gain/loss to a designated ledger account. 5. Payment mode (Letter of Credit or Telegraphic Transfer) and associated banking references are captured and linked to the PO.

Outputs: Multi-currency PO with locked exchange rate, INR-equivalent valuation, exchange gain/loss entry, LC/TT payment reference tracking.

Postconditions: 1. Every import PO has a clear, non-ambiguous INR valuation for budget and accounting purposes at every stage of its lifecycle. 2. Currency fluctuation impact is isolated and reportable rather than buried in the item cost.

Business Rules: 1. The exchange rate source (RBI reference rate, bank-quoted rate, or manual entry) must be selected and is recorded for audit. 2. Exchange gain/loss beyond a configurable materiality threshold is separately reported to Finance monthly. 3. LC-based POs require the LC reference and validity to be captured before the PO can be marked 'Ready for Shipment'.

Exception Handling: 1. If the exchange rate is not available from the configured source at the time of PO creation, the system allows manual entry with a mandatory note, subject to Finance verification before payment release. 2. If the vendor invoices in a different currency than the PO, the transaction is flagged for Finance review before GRN can proceed. 3. If payment is delayed beyond the LC validity, the system alerts Finance and Purchase to initiate LC extension or renegotiation.

**REQ-P011 – PO Addendums**

Priority: Medium | Status: Proposed

Description: Multiple addendums for shipping, customs, clearing, transportation. Separate line items. Same approval workflow as PO. Included in landed cost. Sequential numbering.

Preconditions: An original PO is already approved and issued, and additional cost elements (shipping, customs, clearing, transportation) are incurred subsequent to the original PO.

Inputs: Original PO reference, addendum type (shipping/customs/clearing/transportation/other), addendum value, supporting document (invoice/bill).

Processing Logic: 1. Purchase Executive creates an Addendum against the original PO, selecting the cost category. 2. Each addendum is captured as a distinct, sequentially numbered line item (Addendum-1, Addendum-2, etc.), never overwriting the original PO. 3. The addendum is routed through the same value-based approval workflow as a standard PO, computed on the addendum value (or the revised cumulative PO value, per policy). 4. Approved addendum values automatically feed into the Landed Cost Calculation (REQ-P012) for the underlying items. 5. The original PO document remains unaltered; a consolidated PO-plus-addendum summary is available for reference.

Outputs: Sequentially numbered addendum records, updated landed cost, consolidated PO-with-addendum summary, approval trail per addendum.

Postconditions: 1. The complete cost history of an import/complex PO is reconstructable from the original PO plus its full addendum chain. 2. Landed cost always reflects the latest approved addendum values.

Business Rules: 1. An addendum cannot be created against a PO that has already been fully closed/paid without reopening it through a documented exception process. 2. Each addendum requires its own supporting document; addenda without evidence are held pending. 3. Cumulative addendum value crossing into a higher approval tier automatically routes to that tier, even if the original PO alone did not require it.

Exception Handling: 1. If an addendum is later found to be erroneous, it is reversed via a compensating (negative) addendum rather than deletion, preserving the audit trail. 2. If multiple departments share the cost of an addendum (e.g., shared shipment), the value is apportioned per a documented cost-sharing rule and each department's budget is debited accordingly.

## REQ-P012 – Landed Cost Calculation

Priority: Medium | Status: Proposed

Description: Base PO + Shipping + Customs + Insurance + Transportation + Other charges. Landed cost per unit. Separate reporting. Impact on depreciation.

Preconditions: A PO (and any addenda) with all applicable cost components – base price, shipping, customs, insurance, transportation, other charges – is available in the system.

Inputs: Base PO value, shipping cost, customs duty, insurance, transportation cost, other charges, item-wise quantity.

Processing Logic: 1. System sums the base PO value and all approved addenda cost components to arrive at total landed cost. 2. Total landed cost is apportioned to item lines using a configurable basis (value-weighted, quantity-weighted, or weight/volume-weighted, depending on cargo type). 3. System computes and stores landed cost per unit for each item, distinct from the base purchase price. 4. Landed cost per unit becomes the basis for inventory valuation and, where the item is capitalized as an asset, the basis for depreciation computation. 5. A separate Landed Cost Report is available showing the base-vs-landed cost variance per item and per shipment.

Outputs: Item-wise landed cost per unit, landed cost report, updated inventory/asset valuation basis.

Postconditions: 1. Inventory and asset values reflect true total cost of acquisition, not just the base PO price. 2. Depreciation for capitalized imported assets is computed on the correct landed cost basis.

Business Rules: 1. The apportionment basis (value/quantity/weight) must be selected at shipment level and applied consistently across all items in that shipment. 2. Landed cost cannot be finalized until all expected addenda for the shipment are either recorded or explicitly marked as 'no further charges expected' by Purchase. 3. Any post-finalization addendum triggers automatic recomputation and a variance report.

Exception Handling: 1. If an expected charge (e.g., customs duty) is not yet known at the time inventory must be received into stock, the system allows a provisional landed cost with a mandatory later true-up once actuals are available. 2. If apportionment produces a materially different unit cost than the vendor's quoted price, the variance is flagged for Finance review before the valuation is finalized.

## REQ-P013 – Online Platform Integration

Priority: Medium | Status: Proposed

Description: Integrate Amazon Business, Flipkart Wholesale, IndiaMART. Import orders via API or email parsing. Auto-create PO. Track credit card purchases. Reconcile payments.

Preconditions: The organization has an active account/API access with the relevant online marketplace (Amazon Business, Flipkart Wholesale, IndiaMART) and marketplace purchases are within policy-approved categories.

Inputs: Marketplace order ID, order details (item, quantity, price), payment method (credit card/marketplace wallet), order confirmation email/API payload.

Processing Logic: 1. Authorized user places an order on the approved marketplace within their delegated limit and category policy. 2. System imports the order via API integration (preferred) or automated email parsing where API access is unavailable. 3. Imported order auto-creates a corresponding internal PO record, tagged as 'Marketplace-Sourced', pre-populated with item, quantity, and price. 4. If paid by credit card, the transaction is tagged for credit-card reconciliation (REQ-G013). 5. On delivery, a GRN is created against the marketplace PO in the normal manner. 6. Marketplace invoice/tax document is auto-attached where the marketplace provides one via API/email, else uploaded manually.

Outputs: Marketplace-sourced PO, imported order data, GRN, credit-card reconciliation tag, attached marketplace invoice.

Postconditions: 1. Marketplace purchases are fully integrated into the standard PO/GRN/payment lifecycle rather than existing as untracked external transactions. 2. Marketplace spend is separately reportable for category and vendor-diversity analysis.

Business Rules: 1. Marketplace purchases above a configured value ceiling require standard PO approval before the order is placed, not after. 2. Only pre-approved item categories (e.g., stationery, general consumables) are permitted via marketplace purchase; clinical/pharmacy items are excluded by policy. 3. Marketplace vendor GST details are validated the same way as any other vendor for ITC eligibility.

Exception Handling: 1. If API import fails, the system falls back to manual entry from the order confirmation, with the source flagged as 'Manually Imported' for audit distinction. 2. If a marketplace order is cancelled/returned after PO creation, the linked PO is auto-cancelled and the credit-card/wallet reversal is tracked to closure. 3. If the marketplace invoice does not carry a valid GSTIN, the purchase is flagged as GST-non-compliant for Finance review of input credit eligibility.

## REQ-P014 – Multiple PO Consolidated Approval Email

Priority: Medium | Status: Proposed

Description: Queue multiple POs. Single consolidated approval email with summary. Approve all or individual. List unapproved POs.

Preconditions: Multiple POs are pending approval from the same approver at the same time.

Inputs: List of pending POs for the approver, PO summary data (vendor, value, department, requester), approver ID.

Processing Logic: 1. System queues all POs pending a given approver's action rather than sending a separate email per PO. 2. At a configurable interval (e.g., twice daily) or on demand, system compiles a single consolidated email/portal notification listing every pending PO with key summary fields. 3. Approver can approve individual POs, approve all listed POs in one action (where policy permits bulk approval), or open any PO for full detail before deciding. 4. Any PO not acted upon remains visible in the next consolidated notification cycle. 5. System maintains a running list of POs pending beyond a configurable aging threshold for escalation.

Outputs: Consolidated approval email/notification, per-PO or bulk approval action log, ageing list of unapproved POs.

Postconditions: 1. Approvers are not overwhelmed by high email volume during busy procurement periods, while every PO still receives an individually logged approval decision. 2. Aging unapproved POs are visible for proactive follow-up.

Business Rules: 1. Bulk 'approve all' is only permitted for POs individually within the approver's configured limit; any PO exceeding the limit is excluded from bulk action and must be opened individually. 2. High-priority/urgent POs (REQ-P017) bypass the consolidated queue and are sent immediately regardless of the batching schedule. 3. Bulk approval still generates one audit-log entry per PO, not a single blanket entry.

Exception Handling: 1. If the approver attempts to bulk-approve a list that includes an over-limit PO, the system silently excludes it from the bulk action and displays a notice rather than blocking the entire batch. 2. If the consolidated notification fails to send (email server issue), the system falls back to individual notifications after a configurable retry window.



## REQ-P015 – Rejected/On-Hold PO with Reason

Priority: Medium | Status: Proposed

Description: Mandatory reason text box on rejection/hold. Sent to creator via email. Track revision history. Report frequently rejected POs.

Preconditions: A PO or PR is under review by an approver who intends to reject or place it on hold rather than approve it.

Inputs: PO/PR ID, decision (Reject/Hold), reason text, approver ID.

Processing Logic: 1. Approver selects Reject or Hold and is presented with a mandatory reason text box; the action cannot be submitted with the field blank. 2. System notifies the originator by email/portal with the full reason text and a direct link to the transaction. 3. Rejected/held transactions retain full revision history if the originator edits and resubmits, so reviewers can see what changed. 4. System maintains a report of frequently rejected/held POs, grouped by department, item category, and reason, for process-improvement review.

Outputs: Rejection/hold notification with reason, revision history, frequently-rejected-PO analytical report.

Postconditions: 1. No PO/PR is rejected or held without a documented, retrievable reason. 2. Recurring rejection patterns are visible to Purchase Management for root-cause correction (e.g., training gaps, unclear item specifications).

Business Rules: 1. A held (not rejected) PO retains its position in the workflow and resumes at the same approval tier once the hold is cleared, rather than restarting the full approval chain. 2. If a PO is rejected and resubmitted three or more times, it is auto-flagged for Purchase Manager review before further routing. 3. Reason text is a mandatory free-text field but should reference a configurable reason-code list where possible, to support the analytical report.

Exception Handling: 1. If the originator does not respond to a hold within a configurable SLA, the transaction is auto-escalated to the originator's manager as a reminder. 2. If a PO is rejected at a later approval tier after already being approved at an earlier tier, the earlier approver is also notified of the reversal for transparency.

## REQ-P016 – PO Cancellation/Modification Notification

Priority: Medium | Status: Proposed

Description: Notify original approver for post-approval changes. Approver can approve/reject. Audit trail. Vendor notification for cancellations.

Preconditions: A PO has already received at least one level of approval and is subsequently modified (quantity, price, terms) or cancelled.

Inputs: PO ID, change type (modification/cancellation), changed fields (before/after), user making the change, original approver(s).

Processing Logic: 1. System detects any edit or cancellation attempt on a PO that has passed at least one approval stage. 2. The original approver(s) at every stage already completed are re-notified with a clear before/after comparison of what changed. 3. Each notified approver can Approve (accept the change) or Reject (revert to original / block cancellation) independently. 4. All approvals must be re-obtained before the modified PO becomes active again; until then, the PO is held in a 'Pending Re-approval' state and cannot proceed to GRN. 5. For cancellations, the vendor is automatically notified once the cancellation is fully approved, and any linked GRN/payment activity is checked for conflicts before finalizing. 6. Every change is written to the audit trail with old and new values side by side.

Outputs: Re-approval notifications, before/after change log, 'Pending Re-approval' PO status, vendor cancellation notice, audit trail entry.

Postconditions: 1. A PO can never be silently altered after approval without every relevant approver being re-informed and re-consenting. 2. Vendors are never left unaware of a cancellation that affects a PO they have already acted upon.

Business Rules: 1. Minor administrative corrections (e.g., typo in a narration field) that do not affect value, quantity, or vendor may be configured as exempt from re-approval, subject to Finance policy. 2. If a GRN or payment has already been processed against the PO, cancellation is blocked and the user is directed to the appropriate return/credit-note workflow instead. 3. Re-approval follows the same value-based matrix as the original approval, recalculated on the revised value.

Exception Handling: 1. If an approver who approved the original PO has since left the organization or changed roles, the re-approval routes to their designated successor. 2. If the vendor has already dispatched goods before a cancellation is approved, the system flags this conflict for Purchase to negotiate a partial acceptance or return arrangement rather than blindly cancelling.

**REQ-P017 – Urgent PO Immediate Email**

Priority: Medium | Status: Proposed

Description: Urgent POs bypass notification queue. Simultaneous email to all approvers. SMS backup. Track approval time. Report urgent PO patterns.

Preconditions: A PO is marked Urgent at creation, based on a genuine operational/clinical need for expedited approval.

Inputs: PO ID, urgency flag, urgency justification, list of eligible approvers, approver contact details (email/mobile).

Processing Logic: 1. Purchase Executive marks the PO as Urgent with a mandatory justification. 2. System bypasses the standard batched/consolidated notification queue (REQ-P014) and sends an immediate, individual email to every eligible approver in the chain simultaneously. 3. An SMS is sent in parallel as a backup channel, given the time-sensitivity. 4. System tracks the time from urgent-flag to each approval action, computing an approval-turnaround metric. 5. A monthly report analyzes urgent-PO patterns by department and item category to identify whether 'urgent' is being used appropriately or as a workaround for poor planning.

Outputs: Immediate multi-channel approval notification, approval-turnaround time metric, urgent-PO pattern report.

Postconditions: 1. Genuinely urgent POs receive expedited attention without waiting for the standard batching cycle. 2. Overuse of the urgent flag is visible to Management for corrective action.

Business Rules: 1. The urgent flag still routes through the full value-based approval hierarchy; it changes notification speed, not the approval requirement. 2. Repeated urgent-flagging by the same department/user beyond a configurable frequency triggers a review by the Purchase Manager. 3. Urgent justification is a mandatory field and is retained in the audit trail alongside the approval turnaround time.

Exception Handling: 1. If an approver does not act on an urgent PO within a tightened SLA (e.g., 2 hours), the system automatically escalates to an alternate/backup approver at the same tier, if configured. 2. If urgency is later found unwarranted, the PO is still processed normally but is included in the monthly pattern report with a 'disputed urgency' tag for review.

## REQ-P018 – PO/GRN Cloning

Priority: Medium | Status: Proposed

Description: Clone existing POs/GRNs for repeated purchases. Copy all except invoice number/date/narration. Allow quantity/value modification. Link to original.

Preconditions: An existing PO or GRN of a similar/repeated nature exists in the system and the user wishes to create a new transaction based on it.

Inputs: Source PO/GRN ID, fields to retain (vendor, items, quantities, rates, terms), fields to exclude (invoice number, invoice date, narration).

Processing Logic: 1. User selects 'Clone' on an existing PO or GRN. 2. System creates a new draft transaction copying all fields except invoice number, invoice date, and free-text narration, which are cleared for fresh entry. 3. User reviews and modifies quantity, rate, delivery date, or any other field as needed for the new transaction. 4. The cloned transaction is linked to its source for traceability ('Cloned from PO-XXXX') but is otherwise a fully independent record requiring its own approval workflow. 5. Cloning does not copy approval status; the new transaction starts fresh in the approval queue.

Outputs: Cloned draft PO/GRN linked to source, independent approval workflow trigger.

Postconditions: 1. Repetitive, similar purchases can be created quickly without re-keying standard details, while still being independently reviewed and approved. 2. Every cloned transaction is traceable to its origin for pattern analysis.

Business Rules: 1. A cloned PO must still pass the full three-competitive-quotation check (or documented exception) if it is a fresh open-market purchase, even though item/vendor details are pre-filled. 2. Rates on a cloned PO should be verified against any active rate contract before submission, since market rates may have changed since the source transaction. 3. Cloning is only available to users with Purchase Executive role or above.

Exception Handling: 1. If the source PO's vendor has since been suspended/blacklisted, cloning is blocked with a notice, and the user is prompted to select an alternate approved vendor. 2. If the source item has been discontinued or superseded in the product master, the system flags this during cloning and requires the user to select the current active item code.

**REQ-P019 – Advance PO Consolidation for Finance**

Priority: Medium | Status: Proposed

Description: Weekly/monthly summary of advance POs by department. Total advance for fund planning. Vendor-wise breakdown. Expected delivery dates. Auto-email to Finance.

Preconditions: One or more advance-payment POs have been raised by departments during the reporting week/month.

Inputs: PO ID, department, vendor, advance amount, expected delivery date, PO status.

Processing Logic: 1. System identifies all POs flagged as requiring advance payment within the reporting period. 2. A weekly and monthly consolidated summary is compiled, grouped by department and by vendor, showing total advance commitment. 3. Expected delivery dates are included to help Finance plan cash outflow timing against expected value realization. 4. The summary is auto-emailed to Finance on a fixed schedule (e.g., every Monday for weekly, first working day for monthly). 5. Finance can drill down from the summary to the individual PO for detail.

Outputs: Weekly/monthly advance-PO summary report, vendor-wise and department-wise breakdown, auto-emailed Finance report.

Postconditions: 1. Finance has proactive visibility into upcoming advance-payment cash requirements rather than discovering them at payment time. 2. Advance exposure by vendor is visible for credit-risk awareness.

Business Rules: 1. A PO is included in the advance-consolidation report from the point its advance-payment flag is set, regardless of approval stage, so Finance sees commitments early. 2. Advance POs significantly overdue against their expected delivery date are highlighted for follow-up in the same report. 3. The report excludes cancelled or fully-settled advance POs automatically.

Exception Handling: 1. If expected delivery date is not provided at PO creation, the system defaults to a standard lead time estimate based on item category and flags it as 'estimated' in the report. 2. If a vendor's cumulative advance exposure crosses a configurable risk threshold, the system adds a warning flag to the vendor's profile for Purchase/Finance attention before further advance POs are raised.

## REQ-P020 – Supplier Price Comparison Chart

Priority: Medium | Status: Proposed

Description: Side-by-side price for same product across suppliers. Historical trend. Alert on >10% variation. Management approval for higher-priced selection.

Preconditions: Two or more quotations for the same or comparable product exist in the system, whether from the current RFQ cycle or historical records.

Inputs: Item code, vendor-wise quoted price, quotation date, historical price records for the same item.

Processing Logic: 1. System retrieves all quotations for the selected item across the shortlisted vendors and displays them side by side. 2. A historical price trend line is plotted using past quotations/PO rates for the same item over a configurable look-back period. 3. If the lowest available quote is not selected, or if the selected price varies more than a configurable percentage (e.g., >10%) from the historical average or the lowest quote, the system raises an alert. 4. Purchase Executive must provide justification for selecting a higher-priced vendor (e.g., quality, delivery, past performance), which routes for Management approval before the PO can proceed. 5. The comparison chart and justification are retained against the PO for audit.

Outputs: Side-by-side price comparison chart, historical trend visualization, high-price-selection alert, justification and approval record.

Postconditions: 1. Every non-lowest-price vendor selection is documented and independently reviewed rather than assumed to be at the buyer's discretion. 2. Price trend visibility helps identify vendors with abnormal or escalating pricing over time.

Business Rules: 1. The variation threshold that triggers mandatory justification is configurable by item category (e.g., tighter tolerance for high-value capital items). 2. Justification reasons should reference the weighted vendor evaluation criteria (Section 5.3) rather than being unstructured free text alone. 3. Historical trend data older than a configurable retention period is archived but remains retrievable on request.

Exception Handling: 1. If insufficient historical data exists for a newly introduced item, the system compares only across the current RFQ's quotations and does not attempt a trend alert. 2. If all quotations for an item are unusually high compared to a benchmark/reference price (e.g., MRP-linked items), the system flags this as a possible market-wide price movement rather than a vendor-specific anomaly.

## REQ-P021 – Price Variation Report with Reason

Priority: Medium | Status: Proposed

Description: If invoice value differs from PO: request reason from vendor. MRP change requires uploaded letterhead. Item/supplier/day-wise report. Management approval for threshold breaches.

Preconditions: A GRN/invoice has been received where the invoiced value or MRP differs from the value on the corresponding PO.

Inputs: PO value, invoice value, item-wise rate variance, vendor reason for variance, supporting document (revised price letterhead where applicable).

Processing Logic: 1. System automatically compares invoice value against PO value at the item level during GRN/invoice entry. 2. Any variance triggers a mandatory 'Reason for Variance' request to the vendor via the vendor portal or manual entry by Purchase. 3. If the variance is due to an MRP change, the vendor is required to upload a formal letterhead communication confirming the revised MRP before the variance can be accepted. 4. System compiles a Price Variation Report, filterable by item, supplier, and day, showing all variances and their accepted reasons. 5. Variances exceeding a configurable threshold require Management approval before the invoice can be passed for payment.

Outputs: Price variation report (item/supplier/day-wise), vendor reason capture, MRP-change letterhead attachment, Management approval for threshold breaches.

Postconditions: 1. No invoice with an unexplained price variance is passed for payment. 2. Systemic vendor-side pricing issues are visible in the aggregated report for renegotiation or vendor performance action.

Business Rules: 1. MRP-based items require documentary evidence (letterhead) for any upward revision; verbal or email confirmation alone is not sufficient. 2. Downward variances (invoice lower than PO) still require a reason to be captured, though they typically do not require the same escalation as upward variances. 3. Repeated variance from the same vendor beyond a configurable frequency is factored into the vendor performance score (Section 5.3 / REQ-V010).

Exception Handling: 1. If the vendor does not respond with a reason within a configurable SLA, the invoice is held and escalated to the Purchase Manager for direct vendor follow-up. 2. If the variance is due to a GST rate change rather than a base-price change, the system distinguishes this in the report so it is not conflated with genuine price escalation.

## 3.2 VENDOR MANAGEMENT & PORTAL

### 3.2 Vendor Management & Portal Process Flow

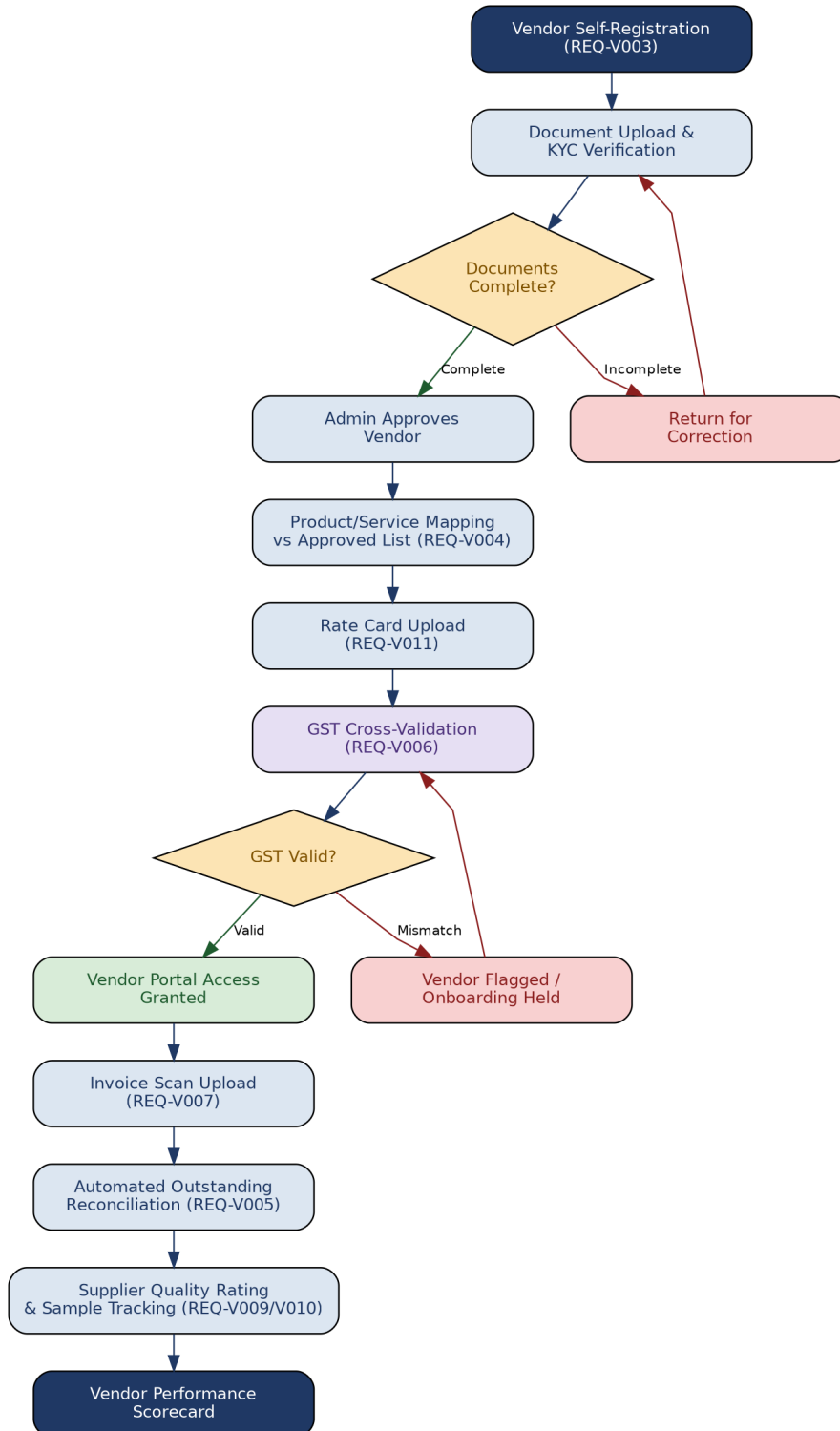


Figure: 3.2 Vendor Management & Portal – Process Flow



## REQ-V001 – Vendor Portal with Excel Import

Priority: High | Status: Proposed

**Description:** The system shall provide a secure, web-based vendor portal accessible via standard internet browsers. Vendors can upload product catalogs, price lists, and rate cards via pre-formatted Excel templates, or key-in product details directly through web forms. The portal serves as the primary communication channel between the hospital and its suppliers.

**Preconditions:** Vendor must be registered and approved by hospital. Portal must be accessible via HTTPS. Excel template must be current version.

**Inputs:** Vendor login credentials, Excel file (product catalog, price list), product details (name, code, rate, GST%, HSN, UOM), rate card data, document uploads (GST certificate, PAN, bank details).

**Processing Logic:** 1. Vendor receives portal activation email with temporary credentials. 2. Vendor logs in and completes profile setup (password change, contact verification). 3. Vendor downloads Excel template from portal (pre-formatted with validation rules). 4. Vendor fills template with product details and uploads via drag-and-drop or file picker. 5. System validates uploaded Excel: mandatory fields, data types, duplicate checks, rate format. 6. System displays validation report: valid rows (green), error rows (red with reason). 7. Vendor corrects errors and re-uploads or keys-in directly for small catalogs. 8. Valid data is staged for hospital approval (Store → Management). 9. Upon approval, products are added to vendor's approved item list. 10. Vendor can view their POs, GRNs, payment status, and outstanding statements. 11. Portal supports multi-language for regional vendors (English, Hindi, Tamil, Telugu).

**Outputs:** Vendor portal account, uploaded product catalog, validation report, approval workflow, approved item list, PO/GRN/Payment visibility.

**Postconditions:** 1. Vendor portal access is granted only after hospital approval of vendor registration. 2. Excel template version is checked; outdated templates are rejected with update instructions. 3. Maximum Excel file size: 10MB. Maximum rows: 5,000 per upload. 4. Vendor can have maximum 3 active upload sessions per day. 5. All portal activities are logged with IP address and timestamp. 6. Vendor portal session expires after 30 minutes of inactivity. 7. Failed login attempts (>5) trigger account lockout for 1 hour.

**Business Rules:** 1. If Excel validation fails, detailed error report is emailed to vendor. 2. If vendor forgets password, self-service reset via email OTP. 3. If portal is down, vendor can email files to designated email for manual processing. 4. If vendor uploads duplicate products, system flags and requests clarification.

Exception Handling: 1. Vendor is registered and approved. 2. Portal is accessible. 3. Excel template is available.

## REQ-V002 – Vendor Payment Status Visibility

Priority: High | Status: Proposed

Description: Vendors shall have real-time visibility into the payment status of their invoices through the vendor portal. This transparency reduces payment inquiry calls to the hospital accounts department and improves vendor satisfaction.

Preconditions: Vendor must be logged in. Invoices must be processed through system (GRN created). Payment data must be updated by accounts.

Inputs: Vendor ID, invoice list, payment status filter, date range filter, PO number filter.

Processing Logic: 1. Vendor logs into portal and navigates to 'Payment Status' section. 2. System retrieves all invoices linked to vendor from GRN and payment tables. 3. System displays invoice list with columns: Invoice No, Invoice Date, PO No, GRN No, Invoice Value, Status (Pending/Processing/Paid), Payment Date (if paid), Payment Ref No, Payment Mode, Outstanding Amount. 4. Vendor can filter by: Date range, Status, PO Number, Invoice Number. 5. Vendor can sort by any column. 6. For paid invoices: Download payment advice note (PDF). 7. For pending invoices: View expected payment date (based on credit period from PO). 8. For processing invoices: View current stage (Accounts verification / Management approval / Bank processing). 9. Outstanding summary at top: Total Invoiced, Total Paid, Total Pending, Overdue Amount. 10. Vendor can download statement of account (Excel/PDF) for any date range. 11. System updates payment status in real-time as accounts team processes payments.

Outputs: Payment status dashboard, invoice list with filters, payment advice download, statement of account, outstanding summary.

Postconditions: 1. Payment status updates within 1 hour of accounts confirmation. 2. Overdue invoices (>credit period) are highlighted in red. 3. Vendor cannot view other vendors' payment data (strict data isolation). 4. Payment advice includes TDS deduction details if applicable. 5. Historical payment data retained for 3 years. 6. Vendor can raise payment query through portal (creates ticket in CRM).

Business Rules: 1. If invoice not yet GRN'd, status shows 'Awaiting GRN'. 2. If payment data not updated, system shows last known status with timestamp. 3. If vendor disputes payment amount, query is routed to accounts team via CRM ticket.

Exception Handling: 1. Vendor is logged in. 2. Invoices exist in system. 3. Payment data is available.

## REQ-V003 – Vendor Registration Form

Priority: High | Status: Proposed

Description: The system shall provide a comprehensive vendor self-registration form capturing all essential business, tax, and banking details. This eliminates manual data entry by hospital staff and ensures data accuracy from the source. All captured data synchronizes automatically to the Finance module.

Preconditions: Registration link must be active. Vendor must have valid email and phone. Mandatory fields must be completed.

Inputs: Company name, registered address, correspondence address, contact person name, designation, phone, email, PAN number, PAN card image (PDF/JPG), GSTIN, GST registration certificate (PDF), TAN number, bank account number, account type, IFSC code, bank name, branch name, cancelled cheque image, MSME registration (if applicable), product categories offered, service categories offered, years in business, major clients, references.

Processing Logic: 1. Vendor accesses registration link (public URL or invitation email). 2. Vendor fills multi-step registration form: Step 1 (Basic Details), Step 2 (Tax Details), Step 3 (Bank Details), Step 4 (Products/Services). 3. System validates each field in real-time: PAN format (ABCDE1234F), GSTIN format (22ABCDE1234F1Z5), email format, phone format. 4. System verifies PAN and GSTIN against government databases (API integration where available). 5. Vendor uploads required documents: PAN card, GST certificate, cancelled cheque. 6. System validates document format and size. 7. Upon completion, vendor submits registration. 8. System generates vendor application ID and sends confirmation email. 9. Application routes to Store team for initial verification. 10. Store verifies business legitimacy and product relevance. 11. Finance verifies tax and bank details. 12. Management gives final approval. 13. Upon approval, vendor receives activation email with portal credentials. 14. All approved vendor data syncs to Finance module automatically. 15. Rejected applications receive reason and option to correct and resubmit.

Outputs: Vendor application, verification workflow, approval/rejection notification, portal credentials, synchronized finance record, audit trail.

Postconditions: 1. Registration is free for vendors. 2. All tax details are mandatory for payment processing. 3. Bank details are verified before first payment. 4. Vendor can save draft and resume later (valid for 7 days). 5. Same PAN/GSTIN cannot be registered twice (duplicate

prevention). 6. Registration approval SLA: 5 business days. 7. Vendor status: Draft → Submitted → Under Verification → Approved / Rejected.

Business Rules: 1. If PAN/GSTIN already exists, system alerts and links to existing vendor record. 2. If document upload fails, vendor can retry or email documents. 3. If verification fails, detailed reason provided with correction instructions. 4. If approval delayed beyond SLA, auto-escalate to management.

Exception Handling: 1. Registration link is valid. 2. Vendor has required documents. 3. Internet connection is available.

## REQ-V004 – Vendor Product/Service Mapping with Approved List

Priority: High | Status: Proposed

Description: Each supplier can only supply items that have been explicitly approved by hospital management. The system maintains an approved item list per vendor and restricts PO creation to approved items only.

Preconditions: Vendor must be approved. Product must exist in master (or creation initiated). Management approval must be obtained.

Inputs: Vendor ID, product ID, vendor product code, vendor product name, rate, GST%, HSN code, UOM, approval status, approved by, approved date, effective date, expiry date.

Processing Logic: 1. Vendor proposes items through portal or hospital staff adds items. 2. System captures: Internal product mapping, vendor's product code, vendor's product name, proposed rate. 3. Item proposal routes to Store team for initial review. 4. Store checks: Product exists in master? If not, initiate product master creation. 5. Department HOD reviews clinical/operational relevance. 6. Management gives final approval with rate negotiation if needed. 7. Approved items appear in PO generation dropdown when vendor is selected. 8. Rejected items are not visible in PO dropdown; vendor notified with reason. 9. Items can be deactivated (not deleted) if no longer required. 10. Rate changes require re-approval (or fall under rate contract). 11. System prevents duplicate product creation by cross-checking names, aliases, and vendor codes.

Outputs: Approved vendor item list, PO generation filter, rejection reason, item activation/deactivation log, duplicate prevention alert.

Postconditions: 1. Only approved items appear in PO dropdown. 2. PO creation is blocked if no approved items exist for selected vendor. 3. Item rate in PO defaults to approved rate but can be modified with justification. 4. Vendor can propose maximum 50 items per month

(configurable). 5. Item approval SLA: 3 business days. 6. Emergency item approval: Store manager can approve with post-hoc management notification (24 hours).

Business Rules: 1. If product not in master, system initiates fast-track product creation workflow. 2. If vendor proposes duplicate, system suggests existing product mapping. 3. If management rejects, vendor can propose alternative or revised rate.

Exception Handling: 1. Vendor is approved. 2. Product master exists or creation is initiated. 3. Approval workflow is configured.

## REQ-V005 – Automated Supplier Outstanding Reconciliation

Priority: High | Status: Proposed

Description: The system shall automate the monthly reconciliation of supplier outstanding amounts by comparing the hospital's accounted invoices against the supplier's uploaded outstanding statement. This eliminates manual reconciliation effort and reduces disputes.

Preconditions: Supplier must have accounted invoices in the period. Supplier must upload statement in prescribed format. Period must be closed for accounting.

Inputs: Hospital's accounted invoice data (invoice no, date, amount, payment status, payment ref), supplier's uploaded Excel statement (invoice no, date, amount, payment received), reconciliation period.

Processing Logic: 1. System generates monthly outstanding statement per supplier from accounted GRN data. 2. System sends email to supplier with: (a) Hospital's statement (Excel), (b) Request to upload their statement. 3. Supplier logs into portal and uploads their outstanding statement in prescribed Excel format. 4. System parses supplier's Excel and extracts: Invoice Number, Invoice Date, Invoice Amount, Payment Received, Balance. 5. System performs three-way matching: Invoice Number + Date + Amount. 6. For matched items: System marks as 'Reconciled' and auto-populates hospital's payment reference number. 7. For unmatched items: System categorizes: (a) Invoice in hospital records but not in supplier statement, (b) Invoice in supplier statement but not in hospital records, (c) Amount mismatch. 8. System generates reconciliation report with: Matched items (green), Mismatched items (red with details), Action required. 9. Report shared with supplier and hospital accounts team. 10. Accounts team investigates mismatches and updates records. 11. System tracks reconciliation status: Pending → In Progress → Reconciled → Disputed → Resolved. 12. Monthly reconciliation certificate generated upon completion.

Outputs: Reconciliation report, matched/unmatched item list, payment reference auto-population, reconciliation certificate, dispute tracking.

Postconditions: 1. Reconciliation is performed monthly for all active suppliers. 2. Supplier has 7 days to upload their statement after receiving hospital's statement. 3. Matching is performed on: Invoice Number (exact, case-insensitive, whitespace-trimmed), Invoice Date ( $\pm 1$  day tolerance), Amount (exact). 4. Partial payments are matched proportionally. 5. Reconciliation data retained for 3 years. 6. Unreconciled items after 30 days escalate to management.

Business Rules: 1. If supplier does not upload statement within 7 days, system sends reminder and then escalates. 2. If supplier's format is incorrect, system rejects with format instructions. 3. If significant mismatch ( $>5\%$  of total outstanding), management review triggered. 4. If invoice number formats differ (e.g., leading zeros), system attempts fuzzy matching with confidence score.

Exception Handling: 1. Supplier has transactions in period. 2. Hospital statement is generated. 3. Supplier uploads valid format.

## REQ-V006 – GST Cross-Validation

Priority: High | Status: Proposed

Description: The system shall cross-validate the hospital's accounted GST data with the supplier's uploaded GST returns data. This identifies discrepancies in GST percentages, invoice values, and missing invoices from either party's records.

Preconditions: Supplier must upload GSTR data. Hospital must have accounted invoices for the supplier in the period. GST configuration must be active.

Inputs: Hospital's GST data: Invoice number, invoice date, taxable value, GST amount, GST% (CGST/SGST/IGST), HSN code. Supplier's uploaded GSTR data: Invoice number, invoice date, taxable value, GST amount, GST%, HSN code.

Processing Logic: 1. Supplier uploads monthly GSTR-1 data (Excel) to vendor portal. 2. System parses and structures supplier's GST data. 3. System retrieves hospital's accounted GST data for the same period and supplier. 4. System performs matching on: Invoice Number + Taxable Value + GST%. 5. For matched items: Mark as 'GST Verified'. 6. For mismatched items, system categorizes discrepancies: a) GST% mismatch (e.g., hospital accounted 12%, supplier filed 18%) b) Taxable value mismatch (difference  $>$  Rs. 10) c) Invoice in hospital records but missing in supplier's GSTR d) Invoice in supplier's GSTR but missing in hospital records 7. System generates GST reconciliation report with discrepancy details. 8. Report shared with accounts team for correction. 9. System tracks correction status: Pending  $\rightarrow$  Corrected  $\rightarrow$  Verified. 10. Monthly GST validation summary for management review.

Outputs: GST reconciliation report, discrepancy list, correction tracking, monthly validation summary, input credit protection alert.

Postconditions: 1. GST cross-validation is performed monthly for all GST-registered suppliers. 2. Tolerance for taxable value mismatch: Rs. 10 (configurable). 3. GST% mismatch is flagged regardless of value difference. 4. Missing invoices in supplier's GSTR may indicate non-filing or delayed filing. 5. Missing invoices in hospital records may indicate unaccounted purchases. 6. Both scenarios require investigation and correction before GST filing. 7. Data retained for 5 years (GST audit requirement).

Business Rules: 1. If supplier does not upload GSTR data, system sends reminders and flags for manual verification. 2. If GSTR format is incorrect, system provides template and rejects upload. 3. If significant discrepancies found (>5 invoices or >Rs. 10,000 value), escalate to GST consultant.

Exception Handling: 1. Supplier is GST-registered. 2. GSTR data is uploaded. 3. Hospital has accounted invoices.

## REQ-V007 – Supplier Invoice Scan Upload

Priority: Medium | Status: Proposed

Description: Suppliers shall upload scan copies of their submitted invoices directly through the vendor portal. This eliminates the hospital's manual scanning effort and ensures document quality from the source.

Preconditions: Vendor must be logged in. Invoice must not already exist in system. File must meet format and size requirements.

Inputs: Invoice image/PDF, invoice number, invoice date, PO reference, file naming convention compliance, vendor ID.

Processing Logic: 1. System defines file naming convention: {SupplierCode}\_{InvoiceNo}\_{InvoiceDate}.pdf Example: ABC001\_INV1234\_15JAN2026.pdf 2. Vendor uploads invoice scan through portal. 3. System validates file name against convention. 4. If naming is incorrect, system rejects with correct format example. 5. System validates file format (PDF, JPG, PNG) and size (<5MB). 6. System performs OCR (Optical Character Recognition) to extract: Invoice Number, Date, Amount, Vendor Name. 7. OCR data is displayed for vendor verification. 8. Vendor confirms or corrects OCR data. 9. Store team verifies uploaded invoice against physical copy during GRN. 10. Verified invoices are stored in document management system with indexing. 11. System eliminates need for hospital staff to scan and rename invoices.

Outputs: Uploaded invoice document, OCR extraction data, verification status, document storage path, store verification record.

Postconditions: 1. File naming convention is mandatory and enforced. 2. OCR confidence score must be >80%; otherwise manual verification required. 3. Vendor can upload maximum 20 invoices per day. 4. Uploaded invoices are visible to store team within 1 hour. 5. Document storage is organized by year/month/vendor for easy retrieval. 6. Invoices are retained for 7 years (tax compliance).

Business Rules: 1. If file name incorrect, system provides rename tool before upload. 2. If OCR fails, system allows manual data entry with upload. 3. If duplicate invoice detected, system blocks upload and alerts vendor. 4. If file size exceeds limit, system suggests compression or split upload.

Exception Handling: 1. Vendor is logged in. 2. Invoice is new. 3. File meets requirements.

## REQ-V008 – Supplier Address/Bank Change Management

Priority: Medium | Status: Proposed

Description: The system shall provide an online interface for suppliers to manage their addresses and bank account details, similar to e-commerce sites. This prevents duplicate vendor creation when minor details change.

Preconditions: Vendor must be logged in. New address/bank details must be valid. Change reason must be provided.

Inputs: Vendor ID, address type (Registered/Correspondence/Warehouse), address details, bank account type (Primary/Secondary), account number, IFSC code, bank name, branch, cancelled cheque image, change reason, effective date.

Processing Logic: 1. Vendor logs into portal and navigates to 'Profile Management'. 2. Vendor can: Add new address, Edit existing address, Mark address as primary, Deactivate old address. 3. For bank changes: Add new account, Set as primary, Deactivate old account. 4. System validates: IFSC code format, account number (numeric), bank name match with IFSC. 5. For bank changes, vendor uploads cancelled cheque of new account. 6. Change request routes to Finance team for verification. 7. Finance verifies bank details (optional: penny drop verification). 8. Upon approval, new address/bank becomes effective from specified date. 9. Old address/bank marked as inactive but retained for historical transactions. 10. System prevents creation of duplicate vendor records for address/bank changes.



Outputs: Updated vendor profile, change approval workflow, historical address/bank record, duplicate prevention alert, effective date tracking.

Postconditions: 1. Address/bank changes require Finance approval before becoming effective. 2. Changes effective date can be future-dated (e.g., new bank effective from next month). 3. At least one active address and one active bank account must exist. 4. Deactivated records are retained for 3 years for audit. 5. System checks for duplicate PAN/GSTIN before allowing new vendor creation. 6. If duplicate detected, system suggests updating existing vendor instead.

Business Rules: 1. If Finance rejects change, vendor is notified with reason and can resubmit. 2. If bank verification fails, vendor must provide corrected details. 3. If vendor tries to create new account instead of updating, system detects duplicate PAN and redirects to update.

Exception Handling: 1. Vendor is logged in. 2. New details are valid. 3. Change reason is provided.

## REQ-V009 – Sample Follow-up Tracking

Priority: Medium | Status: Proposed

Description: The system shall track sample requests, submissions, evaluations, and approvals between suppliers, departments, and stores. This ensures proper evaluation before committing to new suppliers or products.

Preconditions: Supplier must be registered (or fast-track registration). Management approval must be obtained before sample acceptance. HOD must be assigned.

Inputs: Sample request ID, supplier ID, product details, sample quantity, requesting department, HOD ID, management approval reference, sample submission date, evaluation criteria, evaluation result.

Processing Logic: 1. Supplier or hospital staff initiates sample request. 2. System captures: Supplier details, Product details, Sample quantity, Requesting department, Clinical justification. 3. Management approval required before sample acceptance (prevents unsolicited samples). 4. Upon approval, supplier dispatches sample with tracking reference. 5. Stores receives sample and logs receipt with condition notes. 6. Sample forwarded to requesting department HOD for evaluation. 7. HOD evaluates sample against criteria: Quality, Suitability, Packaging, Expiry (if applicable). 8. HOD submits evaluation through software or SMS link (for convenience). 9. Evaluation options: Approved / Rejected / Approved with Conditions. 10. If approved, supplier is authorized for first purchase order. 11. If rejected, reason is recorded and supplier notified. 12. System generates sample pipeline report for management. 13. Approved samples linked to first PO for traceability.

Outputs: Sample request record, approval workflow, evaluation form, HOD feedback, sample status report, first PO linkage.

Postconditions: 1. Samples are accepted only after management approval. 2. HOD evaluation is mandatory within 7 days of sample receipt. 3. Rejected samples are returned to supplier at supplier's cost. 4. Approved samples trigger vendor notification for rate negotiation and PO readiness. 5. Sample tracking is separate from regular inventory (not stock-accounted). 6. Maximum 5 active sample requests per supplier at any time.

Business Rules: 1. If management does not approve within 3 days, request auto-expires. 2. If HOD does not evaluate within 7 days, escalate to management. 3. If sample damaged in transit, stores logs condition and requests re-submission.

Exception Handling: 1. Supplier is registered. 2. Management approves sample acceptance. 3. HOD is available.

## REQ-V010 – Supplier Quality Rating

Priority: Medium | Status: Proposed

Description: The system shall capture department-based quality feedback for products delivered by suppliers. This rating influences future supplier selection and price comparison rankings.

Preconditions: GRN must be confirmed. Product must be issued to department. HOD must be active. Rating must be within 1-5 scale.

Inputs: Supplier ID, product ID, department ID, delivery date, rating parameters (Quality, Timeliness, Packaging, Documentation), rating scale (1-5), feedback text, rater ID (HOD/Department staff).

Processing Logic: 1. After first order delivery from new supplier, system prompts department HOD for quality rating. 2. HOD receives email with rating link 3 days after GRN confirmation. 3. HOD rates on parameters: Product Quality (1-5), Delivery Timeliness (1-5), Packaging Condition (1-5), Documentation Accuracy (1-5). 4. HOD can add free-text feedback. 5. For repeat orders, system prompts for rating every 3 months or after every 5 deliveries (configurable). 6. System calculates weighted average rating per supplier-product combination. 7. System calculates overall supplier rating across all products. 8. Rating is displayed in price comparison chart (weight: 30%). 9. If rating drops below 3.0, system alerts store manager and management. 10. Complaint history is tracked separately and linked to ratings. 11. Management can view rating trends and make vendor retention decisions.

Outputs: Quality rating record, supplier scorecard, rating trend chart, low-rating alert, complaint history, management dashboard.

Postconditions: 1. Rating is mandatory for first order from new supplier. 2. Anonymous ratings are not allowed; rater identity is recorded. 3. Ratings can be updated within 7 days of submission. 4. Supplier cannot view individual ratings (only aggregate score). 5. Rating data is retained for supplier relationship duration + 2 years. 6. Low-rated suppliers (<3.0 for 2 consecutive quarters) are flagged for review.

Business Rules: 1. If HOD does not rate within 7 days, system sends reminder to HOD and notification to store manager. 2. If rating seems inconsistent with complaint history, system flags for verification. 3. If supplier disputes rating, management reviews and can adjust if justified.

Exception Handling: 1. GRN is confirmed. 2. Product is issued. 3. HOD is active.

## REQ-V011 – Rate Card and Vendor Details Upload Format

Priority: Medium | Status: Proposed

Description: The system shall provide standardized Excel templates for vendors to upload rate cards and business details, ensuring data consistency and reducing manual entry errors.

Preconditions: Template must be current version. Vendor must be logged in. File must be in .xlsx format.

Inputs: Excel template with columns: Item Code, Item Name (as per vendor), Unit, Rate (per unit), GST%, HSN Code, Effective Date, Expiry Date, Minimum Order Quantity, Remarks. Vendor details template: Company info, contact, tax, bank.

Processing Logic: 1. System generates and publishes Excel templates with: (a) Rate Card Template, (b) Vendor Details Template. 2. Templates include: Column headers, Data type hints, Example rows, Validation rules sheet. 3. Vendor downloads template from portal. 4. Vendor fills data and uploads. 5. System validates: Mandatory fields populated, Numeric fields contain numbers, Date fields valid, GST% in valid range (0/5/12/18/28). 6. System shows preview of valid data before commit. 7. Vendor can edit preview data or commit. 8. Committed data routes to hospital approval workflow. 9. Error report generated for invalid rows with specific error descriptions. 10. Version control: Each upload creates new version; previous versions archived.

Outputs: Standardized template, uploaded data preview, validation report, approval workflow, version history, error correction guidance.

Postconditions: 1. Template version is embedded in file; outdated templates rejected. 2. Maximum 5,000 rows per upload. 3. Rate effective date cannot be in the past (unless admin override). 4. Duplicate item codes within same upload are flagged. 5. Rate changes >20% from previous rate require management approval. 6. Template updates are versioned and vendors notified of new versions.

Business Rules: 1. If template version outdated, system provides download link for latest version. 2. If validation errors exist, vendor can download error report, correct, and re-upload. 3. If upload fails due to network, system supports resume from last successful row.

Exception Handling: 1. Vendor is logged in. 2. Template is current version. 3. File is valid .xlsx.

## REQ-V012 – Supplier Product Images

Priority: Medium | Status: Proposed

Description: Suppliers shall upload product images (front and back views) through the vendor portal. These images help hospital staff identify correct products across different supplier naming conventions.

Preconditions: Product must be approved. Vendor must be logged in. Images must meet format and size requirements.

Inputs: Product ID, image files (front view, back view), image format (JPG/PNG), image size (<2MB per image), vendor ID.

Processing Logic: 1. System requests suppliers to upload product images within defined deadline (e.g., 30 days from product approval). 2. Supplier logs into portal and navigates to 'Product Images' section. 3. Supplier selects product and uploads front view image (mandatory) and back view image (optional). 4. System validates: Format (JPG/PNG), Size (<2MB), Dimensions (minimum 500x500 pixels). 5. System generates thumbnail (150x150) and full-size view. 6. Images are displayed in: Product master, PO creation screen, Indent screen, GRN verification screen. 7. Store team can verify physical goods against uploaded images during receipt. 8. Images help resolve naming confusion (e.g., same product called differently by different wards). 9. System tracks image upload compliance per supplier. 10. Non-compliant suppliers receive reminders and may be restricted from new POs.

Outputs: Product image gallery, thumbnail view, full-size view, upload compliance report, image-based product search.

Postconditions: 1. Front view image is mandatory; back view is recommended. 2. Images must be clear and representative of actual product. 3. Watermarked images are acceptable if

watermark does not obscure product details. 4. Image upload deadline: 30 days from product approval (configurable). 5. Non-compliance after 2 reminders restricts new PO creation for non-imaged products. 6. Images are stored in cloud storage with CDN for fast loading.

Business Rules: 1. If image upload fails, vendor can retry or email images to stores. 2. If image quality is poor, store team can request re-upload with guidelines. 3. If supplier does not have images, system allows placeholder with 'Image Pending' status.

Exception Handling: 1. Product is approved. 2. Vendor is logged in. 3. Images meet requirements.

## REQ-V013 – Supplier Referral Tracking

Priority: Low | Status: Proposed

Description: The system shall track supplier referrals made by doctors, ward staff, HODs, or management for alternative items or new suppliers. This creates an audit trail for future reference and supplier evaluation.

Preconditions: Referrer must be registered user. Supplier details must be provided. Reason must be at least 30 characters.

Inputs: Referrer ID, referrer department, supplier details, product details, referral reason, referral date, ticket ID, management approval.

Processing Logic: 1. Doctor/Ward/HOD/Management identifies alternative supplier or product. 2. User creates referral entry through CRM ticket or dedicated referral form. 3. System captures: Referrer name, Department, Supplier details, Product details, Reason for referral, Expected benefits. 4. Referral is logged as a ticket in CRM module with unique reference number. 5. Store team reviews referral and initiates vendor evaluation if new supplier. 6. If existing supplier, store checks if product is already in approved list. 7. Management approval required for referral-based supplier onboarding. 8. System tracks referral outcome: Approved/Rejected/Under Evaluation. 9. Report on referral patterns: Who referred, What was referred, Outcome, Time to decision. 10. Future reference: When evaluating same supplier/product, system shows previous referral history.

Outputs: Referral ticket, supplier evaluation record, approval workflow, referral outcome report, historical reference.

Postconditions: 1. Referral does not guarantee supplier approval; standard evaluation applies. 2. Self-referral (referring own relative's company) must be disclosed and requires management approval. 3. Referral data is confidential; only store manager and management

can view full details. 4. Successful referrals are acknowledged in annual review. 5. Referral tracking is for audit and transparency, not for commission or incentive.

Business Rules: 1. If supplier already exists, system suggests updating existing record instead of new referral. 2. If referrer does not provide sufficient details, system requests clarification before processing. 3. If referral is rejected, referrer is notified with reason.

Exception Handling: 1. Referrer is registered user. 2. Supplier details are provided. 3. Reason is documented.

## 3.3 INVENTORY & STOCK CONTROL

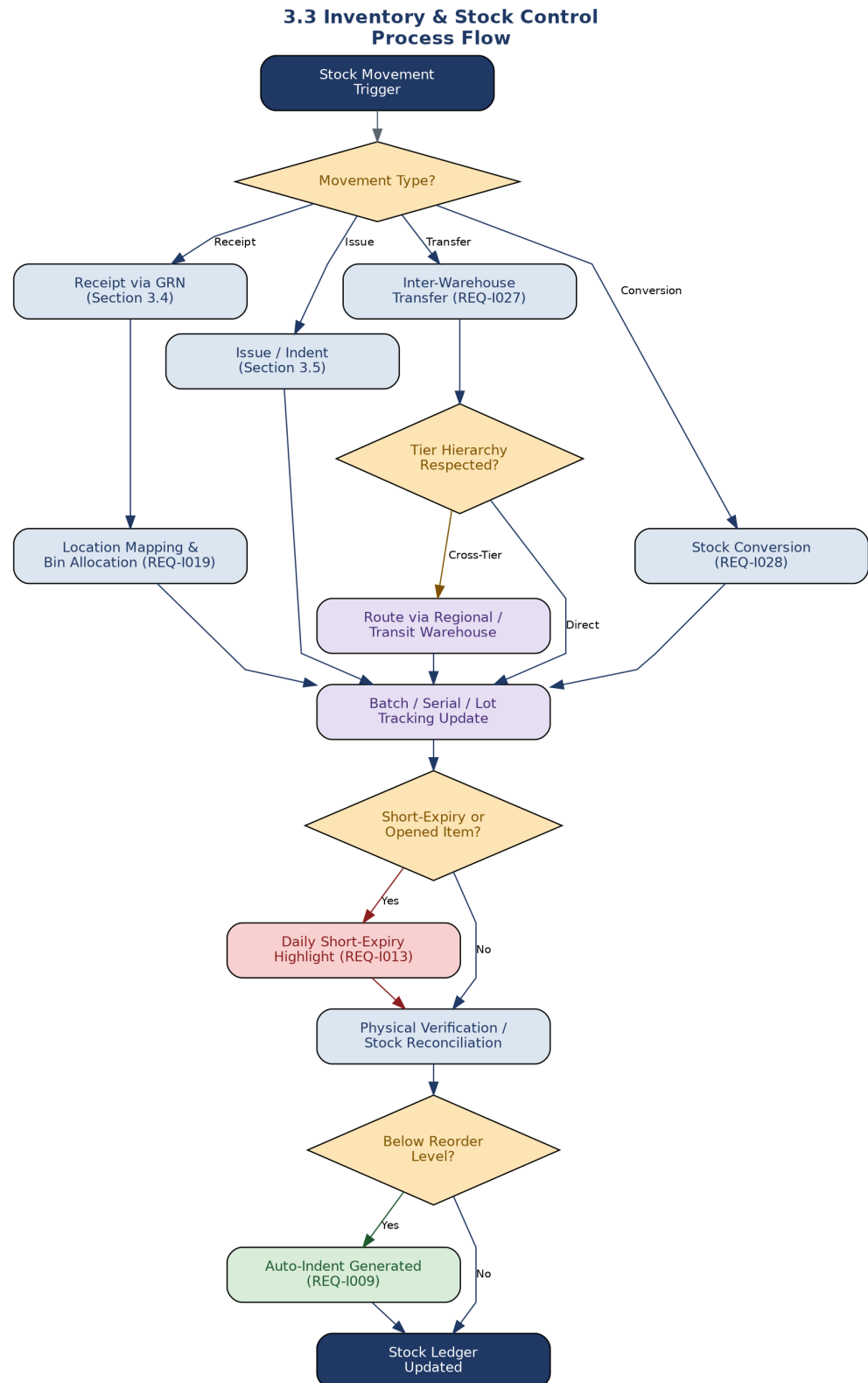


Figure: 3.3 Inventory & Stock Control – Process Flow

## REQ-I001 – Barcode Printing for Assets

Priority: High | Status: Proposed

Description: The system shall generate and print barcode labels for all assets to enable accurate tracking throughout their lifecycle. Barcodes are used during receipt, issue, transfer, maintenance, and disposal operations.

Preconditions: Asset must be registered in system. Printer must be configured and accessible. User must have label printing privilege.

Inputs: Asset ID, asset code, product name, department, purchase date, label format (Code 128 / QR Code), printer selection.

Processing Logic: 1. During asset receipt (GRN), system auto-generates unique asset barcode. 2. System supports Code 128 (1D) for standard scanning and QR Code (2D) for detailed information embedding. 3. Label content includes: Asset Code (human-readable), Product Name (abbreviated), Department Code, Purchase Date (MM/YY), Barcode/QR image. 4. User selects label format and printer from configured list. 5. System generates print-ready label image (PDF). 6. User can print single label or bulk print (up to 100 labels per batch). 7. For damaged/lost labels, user can search by asset code and reprint. 8. QR Code contains: Asset Code, Serial Number, Product ID, Department ID (encoded URL for mobile scanning). 9. Barcode scanning integration: During issue/return/transfer, scan populates asset details automatically. 10. Compatible with Zebra ZT230, TSC TTP-244 Pro, and other standard thermal transfer printers. 11. Label size options: 50x25mm, 75x50mm, 100x50mm (configurable).

Outputs: Printed barcode labels, print log, reprint history, asset scan records, label inventory report.

Postconditions: 1. Each asset receives exactly one unique barcode at time of receipt. 2. Barcode is immutable; asset code never changes even if asset is transferred between departments. 3. Reprint requires reason and is logged for audit. 4. Label material recommendation: Polyester for durability (assets), Paper for consumables. 5. Barcode scanning updates asset location in real-time. 6. Lost label reprint requires supervisor approval if >2 reprints per asset.

Business Rules: 1. If printer offline, system queues print job and retries every 5 minutes. 2. If label format not supported by printer, system alerts and suggests compatible format. 3. If asset code not found, system suggests search by product name or serial number.

Exception Handling: 1. Asset exists in system. 2. Printer is configured. 3. User has printing privilege.



## REQ-I002 – Vendor Code Binding with Products

Priority: High | Status: Proposed

Description: The system shall map multiple vendor-specific product codes (SKUs) to a single internal hospital product code. This enables consistent inventory tracking regardless of which supplier provides the item.

Preconditions: Internal product must exist in master. Vendor must be approved. Vendor product code must be unique per vendor.

Inputs: Internal product ID, internal product code, internal product name, vendor ID, vendor product code, vendor product name, vendor UOM, conversion factor (if UOM differs), packaging details, alias names/tags.

Processing Logic: 1. During product master creation or vendor item approval, system captures vendor's product code and name. 2. System stores mapping: Internal Product Code → Multiple Vendor Product Codes. 3. Each vendor mapping includes: Vendor Product Code, Vendor Product Name, Vendor UOM, Conversion Factor, Packaging (e.g., Box of 10). 4. During PO creation, when vendor is selected, system shows vendor's product name but links to internal product. 5. During GRN, system accepts vendor's product code scan and maps to internal product automatically. 6. System maintains cross-reference table searchable by any code (internal or vendor). 7. Alias names/tags are captured for products known by different names in different wards (e.g., 'Betadine' vs 'Povidone Iodine'). 8. System prevents duplicate product creation by checking: Internal name, All aliases, All vendor codes. 9. If potential duplicate detected, system shows similarity score and existing product details. 10. Admin can merge duplicate products with history preservation.

Outputs: Vendor-product mapping table, cross-reference search, duplicate detection alert, product merge record, alias dictionary.

Postconditions: 1. One internal product can have unlimited vendor mappings. 2. Vendor product code must be unique within that vendor (cannot map two vendor codes to same internal product for same vendor). 3. Alias search is case-insensitive and supports partial matching. 4. Duplicate prevention uses fuzzy matching (Levenshtein distance) with 85% similarity threshold. 5. Product merge preserves all historical transactions with original and new product references. 6. Only admin can perform product merge; requires management approval for merged products with transaction history.

Business Rules: 1. If vendor code already mapped to different internal product, system alerts and requests clarification. 2. If alias conflicts with existing product, system suggests adding

as alias instead of new product. 3. If merge attempted on products with active stock, system requires stock transfer before merge.

Exception Handling: 1. Internal product exists. 2. Vendor is approved. 3. Vendor code is unique for vendor.

## REQ-I003 – Warranty/AMC/CMC Reminders

Priority: High | Status: Proposed

Description: The system shall track warranty periods, Annual Maintenance Contracts (AMC), and Comprehensive Maintenance Contracts (CMC) for all assets and equipment, sending automated reminders at predefined intervals before expiry.

Preconditions: Asset must be registered. Contract dates must be valid (end > start). Notification recipients must be configured.

Inputs: Asset ID, warranty start date, warranty end date, AMC/CMC type, contract vendor, contract value, coverage details, reminder schedule (90/60/30/15 days), notification recipients (department, stores, management).

Processing Logic: 1. During asset receipt, user captures warranty details: Start Date, End Date, Warranty Terms. 2. For AMC/CMC: User creates contract record linked to asset with: Vendor, Contract Period, Value, Coverage (Labour/Parts/Both), Response Time SLA. 3. System calculates reminder dates based on end date minus reminder schedule. 4. At each reminder interval, system sends: Email to department HOD and stores manager, Dashboard alert, Escalation to management at 15-day mark. 5. Reminder content includes: Asset details, Contract expiry date, Days remaining, Recommended action (Renew/Replace/Evaluate). 6. User can initiate renewal workflow directly from reminder. 7. Renewal workflow: Generate RFQ to existing vendor + alternate vendors, Compare quotes, Management approval, Contract update. 8. If not renewed by expiry date, asset status changes to 'Warranty Expired' and maintenance becomes chargeable. 9. System maintains contract history: All past AMC/CMC with costs and vendor performance. 10. Cost comparison report: AMC vs CMC vs Ad-hoc repair costs per asset type.

Outputs: Reminder emails/dashboard alerts, contract renewal workflow, expiry status update, cost comparison report, contract history.

Postconditions: 1. Reminders are sent at 90, 60, 30, and 15 days before expiry (configurable). 2. 15-day reminder is escalated to management with urgent flag. 3. Contract renewal must be initiated 30 days before expiry for uninterrupted coverage. 4. AMC/CMC contracts cannot overlap; new contract effective date must be after old contract end date. 5. Contract value is

tracked against department budget. 6. Vendor performance during contract period (response time, resolution quality) is recorded for renewal decision.

Business Rules: 1. If asset disposed before contract expiry, contract is pro-rata cancelled with refund tracking. 2. If vendor becomes inactive, system suggests alternate vendors for renewal. 3. If contract value exceeds budget, system blocks renewal and alerts management.

Exception Handling: 1. Asset is registered. 2. Contract dates are valid. 3. Recipients are configured.

## REQ-I004 – Preventive Maintenance by Departments

Priority: High | Status: Proposed

Description: The system shall schedule and track preventive maintenance activities for all assets, assigning responsibility to respective departments and ensuring compliance with maintenance schedules.

Preconditions: Asset must be registered and active. Maintenance schedule must be defined. Responsible person must be assigned.

Inputs: Asset ID, maintenance type, frequency (daily/weekly/monthly/quarterly/half-yearly/yearly), maintenance checklist, assigned department, responsible person, due date, completion date, maintenance cost, spare parts used.

Processing Logic: 1. Admin or store manager defines maintenance schedule per asset type: Frequency, Checklist items, Estimated duration, Estimated cost. 2. System auto-generates maintenance calendar for each asset based on schedule. 3. Maintenance tasks appear on responsible person's dashboard with due date. 4. Responsible person performs maintenance and marks checklist items as completed. 5. System captures: Actual completion date, Time taken, Spare parts consumed (if any), Cost incurred, Next due date (auto-calculated). 6. If maintenance overdue by 3 days, system sends escalation email to HOD and store manager. 7. If maintenance overdue by 7 days, system escalates to management and marks asset as 'Maintenance Overdue'. 8. Department-wise maintenance compliance report: Scheduled vs Completed, On-time %, Overdue count. 9. Maintenance cost is tracked per asset and rolled up to department budget. 10. Critical equipment (life support, OT equipment) has stricter SLA: Overdue by 1 day triggers immediate escalation.

Outputs: Maintenance calendar, completion records, overdue alerts, compliance report, maintenance cost analysis, asset status updates.

Postconditions: 1. Preventive maintenance is mandatory for all assets except consumables. 2. Maintenance frequency is defined per asset category (e.g., AC: quarterly, Generator: monthly, UPS: monthly). 3. Department HOD is accountable for maintenance compliance in their department. 4. Third-party AMC/CMC maintenance is also tracked through this module. 5. Maintenance records are required for warranty claims and audit. 6. Asset cannot be transferred if maintenance is overdue.

Business Rules: 1. If responsible person on leave, system reassigns to department backup or HOD. 2. If spare parts not available for maintenance, system auto-generates indent for parts. 3. If maintenance cost exceeds estimate by >50%, management approval required.

Exception Handling: 1. Asset is active. 2. Schedule is defined. 3. Responsible person is assigned.

## REQ-I005 – Product Lifecycle Management

Priority: High | Status: Proposed

Description: The system shall track the complete lifecycle of products from procurement through active use to final disposal, managing status transitions with proper approvals and audit trails.

Preconditions: Product/asset must exist in system. User must have status change initiation privilege. Approver must be configured for target status.

Inputs: Product/Asset ID, current status, transition request (new status), reason for transition, approver ID, transition date, supporting documents (disposal approval, sale invoice, damage report).

Processing Logic: 1. System defines standard lifecycle statuses: Active → In-Use → Under Maintenance → Idle → Red-Tag → Scrap / Sold. 2. Each status transition requires approval: Store Manager (Idle, Red-Tag), Management (Scrap, Sold). 3. User initiates status change with reason and supporting documents. 4. System routes to approver based on target status. 5. Approver reviews and approves/rejects with comments. 6. Upon approval, system updates status and logs: Old status, New status, Reason, Approved by, Date. 7. For Scrap: System calculates book value (original cost - accumulated depreciation), requires disposal committee approval, generates disposal certificate. 8. For Sold: System captures sale value, buyer details, payment status, generates sale invoice. 9. For Red-Tag: System prints physical red tag label with reason and date, asset is quarantined from issue. 10. Lifecycle duration report: Average time in each status by product category. 11. System prevents issue of assets in 'Under Maintenance', 'Red-Tag', or 'Scrap' status.

Outputs: Status transition record, approval workflow, disposal certificate, sale invoice, red-tag label, lifecycle duration report.

Postconditions: 1. Status transitions are irreversible without management approval (e.g., Scrap cannot revert to Active). 2. Asset in 'Idle' status for >180 days triggers management review for disposal decision. 3. Scrap disposal requires physical verification by stores and accounts team. 4. Sale of assets requires minimum 3 quotations for fair value (if value > Rs. 10,000). 5. All lifecycle data is retained for 7 years for audit and tax compliance. 6. Depreciation stops when asset reaches 'Scrap' or 'Sold' status.

Business Rules: 1. If asset has pending maintenance, status change to Scrap/Sold is blocked until maintenance is completed or cancelled. 2. If asset is under AMC/CMC, contract must be cancelled before Scrap/Sold transition. 3. If book value > Rs. 50,000, management approval is mandatory regardless of status.

Exception Handling: 1. Asset exists. 2. User has privilege. 3. Approver is configured.

## REQ-I006 – Product Status Marking (Scrap/Sold/Red-Tag)

Priority: High | Status: Proposed

Description: The system shall provide distinct status markings for products that are no longer in active use: Scrap (damaged beyond repair), Sold (disposed through sale), and Red-Tag (out of use pending decision). Each status has specific workflows and accounting implications.

Preconditions: Asset must be in Active/In-Use/Idle status. User must have status change privilege. Book value must be calculated.

Inputs: Asset ID, status type (Scrap/Sold/Red-Tag), reason (mandatory), supporting documents, book value, approved by, approved date.

Processing Logic: 1. User selects asset and initiates status change. 2. System displays current status and book value. 3. User selects target status and enters detailed reason (minimum 50 characters). 4. For Scrap: User uploads damage report or inspection report. System calculates remaining book value. 5. For Sold: User enters sale value, buyer name, buyer contact, payment mode. If sale value > Rs. 10,000, 3 quotations required. 6. For Red-Tag: User enters reason for quarantine (e.g., 'Quality complaint from ward', 'Pending calibration'). 7. System routes to approver based on status type and book value. 8. Upon approval: Scrap → Write off book value, generate disposal certificate, update asset register. Sold → Generate sale receipt, record payment, update asset register. Red-Tag → Print red-tag label, quarantine asset, set review date (default 30 days). 9. Red-tagged assets reviewed after 30 days: Decision to Restore,

Scrap, or Keep on Hold. 10. All status changes are posted to Tally with appropriate journal entries (loss on disposal, gain/loss on sale).

Outputs: Status change record, disposal certificate, sale receipt, red-tag label, journal entries, asset register update.

Postconditions: 1. Scrap status requires physical verification by stores and accounts before approval. 2. Sold status requires payment receipt before asset handover. 3. Red-Tag status maximum duration: 90 days (configurable). After 90 days, auto-convert to Scrap if no decision. 4. Assets under loan/guarantee cannot be marked Scrap/Sold without lender approval. 5. Environmental hazardous materials require special disposal documentation. 6. Sale of assets to employees requires management approval and fair value assessment.

Business Rules: 1. If asset has pending issue/transfer, status change is blocked until transactions are completed. 2. If book value is negative (over-depreciated), system alerts accounts for adjustment. 3. If red-tag reason is 'Quality Complaint', system links to supplier quality rating module.

Exception Handling: 1. Asset is active. 2. User has privilege. 3. Reason is provided.

## REQ-I007 – Value Capture While Dispensing

Priority: High | Status: Proposed

Description: The system shall capture the financial value of items at the point of issue/dispensing to departments, enabling accurate consumption tracking, budget utilization analysis, and chargeable item billing.

Preconditions: Stock must be available. Indent must be approved. Department must have active budget (for non-chargeable). Patient must be registered (for chargeable).

Inputs: Indent ID, issue voucher ID, product ID, issued quantity, unit cost (FIFO/FEFO), total issue value, department ID, issue date, issued by, patient ID (if chargeable), billing status.

Processing Logic: 1. When approved indent is converted to issue, system retrieves stock batches for the product. 2. System applies valuation method: FIFO (First In First Out) for standard items, FEFO (First Expired First Out) for expiry-sensitive items. 3. System selects batch(es) based on valuation method and requested quantity. 4. For each batch issued, system captures: Batch Number, Unit Cost (from GRN), Quantity Issued, Line Value. 5. System calculates total issue value: Sum of (Quantity × Unit Cost) across all batches. 6. For chargeable items (patient-specific): System links to Patient ID and marks as 'Pending Billing'. 7. For non-chargeable items (department consumables): Value is charged to department

budget. 8. System updates: Stock quantity (reduce), Department consumption (increase), Budget utilization (increase). 9. Issue value is included in Tally integration as consumption entry. 10. Monthly consumption report: Department-wise, Product-wise, Category-wise with value and quantity.

Outputs: Issue voucher with line-item values, stock reduction record, budget utilization update, consumption report, billing queue (chargeable items).

Postconditions: 1. FIFO is default valuation method; FEFO is mandatory for pharmaceuticals and expiry-sensitive items. 2. If stock exists in multiple locations, system issues from location nearest to requesting department. 3. Partial issue is allowed if full quantity unavailable; balance added to backorder. 4. Issue value cannot be modified manually; it is derived from GRN unit cost. 5. Return of issued items reverses the consumption value at original issue cost. 6. Chargeable item billing integration with HIS requires patient ID validation.

Business Rules: 1. If stock insufficient, system suggests: (a) Partial issue, (b) Emergency transfer from other department, (c) Expedited PO. 2. If no batch with valid cost (e.g., stock received as donation), system uses last purchase price or standard cost. 3. If department budget exhausted, issue is blocked unless management override obtained.

Exception Handling: 1. Stock is available. 2. Indent is approved. 3. Budget/patient validation passed.

## REQ-I008 – Depreciation Handling

Priority: High | Status: Proposed

Description: The system shall calculate and track depreciation for all fixed assets using configurable methods and rates, generating monthly/quarterly/annual depreciation entries for accounts and Tally integration.

Preconditions: Asset must be capitalized (cost > Rs. 5,000 or threshold configured). Depreciation parameters must be configured. Asset must be in Active/In-Use status.

Inputs: Asset ID, asset category, acquisition cost (landed cost), salvage value, useful life (years), depreciation method (SLM/WDV/Units of Production), depreciation rate (%) or per-unit depreciation rate, total estimated production/usage units, actual units produced or used in the period, start date, last depreciation date, accumulated depreciation, book value.

Processing Logic: 1. Admin configures depreciation parameters per asset category: Method (Straight Line / Written Down Value / Units of Production), Rate or per-unit depreciation rate, Useful Life or Total Estimated Production Units. 2. During asset receipt, system captures:

Acquisition Cost (landed cost), Date of Put to Use, Asset Category. 3. System auto-calculates depreciation based on method: SLM:  $(\text{Cost} - \text{Salvage Value}) / \text{Useful Life (per year)}$  WDV:  $\text{Book Value} \times \text{Depreciation Rate (per year)}$  Units of Production:  $(\text{Cost} - \text{Salvage Value}) / \text{Total Estimated Production Units}$ , multiplied by Actual Units Produced or Used in the period. 4. User (Store/Accounts) runs depreciation process monthly/quarterly/annually, or per usage cycle for Units of Production assets. 5. System generates depreciation schedule: Period, Opening WDV, Depreciation Amount, Closing WDV. 6. System posts journal entry to Tally: Debit Depreciation Expense, Credit Accumulated Depreciation. 7. Narration includes: Asset Code, Asset Name, Period, Depreciation Amount. 8. System updates asset register: Accumulated Depreciation, Book Value  $(\text{Cost} - \text{Accumulated Depreciation})$ . 9. Depreciation report by department, category, and individual asset. 10. When asset is sold/scrapped, system calculates gain/loss:  $\text{Sale Value} - \text{Book Value}$ . 11. Gain/Loss posted to Tally with appropriate ledger account.

Outputs: Depreciation schedule, journal entries, asset register updates, depreciation report, gain/loss calculation on disposal.

Postconditions: 1. Depreciation starts from month following date of put to use (Indian GAAP). 2. Depreciation stops in the month asset is sold/scrapped. 3. If asset is idle for >180 days, depreciation continues but is flagged for management review. 4. Change in depreciation method requires management approval and is applied prospectively. 5. Asset categories and rates follow Income Tax Act or company policy (whichever is stricter). 6. Depreciation is calculated to 2 decimal places and rounded per accounting policy.

Business Rules: 1. If depreciation parameters not configured for asset category, system uses default category (General Assets). 2. If asset cost is below capitalization threshold, system marks as 'Expense' instead of 'Capitalize' and skips depreciation. 3. If depreciation run fails for any asset, system continues with others and logs error for retry.

Exception Handling: 1. Asset is capitalized. 2. Parameters are configured. 3. Asset is active.

## REQ-I009 – Auto-Indent Based on Reorder Level

Priority: High | Status: Proposed

Description: The system shall automatically generate indent requests to stores when stock levels fall below predefined reorder levels, ensuring continuous availability of critical items without manual monitoring.



Preconditions: Product must have reorder level configured. Location must be active. Auto-indent must be enabled for product category.

Inputs: Product ID, current stock quantity, reorder level, reorder quantity, department ID, location ID, auto-indent flag, last indent date, consumption trend.

Processing Logic: 1. System runs daily batch job (configurable time, e.g., 6:00 AM) to check stock levels. 2. For each product at each location, system compares: Current Stock vs. Reorder Level. 3. If Current Stock  $\leq$  Reorder Level, system checks: Is there already a pending indent for this product? If yes, skip. 4. If no pending indent, system auto-generates indent with: Product, Reorder Quantity, Department (default: location owner), Reason: 'Auto-generated based on reorder level'. 5. System sends notification to: Stores manager (new auto-indent created), Department HOD (stock below threshold). 6. Department HOD can approve, modify, or cancel auto-indent. 7. If HOD does not respond within 24 hours, auto-indent is treated as approved and routed to stores. 8. Stores processes approved auto-indent through standard procurement workflow. 9. System tracks auto-indent accuracy: Stockout prevented vs. excess stock created. 10. Reorder levels are reviewed quarterly based on consumption trends: If stockouts frequent, increase level; If excess stock, decrease level.

Outputs: Auto-indent document, notification to stores and HOD, approval workflow, procurement initiation, reorder level adjustment recommendation.

Postconditions: 1. Reorder Level = Minimum stock to cover lead time + safety stock. 2. Reorder Quantity = Economic Order Quantity (EOQ) or fixed quantity based on consumption pattern. 3. Auto-indent is enabled by default for all consumables; disabled for capital assets. 4. Seasonal items can have seasonal reorder levels (e.g., higher for monsoon season). 5. Critical items (life support, emergency drugs) have lower reorder levels and faster procurement. 6. Auto-indent does not apply to items under rate contract with vendor-managed inventory (VMI).

Business Rules: 1. If reorder level not configured, system alerts admin and skips product. 2. If supplier lead time increased, system suggests reorder level adjustment. 3. If multiple locations have same product below reorder level, system consolidates into single indent with location-wise distribution.

Exception Handling: 1. Product has reorder level. 2. Location is active. 3. Auto-indent is enabled.

## REQ-I010 – Goods Received but Invoice Not Received

Priority: High | Status: Proposed

Description: The system shall handle scenarios where goods are received before the invoice arrives, allowing GRN creation based on Delivery Challan (DC) or packing list while tracking pending invoices and preventing duplicate GRNs.

Preconditions: DC must be physically received. Supplier must be registered. Store keeper must have GRN creation privilege.

Inputs: GRN form with DC option: DC number, DC date, supplier ID, PO reference (if any), item list with quantities, received by, receipt date, condition notes, invoice pending flag.

Processing Logic: 1. Goods arrive with Delivery Challan (DC) but no invoice. 2. Store keeper creates GRN selecting 'GRN against DC' option. 3. System captures: DC Number, DC Date, Supplier, Items, Quantities, Condition. 4. System checks for duplicate DC: Same DC number + Supplier + Date cannot exist. 5. GRN status is set to 'Invoice Pending'. Stock is updated (available for issue). 6. System starts tracking: Days since GRN without invoice. 7. At 7 days: System sends reminder to store keeper and supplier (via portal/email). 8. At 15 days: System escalates to store manager. 9. At 30 days: System escalates to management and flags for follow-up. 10. When invoice arrives, user links invoice to existing GRN: Updates GRN with invoice details, changes status to 'Invoice Received'. 11. System prevents creation of second GRN for same DC/invoice combination. 12. Report of all GRNs with pending invoices: Aging analysis (0-7 days, 8-15 days, 16-30 days, >30 days).

Outputs: DC-based GRN, invoice pending tracking, reminder/escalation emails, aging report, invoice linkage record.

Postconditions: 1. GRN against DC is allowed only if PO exists or emergency purchase is approved. 2. Stock from DC-based GRN is available for issue immediately (no waiting for invoice). 3. Invoice must be linked within 30 days; otherwise, management approval required for further delay. 4. If invoice never arrives, DC-based GRN is treated as 'Invoice Not Traceable' with management approval. 5. Duplicate DC detection is mandatory; override requires store manager approval with reason. 6. DC-based GRNs are included in monthly reconciliation with supplier.

Business Rules: 1. If DC number is blank or illegible, store keeper can use 'DC-Not-Available' with reason and management approval. 2. If invoice arrives before GRN is created, system suggests creating regular GRN instead of DC-based. 3. If supplier disputes DC, system retrieves gate entry record and security log for verification.

Exception Handling: 1. DC is physically received. 2. Supplier is registered. 3. Store keeper is authorized.

## REQ-I011 – Internal Transfer for Emergency Stock

Priority: High | Status: Proposed

Description: When requested stock is unavailable in stores but available in another department/location, the system shall enable emergency internal transfers to meet urgent requirements while maintaining proper tracking and accountability.

Preconditions: Indent must exist. Stock must be available in another location. Source department HOD must be active.

Inputs: Requesting department ID, required product ID, required quantity, reason for emergency, source department (auto-suggested), transfer quantity, approval workflow.

Processing Logic: 1. Department creates indent for item. Stores checks stock: If unavailable or insufficient, system auto-searches other locations/departments. 2. If stock found elsewhere, system displays: Source Department, Available Quantity, Location. 3. Store keeper initiates internal transfer request with: Source Department, Transfer Quantity, Reason. 4. System sends notification to source department HOD for approval. 5. Source HOD can approve (release stock), reject (with reason), or suggest partial quantity. 6. Upon approval, system generates Internal Transfer Note (ITN) with unique number. 7. Physical transfer occurs with ITN. Receiving department confirms receipt in system. 8. System updates stock: Reduce from source, Add to destination (with transfer cost tracking if applicable). 9. System tracks: Transfer request time, Approval time, Physical transfer time, Receipt confirmation time. 10. Report to management: Emergency transfers by department, Product, Frequency, Reasons. 11. System analyzes patterns: Frequent emergency transfers indicate reorder level issues or stock allocation problems.

Outputs: Internal Transfer Note, stock updates (source and destination), transfer timing report, management alert, reorder level recommendation.

Postconditions: 1. Emergency transfer is allowed only for urgent/indents marked as emergency. 2. Transfer quantity cannot exceed available stock at source minus source's reorder level. 3. Transferred stock must be returned or replenished within 7 days (configurable). 4. Cost of transferred stock is tracked for inter-departmental settlement (if applicable). 5. More than 3 emergency transfers per department per month triggers management review. 6. Source department cannot refuse emergency transfer for critical items (ICU, OT, Emergency) without management approval.

Business Rules: 1. If no stock available anywhere, system initiates expedited PO workflow. 2. If source HOD does not respond within 2 hours (emergency), auto-approve with management

notification. 3. If transferred stock is not returned/replenished within deadline, system sends reminder and blocks further transfers from source.

Exception Handling: 1. Indent exists. 2. Stock available elsewhere. 3. Source HOD is active.

## REQ-I012 – Stock Return Module

Priority: High | Status: Proposed

Description: The system shall manage stock returns from departments to stores with proper reason capture, approval workflows, and accounting adjustments.

Preconditions: Items must have been issued to the department. Return quantity cannot exceed available balance. Department must be active.

Inputs: Return request ID, department ID, product ID, return quantity, return reason (dropdown + text), condition of returned goods, HOD approval, stores verification, management notification (high value).

Processing Logic: 1. Department user initiates return request selecting items from their issued stock. 2. System displays: Product, Original Issue Date, Issued Quantity, Already Returned Quantity, Available for Return. 3. User enters: Return Quantity, Reason (Expired / Excess / Not Required / Damaged / Quality Issue / Other), Condition (Good / Damaged / Expired). 4. For returns > Rs. 5,000 or quality issues, mandatory HOD approval. 5. HOD reviews and approves/rejects with comments. 6. Upon HOD approval, request routes to stores for physical verification. 7. Stores verifies: Physical quantity matches request, Condition matches declaration, Expiry date valid (if applicable). 8. Stores accepts or rejects return with reason. 9. If accepted: System updates stock (adds to stores inventory), reverses department consumption value, generates return receipt. 10. If rejected: System logs rejection reason and returns items to department. 11. For expired items: System auto-generates disposal workflow. 12. For quality issues: System links to supplier quality rating and complaint tracking. 13. Report: Return frequency by department, reason, product, and value.

Outputs: Return request, HOD approval, stores verification record, stock update, consumption reversal, return receipt, disposal workflow (expired).

Postconditions: 1. Return must be initiated within 30 days of issue (configurable). 2. Consumables can be returned if unopened and in good condition. 3. Opened/used consumables cannot be returned (except for quality complaints). 4. Asset returns require asset condition assessment and may trigger depreciation adjustment. 5. Return value is credited back to department budget at original issue cost. 6. Frequent returns (>5% of issues) by a department trigger operational review.

Business Rules: 1. If item not found in department's issued stock, return is blocked. 2. If return quantity exceeds available, system suggests maximum allowed quantity. 3. If stores verification reveals condition mismatch (e.g., declared 'Good' but found 'Damaged'), system logs discrepancy and alerts HOD.

Exception Handling: 1. Items were issued to department. 2. Return quantity is valid. 3. Department is active.

## REQ-I013 – Short-Expiry Items Daily Highlight

Priority: High | Status: Proposed

Description: The system shall identify and highlight items nearing expiry on a daily basis, enabling proactive action to minimize wastage and ensure patient safety.

Preconditions: Batches must have valid expiry dates. Daily batch job must be scheduled. Notification recipients must be configured.

Inputs: Product ID, batch number, expiry date, current stock quantity, location, threshold days (30/60/90), daily batch job trigger.

Processing Logic: 1. System runs daily batch job at 5:00 AM to check all batch expiry dates. 2. For each batch, system calculates: Days to Expiry = Expiry Date - Current Date. 3. System categorizes: Red: Expiry within 30 days (critical) Orange: Expiry within 31-60 days (warning) Yellow: Expiry within 61-90 days (alert) 4. System generates daily expiry alert report with: Product, Batch, Expiry Date, Days Remaining, Stock Quantity, Location, Recommended Action. 5. Recommended actions: Red: Priority issue to departments, stop new procurement, consider return to supplier. Orange: Increase issue priority, notify department HOD. Yellow: Monitor consumption, adjust reorder levels. 6. System sends email to: Stores manager, Department HODs (for their items), Management (for high-value items). 7. Dashboard widget shows: Count of Red/Orange/Yellow items, Total value at risk. 8. System enforces FEFO (First Expired First Out) during issue: Short-expiry batches are issued first. 9. If item expires while in stock, system auto-generates disposal workflow and blocks issue. 10. Monthly wastage report: Value of expired items, Reasons, Department-wise breakdown, Trend analysis.

Outputs: Daily expiry alert report, dashboard widget, FEFO enforcement, disposal workflow for expired items, monthly wastage report.

Postconditions: 1. Expiry date is mandatory for all pharmaceutical and medical consumable batches. 2. Non-medical items (stationery, housekeeping) may skip expiry tracking (configurable per category). 3. Items expired >30 days ago are auto-marked for disposal and

excluded from stock valuation. 4. Supplier is notified of near-expiry items >60 days before expiry for potential return. 5. Wastage value is charged to department budget if caused by poor demand planning. 6. FEFO is mandatory; override requires store manager approval with reason.

Business Rules: 1. If expiry date is blank for expiry-tracked item, system flags for data correction and excludes from FEFO. 2. If batch is already expired when received, GRN is blocked pending supplier resolution. 3. If near-expiry item cannot be issued (no demand), system suggests inter-hospital transfer or donation.

Exception Handling: 1. Batches have expiry dates. 2. Batch job is scheduled. 3. Recipients are configured.

### REQ-I014 – Items Bound to Departments

Priority: High | Status: Proposed

Description: Assign items to departments during receipt. Track ownership. Restrict issue unless authorized transfer. Department-wise valuation. Utilization report.

Preconditions: Item has been received via GRN and is being placed into stock for the first time, or is being formally reassigned.

Inputs: Item code, receiving/owning department, custodian, transfer authorization (for reassignment).

Processing Logic: 1. At GRN, Store Keeper assigns each item/asset to an owning department (default: the requesting department from the PR). 2. System records this ownership against the item/asset record. 3. Any issue of that item to a different department requires an authorized internal transfer (REQ-I011/I015), not a direct re-issue. 4. Department-wise stock valuation reports use this ownership tag to attribute value correctly. 5. Utilization reports (issue frequency, consumption rate) are computed per owning department.

Outputs: Department-tagged stock record, restricted-issue enforcement, department-wise valuation and utilization reports.

Postconditions: 1. Every unit of stock has a single, unambiguous owning department at all times. 2. Inter-department stock movement is always traceable through the transfer workflow, never silent.

Business Rules: 1. Ownership can only be changed through an approved transfer transaction, never by direct edit of the master record. 2. Consumable items issued to a ward are

considered consumed (ownership ends) at the point of issue; durable/returnable items retain ownership tracking through their lifecycle.

Exception Handling: 1. If an item is found physically in a department without a matching ownership record (discovered during audit), it is flagged as an unreconciled asset pending investigation rather than auto-assigned. 2. If two departments dispute ownership of an item, the transfer history is used as the system of record to resolve the dispute.

### REQ-IO15 – Transfer Between Locations

Priority: High | Status: Proposed

Description: Inter-location transfer note. Track status: Requested/In-Transit/Received. Update stock on confirmation. Stores manager approval. Report on frequency and reasons.

Preconditions: Stock is available at the source store/location and a valid business need exists to move it to a different location.

Inputs: Source location, destination location, item, quantity, requesting user, approver.

Processing Logic: 1. Requesting store/department raises an inter-location Transfer Note specifying source, destination, item, and quantity. 2. Transfer routes for Stores Manager approval. 3. On approval, stock is deducted from source location and the transfer status is set to 'In-Transit'. 4. Destination location confirms physical receipt, which updates status to 'Received' and credits stock at the destination. 5. System tracks the elapsed time at each status stage and flags transfers stuck 'In-Transit' beyond a configurable threshold.

Outputs: Transfer note with status history, updated source/destination stock, in-transit ageing report.

Postconditions: 1. Stock quantity is never double-counted or lost between locations; it is deducted at source before being credited at destination. 2. Every location-to-location movement is fully auditable.

Business Rules: 1. A transfer cannot be initiated for quantity exceeding available stock at the source. 2. Batch/expiry attributes travel with the transferred quantity so FEFO discipline is preserved at the destination. 3. Transfers between facilities (not just internal stores) may require an additional gate/security confirmation step.

Exception Handling: 1. If the destination does not confirm receipt within the configured window, the system escalates to both source and destination Stores Managers. 2. If the physically received quantity differs from what was dispatched, the discrepancy is logged and routed for investigation before the transfer can be closed.

### REQ-IO16 – Product Images Upload

Priority: Medium | Status: Proposed

Description: Front/back image per product. Thumbnail in search. Help identify across naming conventions. Image compression. Display in vendor portal, PO, GRN screens.

Preconditions: Product exists in the item master and an image is available for upload.

Inputs: Item code, front/back image file.

Processing Logic: 1. Authorized user uploads front/back images against the item master record. 2. System auto-generates a compressed thumbnail for use in search-result lists and a full-size view for detail screens. 3. Images display consistently across the vendor portal, PO creation, GRN entry, and indent screens so that users identify the correct item despite naming-convention differences across vendors and departments.

Outputs: Stored item images, thumbnails, consistent cross-module image display.

Postconditions: 1. Items with visually similar names or overlapping categories can be reliably distinguished by users at the point of transaction.

Business Rules: 1. Image file size is capped and auto-compressed to avoid degrading page load performance. 2. Only Store/Product Master Admin roles may replace an existing image once set.

Exception Handling: 1. If no image is available, the item displays a placeholder icon rather than a broken image, and is flagged in the 'no image' completeness report. 2. If an uploaded image is later found to be incorrect, any user can flag it for review, which reverts display to the placeholder until corrected.

### REQ-IO17 – Quantities by Weight/Length/Volume

Priority: Medium | Status: Proposed

Description: Weight: mg, g, kg. Length: mm, cm, m. Volume: ml, l. Unit conversion. Base and transaction units. Auto-conversion for reporting.

Preconditions: Item is measured in a non-discrete unit (weight, length, or volume) rather than whole-count pieces.

Inputs: Item code, base unit of measure, transaction unit of measure, conversion factor.



Processing Logic: 1. Item master defines a base unit (e.g., kg, m, l) used for stock valuation and reporting. 2. A transaction-time unit (e.g., g, cm, ml) may differ from the base unit; the system applies the configured conversion factor automatically. 3. All quantity entries (GRN, issue, adjustment) are captured in the transaction unit chosen by the user but stored and reported in the base unit for consistency. 4. Reports can display either unit on demand.

Outputs: Base-unit stock ledger, transaction-unit entry convenience, unit-converted reporting.

Postconditions: 1. Stock quantities remain consistent and comparable regardless of which unit staff used at the point of transaction.

Business Rules: 1. Conversion factors are set once per item and changing them requires Product Master Admin approval, since a change retroactively affects historical unit interpretation if not handled carefully. 2. Fractional quantities are supported and rounded per a configurable precision (e.g., 2 decimal places) rather than truncated.

Exception Handling: 1. If a conversion factor is missing for an item flagged as weight/length/volume-based, the system blocks GRN/issue entry until the Product Master Admin configures it.

## REQ-IO18 – Non-Cash Expenditures Tracking

Priority: Medium | Status: Proposed

Description: Barter/exchange, donated items, free samples, internal transfers at zero value. Separate ledger. Audit trail for compliance.

Preconditions: A stock movement occurs that does not involve a normal cash purchase – e.g., donation, sample, barter, or an internal zero-value transfer.

Inputs: Transaction type (donation/sample/barter/zero-value transfer), item, quantity, source/donor, value for reporting (may be zero or notional), approving authority.

Processing Logic: 1. User selects the applicable non-cash transaction type at entry. 2. System posts the item quantity to stock through a distinct non-cash ledger, separate from purchase-based GRNs, so cash-purchase valuation is never distorted by zero/notional-value entries. 3. Donations and free samples require a documented source/donor and approval before being added to stock. 4. Internal zero-value transfers (e.g., inter-department goodwill transfer) are still tracked with full quantity and location movement, valued at the item's standard cost for internal reporting even though no cash changed hands.

Outputs: Non-cash ledger entries, donor/source documentation, compliance-ready audit trail distinct from purchase transactions.

Postconditions: 1. Inventory valuation based on actual purchase spend is never inflated or distorted by non-cash receipts. 2. Donated/sample stock remains fully traceable through issue and consumption like any other item.

Business Rules: 1. Non-cash receipts above a configurable value require the same approval tier as an equivalent-value purchase, even though no payment is involved, to prevent misuse as a bypass route. 2. Free samples used for evaluation link back to the sample-tracking workflow (REQ-V009) where applicable.

Exception Handling: 1. If a 'donation' is later found to require registration/compliance documentation (e.g., for pharmaceuticals) that is missing, the stock is held pending documentation before it can be issued for use.

### REQ-I019 – Location Mapping Inside Stores

Priority: Medium | Status: Proposed

Description: Rack, shelf, bin numbers. Assign products to locations. Location-based search. Pick list with locations. Space utilization report.

Preconditions: A physical store/warehouse structure with racks, shelves, and bins exists or is being defined.

Inputs: Rack number, shelf number, bin number, item-to-location assignment.

Processing Logic: 1. Stores Manager defines the physical rack/shelf/bin structure for each store. 2. Items are assigned to one or more bin locations, supporting split-location stock for high-volume items. 3. Search and pick-list screens display the assigned location(s) alongside item and quantity, enabling faster and more accurate picking. 4. A space-utilization report shows bin occupancy versus capacity to support layout optimization.

Outputs: Location-tagged item master, location-aware pick lists, space utilization report.

Postconditions: 1. Stores staff can locate and pick items without relying on memorized layout knowledge, reducing search time and picking errors.

Business Rules: 1. An item can be assigned to multiple bins (split stock); the system must present all valid locations at pick time, prioritizing FEFO/FIFO order where the item is batch-tracked. 2. Bin reassignment requires an internal transfer transaction so the location change itself is auditable.

Exception Handling: 1. If an item's assigned bin is found empty during a pick (data-vs-physical mismatch), the discrepancy is logged and routed to the next cycle count for reconciliation rather than silently resolved.

### REQ-I020 – Same Order Multiple Batches Indication

Priority: Medium | Status: Proposed

Description: Same product with different batches in single GRN. Batch-wise quantity and expiry. Separate stock tracking. FEFO across batches. Report on batch distribution.

Preconditions: A single GRN includes the same product supplied against two or more distinct batch numbers.

Inputs: GRN reference, item code, batch numbers, batch-wise quantity, batch-wise expiry date and MRP.

Processing Logic: 1. At GRN entry, the system allows multiple batch lines for the same item within one GRN, each capturing its own quantity, expiry date, and MRP. 2. Each batch is tracked as a distinct stock lot from receipt through issue, never merged with another batch's quantity. 3. Issue transactions apply FEFO logic across all available batches (including those received in the same GRN), always offering the nearest-expiry batch first. 4. A batch distribution report shows, per item, how stock is split across active batches with their respective expiry dates.

Outputs: Batch-wise stock ledger, FEFO-compliant issue suggestions, batch distribution report.

Postconditions: 1. No batch's expiry or MRP is ever conflated with another's, even when received together on the same GRN.

Business Rules: 1. Batch numbers must be unique per item per vendor; duplicate batch numbers for the same item trigger a validation warning. 2. MRP differences between batches of the same item are preserved and reflected accurately at the point of billing/issue where MRP-based charging applies.

Exception Handling: 1. If a batch number is left blank in error for a batch-tracked item, the GRN cannot be completed until it is supplied, preventing untraceable stock from entering the system.

### REQ-I021 – Products Marked 'Opened' Before Use

Priority: Medium | Status: Proposed

Description: Products like Betadine marked 'Opened' before first use. Record opening date and expected usage. Alert if not used within validity. Separate stock: Unopened vs Opened.

Preconditions: Item is a multi-use consumable (e.g., an antiseptic solution) whose shelf life after first use is shorter than its sealed shelf life.

Inputs: Item code, opening date, expected in-use validity period (days), department/location.

Processing Logic: 1. At the point an item marked 'Opened-on-Use' (e.g., Betadine, multi-dose vials) is first opened, the user records the opening date via a simple scan/tap action. 2. System computes the in-use expiry as opening date plus the configured validity period for that item category, which is typically shorter than the printed expiry date. 3. An alert is raised if the item is not used within its in-use validity window, prompting disposal per protocol. 4. Opened-item stock is tracked separately from unopened/sealed stock of the same batch for reporting accuracy.

Outputs: Opening-date-tagged stock record, in-use expiry alert, opened-vs-sealed stock split reporting.

Postconditions: 1. Multi-use items are never used beyond their safe in-use window even when the printed (sealed) expiry date is still valid.

Business Rules: 1. The in-use validity period is configured per item category by the Pharmacy/Infection Control policy owner, not left to individual judgement. 2. An item flagged 'Opened' past its in-use validity is automatically excluded from FEFO issue suggestions and flagged for disposal review.

Exception Handling: 1. If the opening date is not recorded at first use, the system defaults to treating the item as opened on the date of first recorded partial-quantity issue, with a flag noting the date is inferred, not confirmed.

### REQ-I022 – Partial Items Request at Spot During Indent

Priority: Medium | Status: Proposed

Description: Partial issue when full quantity unavailable. Record partial qty and reason. Balance to backorder. Notify department. Track fulfillment rate.

Preconditions: An indent is being processed and the full requested quantity of one or more items is not available in stock.

Inputs: Indent ID, item, requested quantity, available quantity, reason for shortfall.

Processing Logic: 1. At the point of issue, if available stock is less than the requested quantity, the Store Keeper issues the available quantity and records the reason for the shortfall (stock-out, reorder pending, allocation constraint). 2. The unfulfilled balance is automatically converted into a backorder linked to the original indent. 3. The requesting department is notified of the partial issue and the expected fulfillment date for the balance, where known. 4. A fulfillment-rate report tracks the percentage of indents fully met on first issue versus requiring backorder completion, by item and department.

Outputs: Partial issue record, backorder linked to original indent, department notification, fulfillment-rate report.

Postconditions: 1. No indent is silently under-fulfilled without the requesting department being informed and the balance being tracked to closure.

Business Rules: 1. A backorder is automatically prioritized for fulfillment once new stock arrives, ahead of new indents for the same item where policy dictates first-come allocation. 2. Repeated partial fulfillment for the same item beyond a configurable frequency triggers a reorder-level review.

Exception Handling: 1. If the backorder remains unfulfilled beyond a configurable period, it is escalated to the Stores Manager and the requesting department may be prompted to source via emergency purchase instead.

## REQ-I023 – Maintenance/HK Items Batch and Expiry

Priority: Medium | Status: Proposed

Description: Option to skip batch/expiry for non-medical consumables. Configurable per category. Default 'NA' for such items. Report on items without batch tracking.

Preconditions: Item belongs to a non-medical consumable category (e.g., housekeeping/maintenance supplies) where batch/expiry tracking is not clinically relevant.

Inputs: Item code, category, batch/expiry-tracking override flag.

Processing Logic: 1. Product Master Admin configures whether batch/expiry tracking is mandatory, optional, or skipped for a given item category. 2. For categories configured as 'skip', GRN and issue screens do not prompt for batch/expiry fields; the system defaults these fields to 'NA' internally. 3. A report lists all items currently configured without batch/expiry tracking, for periodic review to ensure no item requiring tracking was misclassified.

Outputs: Category-level tracking configuration, simplified GRN/issue entry for non-tracked categories, periodic misclassification-review report.

Postconditions: 1. Data-entry effort is reduced for genuinely non-expiry-relevant items, while medical/pharmacy items always retain mandatory tracking.

Business Rules: 1. The default configuration for any newly created item category is 'batch/expiry tracking required' unless explicitly overridden with justification, to avoid accidental gaps in expiry control. 2. Overriding an existing tracked category to 'skip' requires Stores Manager and Quality sign-off.

Exception Handling: 1. If an item is reclassified from a tracked to a non-tracked category, existing batch-tagged stock retains its original batch/expiry data; only new receipts after the change follow the simplified entry.

### REQ-I024 – Stores vs Stationery Classification

Priority: Medium | Status: Proposed

Description: Distinct categories for medical stores and office stationery. Separate reorder levels and workflows. Different valuation methods. Combined or separate reporting.

Preconditions: Items span both medical/clinical stores and general office stationery, which are managed under different policies.

Inputs: Item code, classification (Medical Stores / Stationery), category-specific reorder level, workflow routing rule.

Processing Logic: 1. Product Master Admin classifies every item as either Medical/Clinical Stores or Stationery (or other defined general-stores categories) at creation. 2. Reorder levels, approval workflows, and valuation methods (e.g., FEFO for medical, straightforward FIFO or weighted-average for stationery) are configured independently per classification. 3. Consolidated reports can present both classifications together for overall spend visibility, while operational screens (indent, issue) filter by classification to keep department workflows uncluttered.

Outputs: Classification-tagged item master, independent reorder/workflow configuration per classification, combined and split reporting views.

Postconditions: 1. Medical stores retain the stricter expiry/FEFO discipline they require, while stationery/general items follow a simpler, lower-overhead process appropriate to their risk profile.

Business Rules: 1. An item's classification, once set, can only be changed by the Product Master Admin with justification, since it affects historical workflow interpretation. 2.

Budget/cost-center reporting distinguishes clinical consumable spend from general administrative spend by default.

Exception Handling: 1. If an item is ambiguous (e.g., used both clinically and administratively), the Product Master Admin assigns it to the classification matching its predominant use, with a note enabling future reclassification if usage patterns change.

## REQ-I025 – ID Scanning While Issuing/Using Product

Priority: Medium | Status: Proposed

Description: Scan product barcode during issue/return/transfer. Scan employee ID for accountability. Scan patient ID for chargeable items. Mobile app support. Offline scanning with sync.

Preconditions: Item, employee, and (where relevant) patient identifiers are all barcode/ID-scannable within the system.

Inputs: Item barcode, employee ID badge, patient ID (for chargeable items), transaction type (issue/return/transfer/usage).

Processing Logic: 1. At issue, return, or transfer, the Store Keeper or ward staff scans the item barcode to identify it unambiguously, avoiding manual code entry errors. 2. The issuing staff member's employee ID is scanned for accountability, recording who performed the transaction beyond just the logged-in user session. 3. For chargeable items linked to a specific patient, the patient ID is additionally scanned to link consumption to the patient's billing record via HIS integration. 4. All scans are validated in real time (recognized item/employee/patient) before the transaction is allowed to complete. 5. Where a handheld/mobile device is used, the same scanning workflow is supported via the mobile app.

Outputs: Scan-validated issue/return/transfer transaction, employee accountability record, patient-linked chargeable-item entry (where applicable).

Postconditions: 1. Every chargeable item issued to a patient is traceable to both the issuing staff member and the specific patient for accurate billing and clinical accountability.

Business Rules: 1. A transaction cannot be completed with an unrecognized barcode; the system requires the item to exist in the master or the scan to be retried/corrected. 2. Patient-ID scanning is mandatory only for items configured as chargeable-to-patient; general ward consumables do not require it.

Exception Handling: 1. If a barcode is damaged/unreadable, the system allows manual code entry as a fallback, but flags the transaction as 'manually entered' for later verification. 2. If

the scanned patient ID does not match an active HIS encounter, the transaction is held pending correction rather than posted against an invalid billing reference.

## REQ-IO26 – Request-Deliver-Return Cycle

Priority: Low | Status: Proposed

Description: Temporary items like plates/tumblers: Request → Deliver → Return. Track date/time at each stage. Usage pattern report. Alert if not returned.

Preconditions: A temporary/loanable item (e.g., crockery, cutlery, wheelchairs) is being lent out for a bounded period rather than consumed.

Inputs: Item, quantity, requesting department, request time, delivery time, expected return time, actual return time.

Processing Logic: 1. Requesting department raises a Request for a temporary item. 2. Store Keeper records the Deliver action with a timestamp once the item is handed over. 3. On return, the Store Keeper records the Return action with a timestamp and verifies item condition/count. 4. System computes elapsed time at each stage (request-to-deliver, deliver-to-return) for usage-pattern analysis. 5. If an item is not returned within a configurable expected-return window, the system sends an automated reminder to the borrowing department, escalating if still outstanding.

Outputs: Request-Deliver-Return transaction record with timestamps, usage-pattern report, overdue-return alert.

Postconditions: 1. Temporary/loanable items are never left as an unresolved open loan; every request is tracked to either a confirmed return or a flagged, escalated overdue status.

Business Rules: 1. Quantity returned must be verified against quantity delivered; shortfalls require a reason (damaged/lost) and, where applicable, a cost recovery note. 2. Items overdue beyond a second, longer threshold are escalated to the Stores Manager for department-level follow-up.

Exception Handling: 1. If an item is returned damaged, it is logged as a condition exception rather than a standard return, and routed to the disposal/write-off workflow if beyond repair.



## REQ-I027 – Multi-Tier Warehouse Hierarchy and Transit Transfer

Priority: High | Status: Proposed

Description: The system shall support a multi-tier warehouse structure -- Central Warehouse, Regional Warehouse, Department Store, and Transit Warehouse -- so that stock movement, ownership, and visibility follow a defined organizational hierarchy instead of treating every storage point as an identical flat location.

Preconditions: Each warehouse must be registered with a tier type (Central/Regional/Department Store/Transit) and, where applicable, a parent warehouse. User must have transfer-initiation rights for the source tier.

Inputs: Source warehouse ID and tier, destination warehouse ID and tier, item ID, transfer quantity, transporter/vehicle reference (for inter-site moves), expected transit time, transfer reason.

Processing Logic: 1. Admin configures the warehouse hierarchy: Department Stores report to a Regional Warehouse; Regional Warehouses report to the Central Warehouse; Transit Warehouses are temporary holding points used only during multi-leg movement. 2. When a transfer is initiated, system determines the applicable route based on the hierarchy, and Department Store to Department Store transfers may be required to route via the Regional Warehouse unless a lateral-transfer rule is configured. 3. Stock leaving the source location is moved to an in-transit state and, for moves that cross sites, is held against the Transit Warehouse until receipt is confirmed at destination. 4. Destination user confirms receipt; system moves stock from in-transit to the destination tier's available stock. 5. System maintains real-time stock visibility at every tier so users can see aggregate stock by item across Central, Regional, and Department Store levels. 6. Discrepancy between dispatched and received quantity raises a transit variance for stores review.

Outputs: Warehouse hierarchy map, in-transit stock register, transfer receipt confirmation, transit variance report, tier-wise stock visibility dashboard.

Postconditions: 1. Stock in transit is excluded from both the source and destination available balance until receipt is confirmed. 2. A transfer cannot be created between two locations that have no defined path in the hierarchy unless an admin explicitly enables a lateral/direct transfer rule. 3. Transit Warehouse balances above a configurable age (default 3 days) are flagged for follow-up.

Business Rules: 1. If a required intermediate tier is skipped without a lateral-transfer rule, the system blocks the transfer and prompts for the correct route. 2. If the destination does not confirm receipt within the configured SLA, system escalates to both source and destination location heads.

Exception Handling: 1. Both warehouses exist and are active. 2. User has transfer rights at the source tier. 3. Sufficient available (non-in-transit) stock exists at source.

## REQ-I028 – Stock Conversion and Lot-Wise Tracking

Priority: Medium | Status: Proposed

Description: The system shall support conversion of stock from one unit of measure or packaging form to another (e.g., bulk carton to issue-unit pieces) and shall track consumable stock at the lot level in addition to existing batch and serial tracking, so that lot-specific quality holds, recalls, and consumption can be traced precisely.

Preconditions: Item must have a defined conversion factor between purchase unit and issue unit, or between two stocking units. Lot tracking must be enabled for the item category.

Inputs: Item ID, source unit and quantity, target unit, conversion factor, lot number, conversion date, converting user.

Processing Logic: 1. User selects an item and initiates a stock conversion, specifying source quantity/unit and target unit. 2. System calculates target quantity using the configured conversion factor and reduces source-unit stock while increasing target-unit stock, preserving lot linkage across the conversion. 3. For items received in multiple lots, the system tracks each lot separately by lot number, receipt date, and source GRN, in addition to any batch/serial numbers already captured. 4. Issue and consumption transactions record which lot was drawn from, FIFO by default or user-selected lot for quality reasons. 5. A lot placed on quality hold blocks issue of that lot specifically without affecting other lots of the same item. 6. Lot-wise consumption and balance reports are available alongside existing batch/serial reports.

Outputs: Conversion transaction record, updated stock in both source and target units, lot register, lot-wise consumption report, quality-hold log.

Postconditions: 1. Conversion transactions are fully reversible only via an equal-and-opposite conversion entry, never by editing the original. 2. Lot number is mandatory for any item flagged as lot-tracked at the time of GRN. 3. A lot on hold cannot be issued until the hold is released by an authorized user.

Business Rules: 1. If the conversion factor is not configured for the item, conversion is blocked until admin configures it. 2. If converted stock would leave a fractional quantity below the item's minimum issuable unit, system rounds per the configured rounding rule and logs the adjustment.

Exception Handling: 1. Item exists and conversion factor is configured. 2. Sufficient source-unit stock is available. 3. Lot number is valid and not already fully consumed or scrapped.

## 3.4 GRN & INVOICE MANAGEMENT

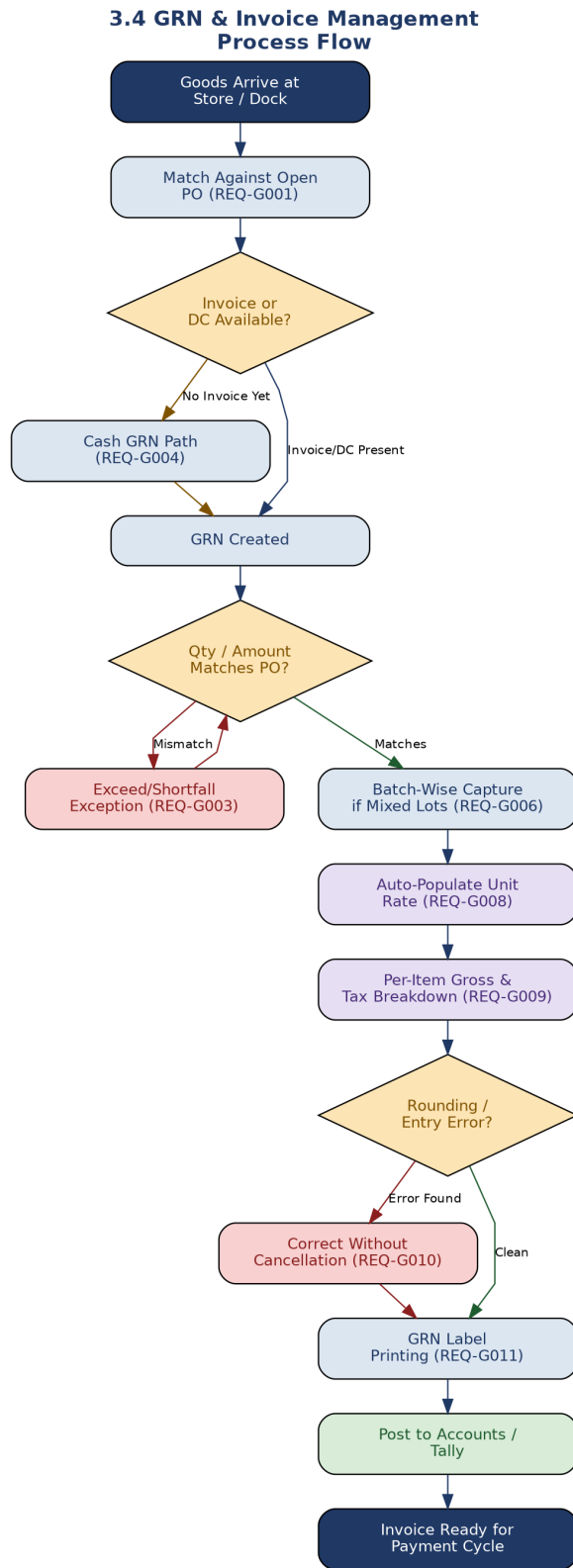


Figure: 3.4 Grn & Invoice Management – Process Flow

## REQ-G001 – GRN with Invoice or DC

Priority: High | Status: Proposed

Description: The system shall support Goods Receipt Note (GRN) creation against either an Invoice or a Delivery Challan (DC). It must prevent duplicate GRNs for the same invoice or DC and ensure proper stock updates upon receipt.

Preconditions: Goods must be physically received. Store keeper must have GRN creation privilege. Supplier must be registered.

Inputs: GRN type (Invoice/DC), invoice number or DC number, invoice date or DC date, PO reference, supplier ID, item list with quantities, batch numbers, expiry dates, MRP, received by, receipt date, condition notes.

Processing Logic: 1. Store keeper selects GRN type: 'Against Invoice' or 'Against DC'. 2. If 'Against Invoice': Enter Invoice Number, Invoice Date, PO Reference (if PO exists). 3. If 'Against DC': Enter DC Number, DC Date, PO Reference (if available). 4. System checks for duplicate: Same Invoice/DC Number + Same Supplier cannot exist in completed GRNs. 5. If duplicate detected, system blocks creation and shows existing GRN details. 6. Store keeper enters received items: Product, Quantity, Batch No, Expiry Date, MRP, Condition. 7. System validates quantities against PO (if linked): Within tolerance ( $\pm 5\%$ ) allowed with reason; beyond tolerance requires management approval. 8. System updates stock: Add quantity to location, Update batch-wise stock. 9. GRN status: Draft → Confirmed → Posted to Tally. 10. For DC-based GRN: Status is 'Invoice Pending' until invoice is linked. 11. System generates GRN document with barcode/QR for physical filing.

Outputs: GRN document, stock update record, duplicate check result, PO fulfillment update, invoice pending tracking (for DC-based).

Postconditions: 1. GRN cannot be created before physical receipt (verified by security gate entry). 2. Duplicate GRN prevention is mandatory; override requires store manager approval with documented reason. 3. PO-linked GRN quantities must match PO within 5% tolerance; otherwise management approval required. 4. Batch and expiry are mandatory for pharmaceutical and medical consumables. 5. GRN confirmation triggers auto-notification to accounts team for invoice processing. 6. DC-based GRN must be converted to invoice-based within 30 days.

Business Rules: 1. If invoice/DC number already exists, system shows existing GRN and blocks duplicate. 2. If PO quantity exceeded, system prompts for management approval workflow. 3. If batch/expiry missing for mandatory items, GRN is saved as draft until completed.

Exception Handling: 1. Goods physically received. 2. Store keeper authorized. 3. Supplier registered.

## REQ-G002 – Trim and Check Invoice Numbers

Priority: High | Status: Proposed

Description: The system shall implement robust duplicate invoice detection by trimming whitespace, normalizing case, and checking against all historical invoices to prevent duplicate payments and accounting errors.

Preconditions: Invoice number must be non-empty. Supplier must be selected. Historical invoice data must be available.

Inputs: Invoice number (raw input), trimmed invoice number, normalized invoice number, supplier ID, existing invoice database.

Processing Logic: 1. User enters invoice number in GRN or vendor portal. 2. System applies normalization: Trim leading/trailing whitespace, Remove extra internal spaces, Convert to uppercase (or lowercase). 3. System checks normalized invoice number against all historical invoices in the database. 4. Check includes: Same normalized number + Same Supplier (different suppliers can have same invoice number). 5. If match found: System displays alert 'Duplicate Invoice Detected' with existing GRN details (GRN No, Date, Supplier, Amount). 6. User options: (a) Correct invoice number if typo, (b) Proceed with override (requires store manager approval + reason), (c) Cancel and investigate. 7. System logs all duplicate detection events: User, Invoice Number, Action Taken, Override Reason (if applicable). 8. Monthly report: Duplicate detection frequency, Override instances, Users with most duplicates. 9. System also checks for visually similar numbers (e.g., 'O' vs '0', 'l' vs '1') and warns.

Outputs: Normalized invoice number, duplicate alert, override approval record, monthly duplicate report, user-wise duplicate frequency.

Postconditions: 1. Invoice number normalization is automatic and cannot be disabled. 2. Duplicate check is case-insensitive and whitespace-insensitive. 3. Override requires store manager approval with mandatory reason (minimum 30 characters). 4. More than 3 overrides per month by same user triggers management review. 5. Credit notes and debit notes are excluded from duplicate check (different document types). 6. Invoice numbers from different financial years are still checked for duplicates.

Business Rules: 1. If invoice number is blank, system blocks GRN creation. 2. If system cannot access historical data (rare), it allows creation with warning and flags for admin review. 3. If vendor uses special characters in invoice number, system preserves them after normalization.

Exception Handling: 1. Invoice number is entered. 2. Supplier is selected. 3. Historical data is accessible.

## REQ-G003 – PO Quantity/Amount Exceed or Shortfall

Priority: High | Status: Proposed

Description: The system shall handle quantity and value deviations between PO and GRN, enforcing tolerance limits and requiring approvals for significant variances.

Preconditions: PO must exist and be approved. GRN must be linked to PO. Variance must be calculable.

Inputs: PO ID, PO quantity, PO value, GRN quantity, GRN value, variance amount, variance percentage, reason for variance.

Processing Logic: 1. During GRN creation (PO-linked), system retrieves PO quantities and values. 2. User enters received quantities and invoice values. 3. System calculates variance:  $\text{Quantity Variance \%} = (\text{GRN Qty} - \text{PO Qty}) / \text{PO Qty} \times 100$ .  $\text{Value Variance \%} = (\text{GRN Value} - \text{PO Value}) / \text{PO Value} \times 100$ . 4. System applies tolerance rules: Within  $\pm 5\%$ : Auto-approve with warning log. 5-10%: Requires store manager approval with reason.  $>10\%$ : Requires management approval with detailed justification. 5. For shortfall (less than PO): User selects reason: Partial Delivery, Supplier Shortage, Quality Rejection, Other. 6. For exceed (more than PO): User selects reason: Bonus Quantity, Supplier Error, Rate Change, Other. 7. System updates PO status: Fully Received / Partially Received / Exceeded. 8. For partial receipt, balance quantity remains open for future GRN. 9. Report on PO fulfillment: % received, Shortfall reasons, Exceed reasons, Supplier-wise fulfillment rate.

Outputs: GRN with variance approval, PO status update, variance reason record, fulfillment report, supplier performance metric.

Postconditions: 1. Tolerance limits are configurable per product category (default  $\pm 5\%$ ). 2. Shortfall does not require payment for un-received quantity. 3. Exceed quantity beyond tolerance is not paid unless management approves. 4. Rate contract POs have stricter tolerance ( $\pm 2\%$ ) due to fixed pricing. 5. Frequent shortfalls from same supplier ( $>20\%$  of orders) trigger supplier review. 6. Exceed quantities are tracked separately for supplier negotiation.

Business Rules: 1. If PO not found, system allows DC-based GRN with manual PO linkage later. 2. If variance reason not provided, system blocks GRN confirmation. 3. If management approver unavailable, system queues for approval and sends reminder.

Exception Handling: 1. PO exists and is approved. 2. GRN is being created. 3. Variance is calculable.

## REQ-G004 – Cash GRN Payment Confirmation

Priority: High | Status: Proposed

Description: The system shall manage payment confirmation workflow for Cash GRNs, ensuring proper documentation and reconciliation of cash purchases.

Preconditions: Cash GRN must exist. Payment proof must be uploaded. Accounts user must have payment confirmation privilege.

Inputs: Cash GRN ID, payment mode (Cash/Cheque/DD/NEFT), payment date, paid amount, paid by (user), payment reference, receipt image, accounts verification status.

Processing Logic: 1. Cash GRN is auto-generated from Cash PO (REQ-P008). 2. Store keeper or purchaser makes physical payment to vendor. 3. User uploads payment proof: Receipt scan, Vendor acknowledgement, Bank statement (for NEFT). 4. User enters payment details: Mode, Date, Amount, Reference Number. 5. System validates: Paid amount = GRN value (or within rounding tolerance). 6. Accounts team reviews payment proof and details. 7. Accounts confirms payment in system: Status changes from 'Payment Pending' to 'Paid'. 8. System generates payment voucher and posts to Tally. 9. If payment proof insufficient, accounts returns to user with clarification request. 10. Monthly Cash GRN reconciliation: Paid vs. Pending, Discrepancies, Outstanding cash advances.

Outputs: Payment confirmation record, payment voucher, Tally entry, reconciliation report, pending payment alert.

Postconditions: 1. Cash GRN payment must be confirmed within 3 days of GRN creation. 2. Payment proof is mandatory; without proof, GRN remains in 'Payment Pending' status. 3. Cash payments > Rs. 10,000 require management approval per Income Tax rules. 4. Petty cash register is updated with each Cash GRN payment. 5. Unconfirmed Cash GRNs after 7 days trigger escalation to store manager and accounts head. 6. Cash GRN payments are excluded from standard payment approval workflow (already approved at PO stage).

Business Rules: 1. If payment amount differs from GRN value, system requires variance justification. 2. If payment proof is unclear, accounts requests re-upload. 3. If petty cash is insufficient, system alerts for replenishment and blocks further Cash POs.

Exception Handling: 1. Cash GRN exists. 2. Payment proof is uploaded. 3. Accounts user is authorized.

## REQ-G005 – Image Upload for Cash GRN

Priority: High | Status: Proposed

Description: The system shall provide reliable image/document upload functionality for Cash GRNs, supporting multiple file formats with validation, preview, and retry capabilities.

Preconditions: Cash GRN must exist. File must meet format and size requirements. User must have upload privilege.

Inputs: Image/PDF files, file format (JPG/PNG/PDF), file size, upload progress, thumbnail preview, metadata (upload date, uploader).

Processing Logic: 1. User clicks 'Upload Document' on Cash GRN screen. 2. System opens file picker with filter: JPG, PNG, PDF. 3. User selects file(s); system validates: Format (allowed types), Size (<5MB per file), Dimensions (if image, minimum 800x600). 4. System shows upload progress bar with percentage and estimated time. 5. Upon upload completion, system generates thumbnail preview. 6. User can view full-size image, rotate, zoom, and delete if incorrect. 7. Multiple files can be uploaded per GRN (receipt, vendor acknowledgement, delivery photo). 8. System stores files in cloud storage with encryption at rest. 9. Files are linked to GRN record and accessible to authorized users (store, accounts, management). 10. System provides retry mechanism for failed uploads (network interruption). 11. Download option for all uploaded files (original quality).

Outputs: Uploaded document(s), thumbnail preview, storage path, access log, download link.

Postconditions: 1. Maximum 5 files per Cash GRN. 2. Total upload size per GRN: 25MB. 3. Uploaded files are retained for 7 years (tax compliance). 4. Files are watermarked with GRN number and upload timestamp for authenticity. 5. OCR is performed on uploaded invoices for data extraction (optional). 6. Virus scan is performed on all uploads before storage.

Business Rules: 1. If upload fails due to network, system retries automatically (3 attempts) and notifies user. 2. If file format is invalid, system suggests conversion tools. 3. If file size exceeds limit, system suggests compression or split upload.

Exception Handling: 1. Cash GRN exists. 2. File meets requirements. 3. User has upload privilege.

## REQ-G006 – Same Item Different Batches in GRN

Priority: High | Status: Proposed

Description: The system shall allow receipt of the same product with different batch numbers in a single GRN, enabling proper batch-wise stock tracking and FEFO issue management.



Preconditions: Product must exist in master. Batch numbers must be unique per product (or same batch for addition). Expiry dates must be valid.

Inputs: GRN ID, product ID, multiple batch entries (batch number, manufacture date, expiry date, MRP, quantity, unit rate per batch).

Processing Logic: 1. During GRN creation, user adds product line item. 2. If product has multiple batches in the delivery, user clicks 'Add Batch' to create additional batch entries for same product. 3. Each batch entry captures: Batch Number, Manufacture Date, Expiry Date, MRP, Quantity, Unit Rate. 4. System validates: Total quantity across all batches = GRN quantity for the product. 5. System stores each batch as separate stock record with unique batch identifier. 6. Stock display shows: Product → Batch-wise breakdown (Batch No, Qty, Expiry, Location). 7. During issue, system enforces FEFO: Shortest expiry batch is suggested first. 8. Batch-wise stock valuation: Each batch valued at its own unit rate (from GRN). 9. Report: Products with multiple batches in single GRN, Batch-wise stock aging, Batch expiry timeline. 10. If same batch received again, system adds quantity to existing batch record.

Outputs: GRN with multiple batch entries, batch-wise stock records, FEFO enforcement, batch valuation, batch report.

Postconditions: 1. Each batch is tracked independently in stock ledger. 2. FEFO is mandatory for batch-tracked items; override requires store manager approval. 3. Batch numbers cannot be modified after GRN confirmation; errors require correction workflow. 4. Expired batches are automatically blocked from issue. 5. Batch-wise stock count is used for physical stock verification. 6. For non-batch items (stationery, etc.), system uses single batch record with 'NA'.

Business Rules: 1. If batch number is duplicate for same product (and not intended as addition), system alerts for verification. 2. If expiry date is past, GRN is blocked pending supplier resolution. 3. If total batch quantities don't match GRN quantity, system prompts for correction.

Exception Handling: 1. Product exists. 2. Batch numbers are valid. 3. Expiry dates are future dates.

## REQ-G007 – Round-Off Error Handling

Priority: High | Status: Proposed

Description: The system shall handle rounding discrepancies between line-item calculations and invoice totals, ensuring accounting accuracy and preventing payment mismatches.

Preconditions: Line items must be entered. Invoice total must be provided. Tax rates must be configured.

Inputs: Line item details (qty, rate, amount), tax calculations, invoice gross total, invoice grand total, rounding difference.

Processing Logic: 1. During GRN/invoice entry, system calculates line amounts:  $\text{Line Amount} = \text{Qty} \times \text{Rate}$ . 2. System calculates tax per line:  $\text{Tax} = \text{Line Amount} \times \text{Tax\%} / 100$ . 3. System sums line amounts and taxes to derive calculated gross total and grand total. 4. System compares calculated total with invoice-stated total. 5. If difference = 0: Proceed normally. 6. If difference  $\leq$  Rs. 1 (configurable tolerance): System auto-adjusts last line item or tax amount to absorb difference. 7. If difference  $>$  Rs. 1: System alerts user and requires resolution: a) Verify line items for data entry errors b) Adjust specific line item with reason c) Create rounding difference entry with management approval 8. Rounding difference is posted to separate ledger account 'Rounding Off' in Tally. 9. System tracks rounding differences by supplier to identify systematic issues. 10. Report: Rounding differences by month, supplier, user, and amount.

Outputs: Adjusted GRN with rounding entry, rounding ledger posting, discrepancy alert, rounding report.

Postconditions: 1. Rounding tolerance is configurable (default Rs. 1). 2. Rounding difference is always posted to 'Rounding Off' account, never absorbed silently. 3. If rounding difference  $>$  Rs. 10, management approval is mandatory. 4. System prefers adjusting tax amount over line amount for rounding (less impact on inventory valuation). 5. Rounding differences are included in monthly reconciliation with supplier. 6. Frequent rounding issues from same supplier ( $>10\%$  of invoices) trigger rate card review.

Business Rules: 1. If line items don't match invoice description, system suggests re-verification. 2. If tax rate mismatch causes rounding, system highlights specific line items. 3. If rounding difference is negative (invoice total  $<$  calculated), system treats same as positive difference.

Exception Handling: 1. Line items are entered. 2. Invoice total is provided. 3. Tax rates are configured.

## REQ-G008 – Auto-Populate Unit Rate

Priority: High | Status: Proposed

Description: When suppliers provide only total quantity and total value (without per-unit rate), the system shall automatically calculate the unit rate, eliminating manual calculation errors.

Preconditions: Total quantity and total value must be positive numbers. Product must be selected. Quantity must be > 0.

Inputs: Total quantity, total value, product ID, calculated unit rate, editable flag, validation check.

Processing Logic: 1. During GRN/invoice entry, user enters total quantity and total value for a line item. 2. System auto-calculates unit rate:  $\text{Unit Rate} = \text{Total Value} / \text{Total Quantity}$ . 3. System rounds unit rate to 2 decimal places (currency precision). 4. System displays calculated rate with editable option (user can modify if needed). 5. System validates: Recalculated total ( $\text{Qty} \times \text{Rate}$ ) must equal original total value (within rounding tolerance). 6. If recalculation mismatch > Rs. 1, system alerts user to verify inputs. 7. Calculated rate is stored in GRN line item and used for inventory valuation. 8. System logs instances where auto-calculation was used (vs. manual rate entry) for analysis. 9. Report: Frequency of auto-calculation usage, Suppliers who commonly omit unit rates.

Outputs: Calculated unit rate, validation result, GRN line item, inventory valuation rate, usage report.

Postconditions: 1. Auto-calculation is triggered only when rate field is left blank. 2. If user enters rate manually, auto-calculation is skipped. 3. Calculated rate is capped at 2 decimal places; remainder is handled as rounding difference. 4. For items with rate contract, system compares calculated rate with contract rate and warns if different. 5. Auto-calculated rates are flagged in reports for audit review.

Business Rules: 1. If total value is zero (free item), system sets rate as zero and flags as 'Free of Cost'. 2. If calculation results in more than 2 decimal places, system rounds and logs rounding difference. 3. If user modifies calculated rate, system requires reason for deviation.

Exception Handling: 1. Total quantity and value are entered. 2. Quantity > 0. 3. Product is selected.

## REQ-G009 – Per-Item Gross and Tax Value Breakdown

Priority: High | Status: Proposed

Description: The system shall display and store detailed breakdown of gross value and tax value for each line item, enabling easy identification of errors in invoice totals.

Preconditions: Line items must be entered. Tax rates must be configured per product/HSN. GST type must be determined (intra-state/inter-state).

Inputs: Line item details (product, qty, rate), gross value per line, tax type (CGST/SGST/IGST), tax rate per line, tax amount per line, total per line, invoice summary.

Processing Logic: 1. During GRN/invoice entry, system captures per line item: Product, Quantity, Unit Rate, Gross Value (Qty × Rate), Tax Type, Tax Rate, Tax Amount, Line Total (Gross + Tax). 2. System displays line-item breakdown in tabular format: | Product | Qty | Rate | Gross | CGST% | CGST Amt | SGST% | SGST Amt | Total | 3. For IGST (inter-state): Single IGST column instead of CGST+SGST. 4. System calculates invoice summary: Total Gross, Total CGST, Total SGST/IGST, Grand Total. 5. If invoice total does not match calculated grand total, system highlights discrepancy with line-item comparison. 6. User can identify error line item by comparing individual line totals with invoice. 7. All line-item details are stored in database for reporting and audit. 8. Tally integration posts each line item separately with tax split. 9. Report: Line-item wise tax analysis, Tax liability by HSN code, GST reconciliation support.

Outputs: Line-item breakdown table, invoice summary, discrepancy highlight, tax report, Tally line-item entries.

Postconditions: 1. Tax split is mandatory: CGST + SGST for intra-state, IGST for inter-state. 2. Tax rate is determined by HSN code and product category (configurable mapping). 3. Line-item gross and tax values are immutable after GRN confirmation. 4. Export to Excel preserves line-item breakdown format. 5. Tax amounts are rounded to 2 decimal places per line; total tax may have minor rounding difference handled separately. 6. Zero-rated and exempt items are clearly marked with reason (e.g., 'Nil GST - Healthcare Exemption').

Business Rules: 1. If tax rate not configured for product, system uses default rate and alerts admin. 2. If HSN code missing, system prompts for entry before tax calculation. 3. If invoice tax structure differs from system calculation, system flags for manual verification.

Exception Handling: 1. Line items are entered. 2. Tax rates are configured. 3. GST type is determined.

## REQ-G010 – Invoice Error Correction Without Cancellation

Priority: High | Status: Proposed

Description: The system shall allow correction of posted invoices/GRNs without requiring full cancellation and re-accounting, reducing processing time and maintaining audit integrity.

Preconditions: GRN/Invoice must be posted (not in draft). User must have correction initiation privilege. Correction reason must be provided.

Inputs: GRN/Invoice ID, error type (Wrong rate/Wrong qty/Wrong tax/Wrong supplier/Wrong date), original values, corrected values, correction reason, approver ID, correction timestamp.

Processing Logic: 1. User identifies error in posted GRN/invoice (e.g., wrong rate entered, quantity typo). 2. User initiates 'Correction Request' from GRN view screen. 3. System displays original values in read-only format. 4. User enters corrected values and selects error type from dropdown. 5. User provides detailed correction reason (minimum 30 characters). 6. System routes correction request to approver (store manager or accounts based on error type). 7. Approver reviews: Original vs. Corrected, Reason, Impact on stock/payment. 8. Approver can approve, reject, or request clarification. 9. Upon approval: System updates GRN/invoice with corrected values, generates correction journal entry, updates stock if quantity/rate changed. 10. System maintains correction history: Original values, Corrected values, Correction reason, Approved by, Date. 11. Correction report for management: Error frequency by user, Error types, Correction turnaround time. 12. Tally integration: Original entry is reversed, new entry posted with correction reference.

Outputs: Corrected GRN/invoice, correction history, journal entries, stock updates, correction report.

Postconditions: 1. Correction is allowed only for data entry errors, not for supplier invoice errors (those require credit note). 2. Correction after payment requires additional payment adjustment workflow. 3. Correction after stock issue requires stock reconciliation if quantities changed. 4. Maximum 2 corrections per GRN/invoice; beyond that requires management approval. 5. Correction history is immutable; no deletion allowed. 6. Monthly correction report is shared with management for process improvement. 7. Users with frequent corrections (>3 per month) trigger training requirement.

Business Rules: 1. If GRN is paid, system initiates payment adjustment workflow before correction. 2. If stock already issued, quantity correction requires stock return/re-issue. 3. If approver rejects, user can appeal to management with additional justification.

Exception Handling: 1. GRN/Invoice is posted. 2. User has correction privilege. 3. Reason is provided.

### REQ-G011 – GRN Label Printing

Priority: Medium | Status: Proposed

Description: Auto-generate GRN label with barcode. Content: GRN number, date, supplier, PO reference. Print and paste on invoice. Standard label sizes. Bulk printing.

Preconditions: A GRN has been created and finalized against a PO and supplier invoice/DC.

Inputs: GRN number, GRN date, supplier name, PO reference, label printer configuration.

Processing Logic: 1. On GRN finalization, the system auto-generates a printable label containing a barcode encoding the GRN number, along with human-readable GRN number, date, supplier name, and PO reference. 2. Label is sent to a configured label printer in a standard supported size. 3. Store Keeper affixes the printed label to the physical invoice/DC for quick future retrieval and scanning. 4. Reprint is available at any time by recalling the GRN number, in case of a lost/damaged label.

Outputs: Printed GRN barcode label, reprint capability, printer configuration profile.

Postconditions: 1. Every physical invoice on file can be instantly linked back to its system GRN record via a single barcode scan.

Business Rules: 1. Reprinting a label does not alter the original GRN data; it only regenerates the physical label from the stored GRN metadata. 2. Label format/size is configurable per store to match available printer hardware.

Exception Handling: 1. If the configured label printer is unavailable, the system offers a PDF fallback that can be printed on any standard printer with the barcode still scannable.

## REQ-G012 – Filters in PO/GRN Modules

Priority: Medium | Status: Proposed

Description: Filter by invoice value range, invoice number (partial), supplier, date range, status. Save frequent filters. Export filtered results.

Preconditions: User needs to locate specific PO or GRN records among a large transactional volume.

Inputs: Invoice value range, invoice number (partial match), supplier, date range, status.

Processing Logic: 1. PO and GRN list screens provide filter controls for invoice value range, partial invoice number search, supplier, date range, and status (e.g., Pending/Approved/GRN Pending/Closed). 2. Filters can be combined (AND logic) to narrow results precisely. 3. Frequently used filter combinations can be saved by the user as a named view for one-click reuse. 4. Filtered result sets can be exported to Excel for offline analysis or sharing.

Outputs: Filtered PO/GRN list view, saved filter presets, exported result set.

Postconditions: 1. Users can retrieve the specific subset of transactions they need without manually scanning long lists.

Business Rules: 1. Saved filters are personal to the user by default but can optionally be shared department-wide by a Stores Manager. 2. Exported data respects the user's role-based access – a user cannot export records outside their permitted scope.

Exception Handling: 1. If a filter combination returns zero results, the system displays a clear 'no matching records' message rather than an empty, ambiguous screen.

### REQ-G013 – Credit Card Purchase Tagging

Priority: Medium | Status: Proposed

Description: Tag credit card purchases (MD/Doctors). Capture last 4 digits, cardholder. Track reimbursement. Reconcile with statement. Alert untagged transactions. Report by cardholder.

Preconditions: A purchase was made using a hospital-linked credit card (e.g., by MD or a Doctor) rather than through the standard vendor-invoice payment route.

Inputs: PO/GRN reference, cardholder name, last four digits of card, transaction amount, statement reference.

Processing Logic: 1. At GRN/invoice entry, the user tags the transaction as a credit card purchase, capturing cardholder name and the last four digits of the card (never the full card number, for PCI compliance). 2. The transaction is flagged for reimbursement/settlement tracking against the corresponding monthly card statement. 3. When the card statement is available, Finance reconciles each tagged transaction against the statement line item, marking it Reconciled or Discrepant. 4. A report lists any credit-card purchase transactions that remain untagged, since these represent a reconciliation gap.

Outputs: Credit-card-tagged transaction, statement reconciliation status, untagged-transaction exception report.

Postconditions: 1. Every credit-card-funded purchase is reconciled to the monthly statement, closing the loop between operational purchase and financial settlement.

Business Rules: 1. Full card numbers are never stored in the system; only the last four digits and cardholder name are captured, per standard PCI-DSS data-minimization practice. 2. Untagged transactions on the card statement are investigated by Finance directly with the cardholder.

Exception Handling: 1. If a tagged transaction's amount does not match the statement line during reconciliation, it is flagged Discrepant and routed to the cardholder/Finance for clarification before being marked closed.

## REQ-G014 – Department Delayed Invoice Submission Tracking

Priority: Medium | Status: Proposed

Description: Track when departments receive goods directly. Compare receipt vs invoice submission date. Alert if >3 days. Report delayed submissions. Escalate repeated delays.

Preconditions: A department has received goods directly (bypassing central Stores receipt) and is expected to submit the corresponding invoice/DC for accounting.

Inputs: Department, item/PO reference, goods receipt date (as reported by department), invoice submission date.

Processing Logic: 1. System captures the date the department reports having received goods directly. 2. It compares this against the date the invoice/DC is actually submitted to Stores/Accounts for processing. 3. If the gap exceeds a configurable threshold (e.g., 3 days), the department is flagged and an automated reminder is sent. 4. A recurring-delay report identifies departments and item categories with a pattern of late invoice submission for management follow-up.

Outputs: Receipt-to-submission gap tracking, automated reminder, recurring-delay report.

Postconditions: 1. Delays in invoice booking caused by department-side direct receipt are visible and addressable rather than silently accumulating as unaccounted goods-received-not-invoiced exposure.

Business Rules: 1. The reminder threshold and escalation path are configurable by Finance policy and may differ by department or item criticality. 2. Persistent late-submission departments are included in a periodic management review alongside REQ-I010 (Goods Received but Invoice Not Received).

Exception Handling: 1. If a department disputes the recorded receipt date, the dispute is logged and the goods-receipt date is corrected only with Stores Manager verification, not by the department unilaterally.



## 3.5 INDENT & ISSUE MANAGEMENT

### 3.5 Indent & Issue Management Process Flow

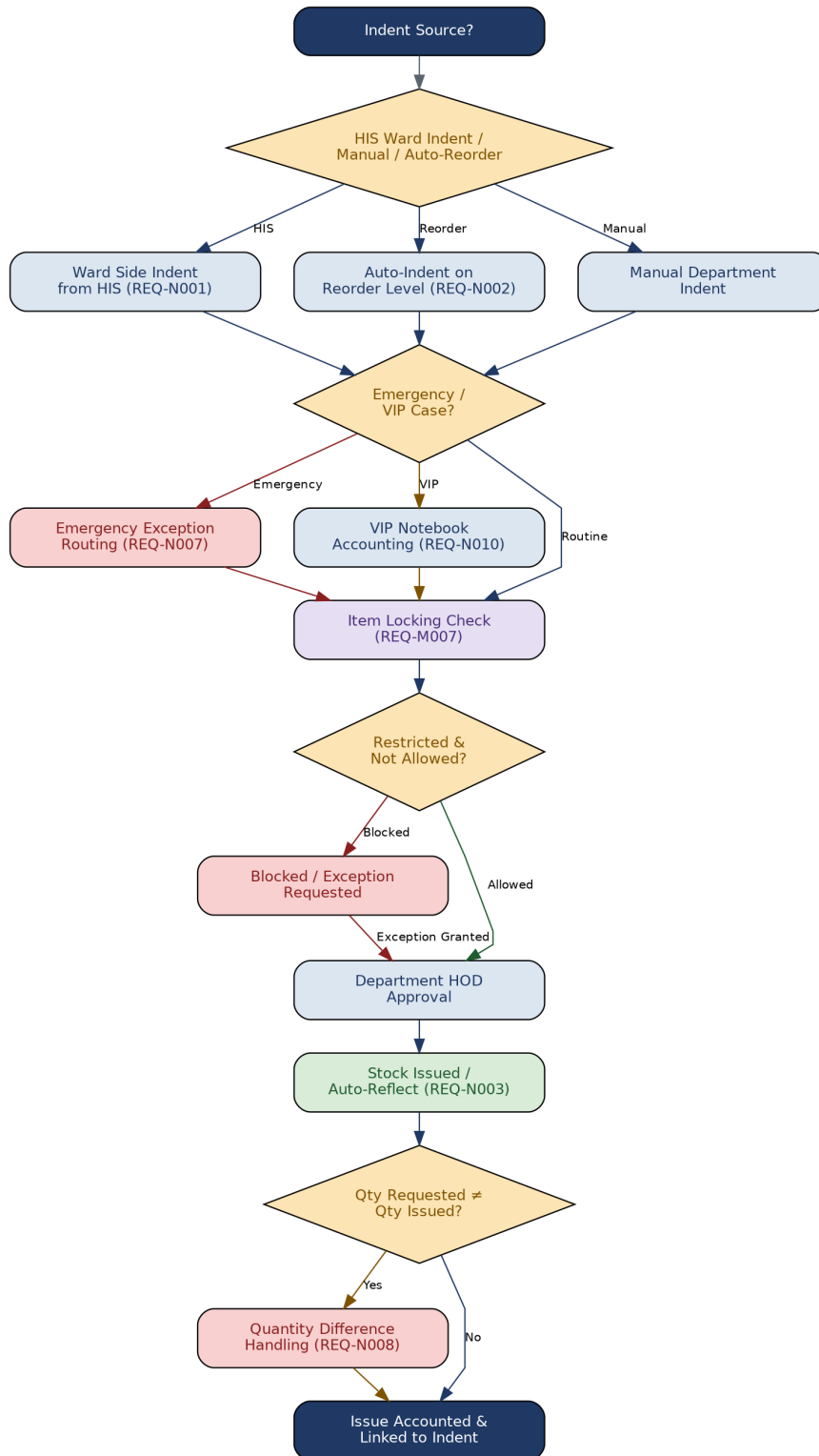


Figure: 3.5 Indent &amp; Issue Management – Process Flow

## REQ-N001 – Ward Side Indent from HIS

Priority: High | Status: Proposed

Description: The system shall integrate with the existing Hospital Information System (HIS) to automatically generate indent requests from ward-level consumption points, eliminating duplicate data entry and ensuring real-time stock demand capture.

Preconditions: HIS API/database must be accessible. Product mapping must be configured. HIS user must have indent creation privilege.

Inputs: HIS indent data: Ward ID, Department ID, Patient ID (if applicable), Item code, Item name, Quantity required, Urgency flag, Requested by (doctor/nurse), Request timestamp, Clinical notes.

Processing Logic: 1. HIS generates indent request based on: Ward stock consumption, Doctor prescription, Nursing requirements, Patient admission/discharge. 2. HIS pushes indent data to IHMS via REST API or database sync (configurable: real-time or batch every 15 minutes). 3. IHMS receives indent data and maps HIS item codes to IHMS internal product codes (via cross-reference table). 4. System creates indent in IHMS with status 'Received from HIS'. 5. System validates: Item exists in IHMS master? If not, flag for product mapping. 6. Indent routes to department HOD for approval (if value > threshold or new item). 7. Upon HOD approval, indent status changes to 'Approved' and routes to stores for processing. 8. Stores processes indent: Check stock, Create issue voucher, Update stock, Deliver to ward. 9. IHMS updates HIS with indent status: Pending / Approved / Issued / Rejected. 10. HIS displays IHMS status in ward dashboard for staff visibility. 11. Error handling: If sync fails, system retries 3 times with exponential backoff, then logs for manual intervention. 12. HIS integration audit log: All push/pull transactions with timestamps, status, error details.

Outputs: IHMS indent created from HIS, status sync back to HIS, product mapping log, error handling record, integration audit log.

Postconditions: 1. HIS indent integration is mandatory for all wards; manual indent is fallback only. 2. Product mapping between HIS and IHMS is maintained by admin; new mappings require approval. 3. HIS indents for new products (not in IHMS master) are held in 'Pending Mapping' queue. 4. Urgent HIS indents (ICU, OT, Emergency) bypass HOD approval and route directly to stores. 5. HIS indent numbers are preserved in IHMS for cross-reference. 6. Batch sync runs every 15 minutes during business hours; real-time for urgent indents.

Business Rules: 1. If HIS API is down, system queues indents and processes when API is restored. 2. If product mapping not found, system alerts admin and holds indent in mapping

queue. 3. If HIS sends duplicate indent, system detects by HIS indent number and ignores duplicate.

Exception Handling: 1. HIS API is accessible. 2. Product mapping exists. 3. HIS user is authorized.

## REQ-N002 – Auto-Indent Based on Reorder Level

Priority: High | Status: Proposed

Description: The system shall automatically generate indent requests when stock levels fall below predefined reorder thresholds, ensuring continuous availability of critical items without manual monitoring.

Preconditions: Product must have reorder level configured. Location must be active. Auto-indent must be enabled for product category.

Inputs: Product ID, current stock, reorder level, reorder quantity, department ID, location ID, consumption trend, last indent date.

Processing Logic: 1. System runs daily batch job (configurable time, e.g., 6:00 AM) to evaluate stock levels. 2. For each product at each location: If Current Stock  $\leq$  Reorder Level, trigger auto-indent evaluation. 3. System checks: Is there already a pending indent or PO for this product at this location? If yes, skip. 4. System calculates reorder quantity based on: Consumption trend (last 3 months average), Lead time, Safety stock. 5. System generates auto-indent with: Product, Reorder Quantity, Department (location owner), Reason: 'Auto-generated based on reorder level'. 6. Auto-indent routes to department HOD for approval (or auto-approves if within routine limits). 7. HOD can modify quantity, add items, or cancel if not required. 8. Approved auto-indents route to stores for procurement action. 9. System sends notification: To stores (new auto-indent), To HOD (stock below threshold). 10. System tracks auto-indent effectiveness: Stockouts prevented vs. excess stock created. 11. Quarterly review: Reorder levels adjusted based on consumption variance analysis.

Outputs: Auto-indent document, approval workflow, procurement initiation, stock level alert, reorder level adjustment recommendation.

Postconditions: 1. Reorder Level = (Average daily consumption  $\times$  Lead time in days) + Safety stock. 2. Auto-indent is enabled by default for consumables; disabled for capital assets. 3. Critical items (emergency drugs, life support) have lower reorder levels and faster processing. 4. Seasonal items can have seasonal reorder levels (higher for monsoon/flu season). 5. Auto-indent does not apply to vendor-managed inventory (VMI) items. 6. More than 5 auto-indents per product per month triggers consumption pattern review.

Business Rules: 1. If reorder level not configured, system alerts admin and skips product. 2. If supplier lead time increased, system suggests reorder level adjustment. 3. If multiple locations have same product below reorder, system consolidates into single indent.

Exception Handling: 1. Product has reorder level. 2. Location is active. 3. Auto-indent is enabled.

## REQ-N003 – Auto-Reflect Indent as Issue

Priority: High | Status: Proposed

Description: The system shall automatically convert approved indents into issue vouchers, reducing manual data entry time and ensuring rapid stock dispatch to requesting departments.

Preconditions: Indent must be approved. Stock must be available (at least partially). Store keeper must have issue privilege.

Inputs: Approved indent ID, indent line items, available stock, issue voucher template, issuer ID, issue date.

Processing Logic: 1. Indent is approved by HOD (or auto-approved). 2. System checks indent status: If 'Approved', trigger auto-issue workflow. 3. System retrieves available stock for each item in the indent. 4. System creates draft issue voucher with: Indent reference number, Items, Requested quantities, Available quantities. 5. Store keeper reviews draft issue voucher: Can modify issued quantities if stock insufficient, Can add notes for partial issue. 6. Store keeper confirms issue voucher. 7. System updates stock: Reduce from stores, Add to department stock (if tracked). 8. System updates indent status: 'Issued'. 9. System generates issue voucher document with barcode for physical delivery verification. 10. Department receives notification: Issue ready for collection/delivery. 11. If partial issue: Balance quantity remains in indent as 'Pending Issue' for future fulfillment. 12. System tracks indent-to-issue conversion time for SLA monitoring.

Outputs: Issue voucher, stock update, indent status update, issue document, delivery notification, partial issue tracking.

Postconditions: 1. Auto-issue is triggered immediately upon indent approval (no batch delay). 2. Store keeper has 2 hours to review and confirm draft issue (configurable SLA). 3. If store keeper does not act within SLA, system escalates to store manager. 4. Partial issue is allowed; balance auto-converts to backorder or new indent. 5. Issue voucher is linked to indent by reference number for traceability. 6. Emergency indents bypass draft review; auto-issue executes immediately.

Business Rules: 1. If no stock available, system generates expedited PO instead of issue voucher. 2. If store keeper rejects issue, indent returns to HOD with reason. 3. If item requires batch selection (FEFO), system auto-suggests batches.

Exception Handling: 1. Indent is approved. 2. Stock is available. 3. Store keeper is authorized.

## REQ-N004 – Auto-Issue Accounting for Direct Orders

Priority: High | Status: Proposed

Description: The system shall automatically account for items and services ordered directly by departments (without stores involvement), ensuring complete usage tracking and budget allocation.

Preconditions: Department must be active. HOD must confirm receipt. Invoice must be available for accounting.

Inputs: Department direct order details, invoice data, HOD confirmation, stores accounting entry, budget charge.

Processing Logic: 1. Department places direct order with vendor (e.g., service repair, specialized equipment). 2. Vendor delivers service/goods and submits invoice directly to department or stores. 3. Department HOD confirms receipt/acceptance in system: 'Goods/Services Received and Accepted'. 4. If invoice comes to stores, store keeper creates accounting entry: Links to department, Captures invoice details, Charges to department budget. 5. If invoice comes directly to accounts, accounts creates entry with department reference. 6. System captures: Department, Product/Service, Value, Vendor, Date, HOD confirmation status. 7. System updates department usage report with direct order values. 8. System updates budget utilization for the department. 9. Monthly report: Direct orders by department, Value, Vendor, Approval status. 10. Direct orders without HOD confirmation are flagged for follow-up.

Outputs: Direct order accounting entry, department usage update, budget charge, HOD confirmation record, monthly direct order report.

Postconditions: 1. Direct orders > Rs. 10,000 require pre-approval by management (even if department-initiated). 2. Direct orders for capital equipment always require management approval. 3. HOD confirmation is mandatory before accounting entry; without confirmation, payment is blocked. 4. Direct orders are included in department budget utilization. 5. Stores maintains visibility of all direct orders for inventory completeness. 6. Frequent direct orders (>20% of department purchases) trigger centralization review.

Business Rules: 1. If HOD does not confirm within 7 days, system escalates to management. 2. If invoice not received within 30 days of service, system alerts department for follow-up. 3. If direct order exceeds budget, system blocks accounting and alerts management.

Exception Handling: 1. Department is active. 2. HOD confirms receipt. 3. Invoice is available.

## REQ-N005 – Indent Timetable Cross-Verification

Priority: Medium | Status: Proposed

Description: Management-approved indent timetable. Cross-verify indent date/time. Alert if off-schedule. Emergency override with reason. Four key departments exempt. Report off-schedule indents.

Preconditions: Management has approved a fixed indenting timetable specifying which days/times each department may submit routine indents.

Inputs: Department, configured indent day/time slot, actual indent submission timestamp.

Processing Logic: 1. System stores the management-approved indent timetable per department. 2. On indent submission, the system compares the submission timestamp against the department's configured slot. 3. If the indent falls within the approved window, it proceeds normally. 4. If it falls outside the window, the system flags it as off-schedule and requires the requester to provide an emergency-override reason before it can be submitted. 5. Off-schedule submissions are still processed (not blocked) but are visibly tagged for management review.

Outputs: Timetable-compliance flag on each indent, emergency-override reason capture, off-schedule indent tagging.

Postconditions: 1. Stores workload is more predictable because most indents arrive within the planned schedule, while genuine exceptions are still accommodated and tracked, not blocked outright.

Business Rules: 1. The four key departments defined in REQ-N007 (ICU, OT, Emergency, NICU) are exempt from timetable enforcement entirely. 2. Timetable changes require Management approval and are versioned so historical compliance can be assessed against the timetable in force at the time.

Exception Handling: 1. If a department has no configured timetable slot (new department, not yet scheduled), all its indents are treated as off-schedule by default until a slot is assigned.

## REQ-N006 – Department Indent Timing Report

Priority: Medium | Status: Proposed

Description: Show departments that indented on schedule. Identify repeated off-schedule requests. Frequency analysis. Trend over weeks/months. Management dashboard.

Preconditions: Indent timetable compliance data (REQ-N005) is being accumulated over time.

Inputs: Department, indent date/time, on-schedule/off-schedule flag, reporting period.

Processing Logic: 1. System aggregates each department's indents over a selected period, computing the proportion submitted on-schedule versus off-schedule. 2. Departments with repeated off-schedule submissions are highlighted distinctly from those with an occasional exception. 3. A trend view shows whether a department's compliance is improving, stable, or worsening over successive weeks/months. 4. The report is available to Stores Management for coaching conversations with repeat-offending departments.

Outputs: Department-wise on-schedule/off-schedule ratio, repeat-offender highlight, week-over-week/month-over-month trend.

Postconditions: 1. Management has an evidence-based basis for engaging with departments about indenting discipline rather than relying on anecdotal impressions.

Business Rules: 1. The exempt key departments (REQ-N007) are excluded from this compliance analysis, since they are not subject to the timetable in the first place.

Exception Handling: 1. If a department's indent volume is too low in a period to be statistically meaningful (e.g., fewer than a configurable minimum count), the report flags the data as 'insufficient volume' rather than displaying a misleading percentage.

## REQ-N007 – Emergency Exception for Key Departments

Priority: Medium | Status: Proposed

Description: Four key departments (ICU, OT, Emergency, NICU) exempt from timetable. Fast-track approval. Priority issue. Track frequency for capacity planning.

Preconditions: The requesting department is one of the four designated key/critical departments: ICU, OT, Emergency, or NICU.

Inputs: Department, indent details, urgency context.

Processing Logic: 1. System recognizes indents from ICU, OT, Emergency, and NICU as exempt from the standard indent timetable at the point of submission. 2. Such indents bypass the off-schedule flagging entirely and are automatically routed through a fast-track approval path. 3. Stock issue for these departments is prioritized ahead of standard-priority indents when stock is limited. 4. System tracks the frequency and volume of exempted indents from each key department over time to inform capacity and stocking-level planning for these critical areas.

Outputs: Fast-tracked, non-flagged indent processing for key departments, priority issue allocation, frequency-tracking data for capacity planning.

Postconditions: 1. Life-critical departments are never delayed by timetable compliance mechanics designed for routine, non-critical departments.

Business Rules: 1. The list of exempt key departments can only be changed by Management approval, not by individual department request. 2. Even though exempt from the timetable, these departments' indents still pass through the standard value-based and stock-availability workflows.

Exception Handling: 1. If a non-exempt department attempts to claim key-department priority handling, the system rejects the override since exemption is tied to the department master flag, not to a self-declared urgency claim.

## REQ-N008 – Quantity Difference Handling During Issue

Priority: Medium | Status: Proposed

Description: If issued qty  $\neq$  requested: record actual qty. Mandatory reason. Balance to backorder. Department notification. Track accuracy rate.

Preconditions: An indent has been approved and is being fulfilled by Stores, but the quantity actually issued may not exactly equal the quantity requested.

Inputs: Indent ID, requested quantity, issued quantity, reason for difference.

Processing Logic: 1. At the point of issue, the Store Keeper records the actual quantity issued, which the system compares against the originally requested quantity. 2. Any difference (over or under) requires a mandatory reason (stock-out, packaging unit constraint, rounding to case size, department over-request correction). 3. Any shortfall is automatically converted to a backorder (per REQ-I022) linked to the same indent. 4. System computes an issue-accuracy rate (percentage of indents issued exactly as requested) by item and department for process-



improvement analysis. 5. The requesting department is notified of any quantity difference and its reason.

Outputs: Recorded actual-vs-requested quantity, reason log, backorder linkage, issue-accuracy rate report, department notification.

Postconditions: 1. Every quantity discrepancy between what was requested and what was issued is explained and traceable, not silently absorbed.

Business Rules: 1. Differences due to standard packaging/case-size rounding may be configured as a pre-approved reason category not requiring case-by-case justification, provided the rounding direction and magnitude are within policy tolerance.

Exception Handling: 1. If a pattern of significant, unexplained quantity differences emerges for a specific item or department, the system flags it for Stores Manager review outside the routine reporting cycle.

### REQ-N009 – Addition/Deletion in Online Indents

Priority: Medium | Status: Proposed

Description: Department can modify indents before stores processing. Track modification count. Report to stores/management. Identify frequent modifiers. Include emergency departments.

Preconditions: An indent has been submitted online but has not yet been processed/picked by Stores.

Inputs: Indent ID, modification type (add item / delete item / change quantity), department, user making the modification.

Processing Logic: 1. While an indent remains in 'Pending Processing' status, the requesting department can add items, remove items, or change requested quantities. 2. Once Stores begins processing (picking/issuing) the indent, further modification is blocked; any additional need must be raised as a new indent. 3. Every modification is counted and logged against the indent and the department. 4. A report aggregates modification counts by department, distinguishing frequent modifiers from departments that typically submit accurate, final indents on first entry.

Outputs: Modification history per indent, modification-count report by department, frequent-modifier identification.

Postconditions: 1. Departments retain reasonable flexibility to correct an indent before it is acted upon, while Stores is protected from being asked to reverse work already in progress.

Business Rules: 1. A modification after processing has started must go through the return/adjustment workflow instead of direct indent editing. 2. Excessive modification frequency by a department is a candidate topic for the indent-timetable and process-training review alongside REQ-N006.

Exception Handling: 1. If Stores begins processing at the exact moment a modification is submitted, the system resolves the race condition by rejecting the modification and notifying the department that processing has already started.

### REQ-N010 – VIP Indent Notebook Accounting

Priority: Medium | Status: Proposed

Description: Record notebook-based indents. Convert to system indents. Track notebook vs online ratio. Reconciliation report for audit.

Preconditions: A VIP or senior stakeholder's indent is initially recorded in a physical notebook rather than the online system, per existing informal practice.

Inputs: Notebook entry details (date, item, requester, quantity), subsequent system indent reference (once converted).

Processing Logic: 1. Store Keeper logs the notebook-based indent details into the system as a distinct 'Notebook-Origin' indent type, preserving the original notebook date and requester. 2. This converted entry then follows the standard indent-to-issue workflow like any other indent. 3. System tracks the ratio of notebook-origin indents versus directly-online indents over time, by requester/department, to monitor the extent of off-system practice still in use. 4. A reconciliation report for audit purposes confirms that every notebook entry for a period has a corresponding system record, with any gaps highlighted.

Outputs: Notebook-origin indent record, notebook-vs-online ratio tracking, audit reconciliation report.

Postconditions: 1. Even informally-initiated indents are captured in the system of record, preserving a complete and auditable picture of all material requests, not just those entered directly online.

Business Rules: 1. Notebook-origin indents are still subject to the same approval, budget, and stock-availability controls as any other indent; the informal origin does not bypass

governance. 2. Persistent, high-volume notebook usage by a particular requester/department is flagged for a digital-adoption follow-up conversation.

Exception Handling: 1. If a notebook entry cannot be matched to a valid item/department in the system (illegible or ambiguous notebook entry), it is logged as an exception pending clarification from the Store Keeper who received the original request.

## 3.6 ASSET MANAGEMENT

### 3.6 Asset Management Process Flow

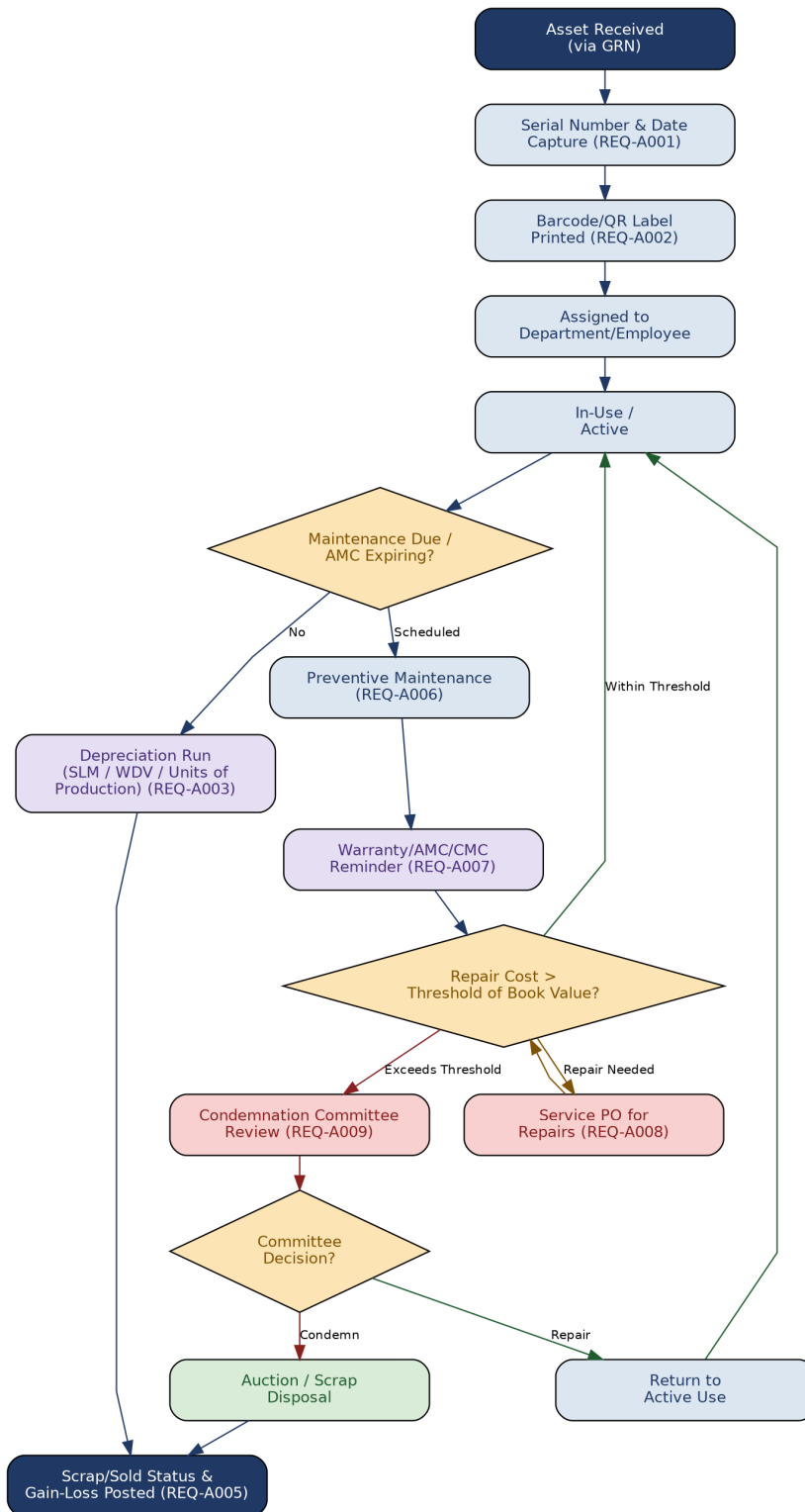


Figure: 3.6 Asset Management – Process Flow

## REQ-A001 – Serial Number and Date Capture

Priority: High | Status: Proposed

Description: The system shall capture comprehensive asset identification details including serial numbers, manufacture dates, expiry dates, and warranty periods to ensure complete traceability and compliance.

Preconditions: Asset must be received through GRN. Serial number must be unique. Department and location must be assigned.

Inputs: Asset ID, serial number, manufacturer name, model number, manufacture date, purchase date, warranty start date, warranty end date, asset tag number, location, department, custodian.

Processing Logic: 1. During asset receipt (GRN), system prompts for asset-specific details. 2. Mandatory fields: Serial Number (unique per asset), Manufacturer, Model Number, Purchase Date. 3. Optional fields: Manufacture Date, Warranty Start Date, Warranty End Date, Asset Tag Number (auto-generated if blank). 4. System validates: Serial number uniqueness across all assets. 5. System calculates warranty period: Warranty End Date - Warranty Start Date. 6. System generates asset tag number if not provided: Format {DeptCode}-{Year}-{Sequence} (e.g., ICU-2026-0156). 7. Asset details are stored in Asset Register with full history. 8. Search capability: By serial number, asset tag, product name, department, location. 9. Asset details are printed on barcode label (REQ-I001). 10. Warranty expiry triggers reminder workflow (REQ-I003).

Outputs: Asset register entry, barcode label data, warranty reminder trigger, search index, asset history log.

Postconditions: 1. Serial number is mandatory and unique across all assets. 2. Asset tag number is auto-generated but can be overridden by admin. 3. Manufacture date cannot be later than purchase date. 4. Warranty end date must be after warranty start date. 5. Asset details can be updated by admin only; changes are logged. 6. Asset register is synchronized with Tally for fixed asset accounting.

Business Rules: 1. If serial number is duplicate, system alerts and requires verification. 2. If manufacture date > purchase date, system blocks and requests correction. 3. If warranty dates not provided, system assumes 1-year standard warranty.

Exception Handling: 1. Asset is received. 2. Serial number is unique. 3. Department is assigned.

## REQ-A002 – Barcode Printing for Assets

Priority: High | Status: Proposed

Description: The system shall generate durable barcode labels for all fixed assets to enable scanning-based tracking during transfers, maintenance, and inventory audits.

Preconditions: Asset must be registered. Printer must be configured. User must have label printing privilege.

Inputs: Asset ID, asset code, label format (Code 128/QR Code), printer selection, label size, print quantity.

Processing Logic: 1. System generates unique barcode for each asset at time of receipt. 2. Barcode formats supported: Code 128 (1D) for standard scanners, QR Code (2D) for mobile scanning. 3. Label content: Asset Code (human-readable), Product Name (abbreviated), Department, Purchase Date (MM/YY), Barcode/QR image. 4. User selects label size: 50x25mm, 75x50mm, 100x50mm. 5. User selects printer from configured list (Zebra, TSC, etc.). 6. System generates print-ready PDF with label layout. 7. Single print or bulk print (up to 100 labels per batch). 8. Reprint option for damaged/lost labels (requires reason and supervisor approval if >2 reprints). 9. QR Code contains URL linking to asset detail page (for mobile access). 10. Label print log: Asset, Date, User, Reason (new/reprint).

Outputs: Printed barcode labels, print log, reprint history, asset scan records, label inventory.

Postconditions: 1. Each asset receives exactly one barcode at receipt; duplicates prevented. 2. Barcode is immutable; asset code never changes. 3. Reprint >2 times requires store manager approval. 4. Label material: Polyester recommended for durability. 5. Barcode scanning updates asset location in real-time. 6. Lost label reprint requires physical verification of asset.

Business Rules: 1. If printer offline, system queues job and retries. 2. If label format incompatible, system suggests alternative. 3. If asset code not found, search by serial number or product name.

Exception Handling: 1. Asset is registered. 2. Printer is configured. 3. User has printing privilege.

## REQ-A003 – Depreciation Handling

Priority: High | Status: Proposed

Description: The system shall calculate and track depreciation for fixed assets using configurable methods and rates, generating periodic depreciation entries for accounts and Tally integration.

Preconditions: Asset must be capitalized. Depreciation parameters must be configured. Asset must be in Active/In-Use status.

Inputs: Asset ID, asset category, acquisition cost, salvage value, useful life, depreciation method (SLM/WDV/Units of Production), rate or per-unit rate, total estimated production/usage units, actual units produced or used, start date, accumulated depreciation, book value.

Processing Logic: 1. Admin configures depreciation parameters per asset category: Method (Straight Line / Written Down Value / Units of Production), Rate or per-unit rate, Useful Life or Total Estimated Production Units. 2. During asset receipt, system captures: Acquisition Cost (landed cost), Date of Put to Use, Asset Category. 3. System auto-calculates depreciation: SLM:  $(\text{Cost} - \text{Salvage Value}) / \text{Useful Life (annual)}$  WDV:  $\text{Book Value} \times \text{Depreciation Rate (annual)}$  Units of Production:  $(\text{Cost} - \text{Salvage Value}) / \text{Total Estimated Production Units}$ , multiplied by Actual Units Produced or Used. 4. User runs depreciation process monthly/quarterly/annually, or per usage cycle for Units of Production assets. 5. System generates depreciation schedule: Period, Opening WDV, Depreciation Amount, Closing WDV. 6. System posts journal entry to Tally: Debit Depreciation Expense, Credit Accumulated Depreciation. 7. Narration: 'Depreciation for {AssetCode} - {AssetName} - {Period}'. 8. System updates asset register: Accumulated Depreciation, Book Value. 9. Depreciation report by department, category, individual asset. 10. On disposal:  $\text{Gain/Loss} = \text{Sale Value} - \text{Book Value}$ , posted to Tally.

Outputs: Depreciation schedule, journal entries, asset register updates, depreciation report, gain/loss calculation.

Postconditions: 1. Depreciation starts month after date of put to use (Indian GAAP). 2. Depreciation stops in month of sale/scraping. 3. Idle assets (>180 days) continue depreciation but flagged for review. 4. Method change requires management approval; applied prospectively. 5. Rates follow Income Tax Act or company policy (whichever is stricter). 6. Depreciation calculated to 2 decimal places.

Business Rules: 1. If parameters not configured, system uses default category. 2. If asset cost below threshold, marked as 'Expense' and skipped. 3. If depreciation run fails, system continues with others and logs error.

Exception Handling: 1. Asset is capitalized. 2. Parameters are configured. 3. Asset is active.

## REQ-A004 – Product Lifecycle Management

Priority: High | Status: Proposed

Description: Procurement → Receipt → Installation → Active → Maintenance → Disposal. Status transitions with approval. Lifecycle cost tracking. Utilization metrics. Retirement planning. Compliance.

Preconditions: An asset has been procured and capitalized, entering the system as a tracked asset record.

Inputs: Asset ID, lifecycle stage, transition date, approver, associated cost at each stage.

Processing Logic: 1. System defines a standard asset lifecycle: Procurement → Receipt → Installation → Active (In-Use) → Maintenance (as needed, non-linear) → Disposal. 2. Each transition between stages requires the appropriate approval (e.g., Biomedical Engineering sign-off for Installation-to-Active, Management approval for Active-to-Disposal). 3. System accumulates lifecycle cost at each stage – acquisition cost, installation/commissioning cost, cumulative maintenance cost – building a total cost of ownership view per asset. 4. Utilization data (where available from usage-metering or manual logging) is tracked during the Active stage to inform replacement planning.

Outputs: Stage-tagged asset record, transition approval log, total cost of ownership per asset, utilization data.

Postconditions: 1. Every asset's current stage and complete stage history is available at a glance, supporting both operational planning and statutory asset audit.

Business Rules: 1. An asset cannot move to Active status until Installation is confirmed complete and, where applicable, calibration/commissioning is certified. 2. Disposal can only be initiated from Active or Maintenance status, never directly from Procurement/Receipt.

Exception Handling: 1. If an asset sits in Receipt or Installation status beyond a configurable threshold without progressing, it is flagged for Biomedical Engineering follow-up to prevent unused capitalized assets from remaining unaccounted for operational purposes.



## REQ-A005 – Scrap/Sold/Red-Tag Marking

Priority: High | Status: Proposed

Description: Scrap: write-off. Sold: track value and buyer. Red-Tag: pending. Management approval. Physical label. Disposal report.

Preconditions: An asset has been identified for scrapping, sale, or has an undecided disposition pending review (Red-Tag).

Inputs: Asset ID, proposed status (Scrap/Sold/Red-Tag), reason, sale value and buyer (if Sold), approving authority.

Processing Logic: 1. Custodian or Biomedical Engineering proposes a status change for the asset with supporting reason. 2. For Scrap: a write-off request is routed for Management approval, referencing the asset's net book value from the depreciation module. 3. For Sold: the sale value and buyer details are captured, and the difference between sale value and net book value is computed as a gain/loss on disposal for posting to the fixed asset accounting module. 4. For Red-Tag: the asset is marked as pending-decision, physically labeled to prevent accidental continued use, while a final disposition (Scrap/Sold/redeploy) is determined. 5. A physical label reflecting the current status is generated for affixing to the asset. 6. A consolidated disposal report tracks all status changes with reasons, approvers, and financial impact.

Outputs: Status-change approval record, physical status label, gain/loss on disposal posting, disposal report.

Postconditions: 1. No asset is scrapped, sold, or left in an undecided (Red-Tag) state without a documented, approved, and financially reconciled record.

Business Rules: 1. Every disposal-related status change follows the same value-based approval matrix used for procurement (Section 5.4 of the Stores/Procurement volume), applied to the asset's book or realizable value. 2. A Red-Tag status has a maximum configurable duration before it is escalated to force a final decision, preventing indefinite limbo status.

Exception Handling: 1. If a proposed sale value is materially below the asset's net book value without clear justification (e.g., genuine obsolescence, market conditions), the transaction is escalated to a higher approval tier before proceeding.

## REQ-A006 – Preventive Maintenance

Priority: High | Status: Proposed

Description: Frequency per asset type. Calendar schedule. Checklist completion. Overdue alerts. Cost tracking. Compliance report.

Preconditions: Asset class has a defined preventive maintenance (PM) frequency, and the asset is in Active status.

Inputs: Asset ID, PM frequency (weekly/monthly/quarterly/yearly), scheduled date, checklist items, performing technician, cost.

Processing Logic: 1. System auto-generates a PM schedule per asset based on its class's configured frequency. 2. On the scheduled date, a work order is created with the relevant checklist for that asset type. 3. Biomedical Engineering/assigned technician completes the checklist, recording pass/fail per item and any parts/consumables used. 4. Completion is signed off, closing the PM work order; cost (labor and parts) is captured against the asset's maintenance cost history. 5. Overdue PM (not completed by scheduled date) is escalated to the Biomedical Engineering Head. 6. A compliance report shows the percentage of PM completed on schedule, by asset class and department, supporting NABH audit readiness.

Outputs: PM schedule, completed checklist records, overdue-PM escalation, PM compliance report, accumulated maintenance cost.

Postconditions: 1. Every asset requiring preventive maintenance has a demonstrable, auditable record of PM performed on schedule, directly supporting NABH FMS chapter compliance.

Business Rules: 1. PM schedules cannot be silently skipped; a missed PM must be either completed late (with the delay recorded) or formally deferred with documented justification and re-approval. 2. Checklist templates are configurable per asset class/model by Biomedical Engineering.

Exception Handling: 1. If an asset is unavailable for PM on its scheduled date (e.g., in active clinical use), the PM is rescheduled within a short configurable grace window, with the reason for delay recorded.

## REQ-A007 – Warranty/AMC/CMC Reminders

Priority: High | Status: Proposed

Description: Contract details. Reminders at 90/60/30/15 days. Email and dashboard. Renewal workflow. Document storage. Cost comparison: AMC vs CMC vs ad-hoc.

Preconditions: An asset has an active or upcoming warranty, AMC, or CMC contract requiring renewal tracking.

Inputs: Asset ID, contract type (Warranty/AMC/CMC), vendor, start/end date, coverage scope, cost, renewal decision.

Processing Logic: 1. Contract details (type, vendor, dates, scope, cost) are captured against the asset record. 2. System sends escalating reminders at 90, 60, 30, and 15 days before expiry via email and dashboard alert to Biomedical Engineering and Finance. 3. As a contract nears expiry, the system prompts a renewal decision, presenting a cost comparison across AMC, CMC, and ad-hoc/pay-per-call maintenance based on the asset's maintenance history, to support an informed choice. 4. Renewed or newly selected contracts are captured with their documents (scanned agreement) stored against the asset. 5. Coverage gaps (asset with no active contract past its warranty) are flagged in a dedicated exposure report.

Outputs: Contract record with documents, tiered expiry alerts, cost-comparison analysis, coverage-gap exposure report.

Postconditions: 1. No critical asset is left without a deliberate, cost-informed maintenance coverage decision once its warranty or existing contract lapses.

Business Rules: 1. The cost-comparison analysis must reference the asset's actual historical breakdown/repair cost, not just list price, to give a realistic AMC-vs-CMC-vs-ad-hoc recommendation. 2. Contracts above a configurable value require Management approval before renewal.

Exception Handling: 1. If a high-criticality asset (per its VED/asset-criticality classification) is approaching expiry without a renewal decision made, the system escalates directly to the Biomedical Engineering Head rather than only the standard reminder cadence.

## REQ-A008 – Service PO for Repairs

Priority: High | Status: Proposed

Description: Service PO for repairs (AC, generator). Track old parts handed to stores. Parts assessment: Repairable/Scrap/Return. Link to asset record. Repair history and costs. Vendor turnaround time.

Preconditions: An asset (e.g., AC unit, generator) requires external repair, and functioning or replaced parts need to be tracked.

Inputs: Asset ID, repair vendor, Service PO reference, parts removed, parts assessment (Repairable/Scrap/Return), cost.

Processing Logic: 1. Biomedical Engineering raises a Service PO for the repair, following the same value-based approval workflow as any purchase (Section 5.4). 2. Parts removed from the

asset during repair are logged and handed to Stores for safekeeping, each tagged with an assessment: Repairable (candidate for future reuse/repair), Scrap (write-off), or Return (to be returned to vendor, e.g., under warranty exchange). 3. Stores maintains a distinct sub-inventory for such removed parts, separate from new-purchase stock. 4. The repair record – vendor, cost, parts handled, and outcome – is linked to the asset's maintenance history for full lifecycle traceability.

Outputs: Service PO, removed-parts inventory with assessment tags, linked asset maintenance history entry.

Postconditions: 1. No removed part is lost or unaccounted for; every part has a documented assessment and disposition, and repair cost is fully attributed to the correct asset.

Business Rules: 1. Parts assessed as 'Return' to the vendor are tracked until the return is confirmed complete; unresolved returns beyond a configurable period are escalated. 2. Parts assessed 'Repairable' remain in the sub-inventory until either reused or reclassified to Scrap after a review period.

Exception Handling: 1. If a vendor disputes the condition/assessment of a part being returned under warranty, the dispute is logged against the Service PO and routed to Biomedical Engineering and Purchase for joint resolution.

## REQ-A009 – Asset Condemnation Committee Workflow

Priority: High | Status: Proposed

**Description:** The system shall support a formal condemnation workflow for assets that are beyond economical repair -- Inspection, Repair Evaluation, Condemnation Committee review, Approval, Auction/Scrap, Disposal, and Accounting Entry -- in addition to the existing simple Scrap/Sold status marking (REQ-A005), for assets where policy requires committee sign-off before disposal.

**Preconditions:** Asset must be in Active, Idle, or Under Maintenance status. Condemnation committee members must be configured in the system.

**Inputs:** Asset ID, inspection report, repair cost estimate, inspector remarks, photographs, committee member decisions, disposal route (Auction/Scrap), disposal value, scrap vendor/auction reference.

**Processing Logic:** 1. User or system, via the idle-asset review already defined for asset status transitions, initiates a condemnation request and records the reason. 2. A technical inspection is logged, capturing condition, estimated repair cost, and at least one supporting photograph. 3. System compares repair cost against current book value; if repair cost exceeds a configurable threshold of book value, the asset is routed to the Condemnation Committee automatically. 4. Each committee member records a recommendation, Repair or Condemn, with remarks; the committee's collective decision is captured with date and signatories. 5. On a Condemn decision, the request routes to management approval per the configured approval matrix. 6. On approval, disposal route is selected: Auction, with auction reference and winning bid value, or Scrap, with scrap vendor and disposal value. 7. On completion, system posts the accounting entry for gain or loss versus book value, per the existing depreciation handling requirement. 8. Asset status moves to Scrap/Sold only after the accounting entry is posted.

**Outputs:** Condemnation case file including inspection report and photographs, committee decision record, disposal certificate, accounting gain/loss entry, condemnation register.

**Postconditions:** 1. An asset cannot bypass the committee step once repair cost exceeds the configured threshold, even if a single approver wants to fast-track it. 2. The condemnation case file, including photographs and committee remarks, is retained for audit for the configured retention period. 3. Disposal value below the committee's approved reserve for auction requires re-approval.

**Business Rules:** 1. If fewer than the configured quorum of committee members respond within the SLA, the case is escalated to the committee chair. 2. If the asset is under an active AMC/CMC or insurance claim, condemnation is blocked until that contract or claim is closed.

Exception Handling: 1. Asset exists and is not already Scrapped/Sold. 2. Committee is configured. 3. Inspection report and at least one photograph are attached before committee review.

## 3.7 BUDGET & FINANCE INTEGRATION

### 3.7 Budget & Finance Integration Process Flow

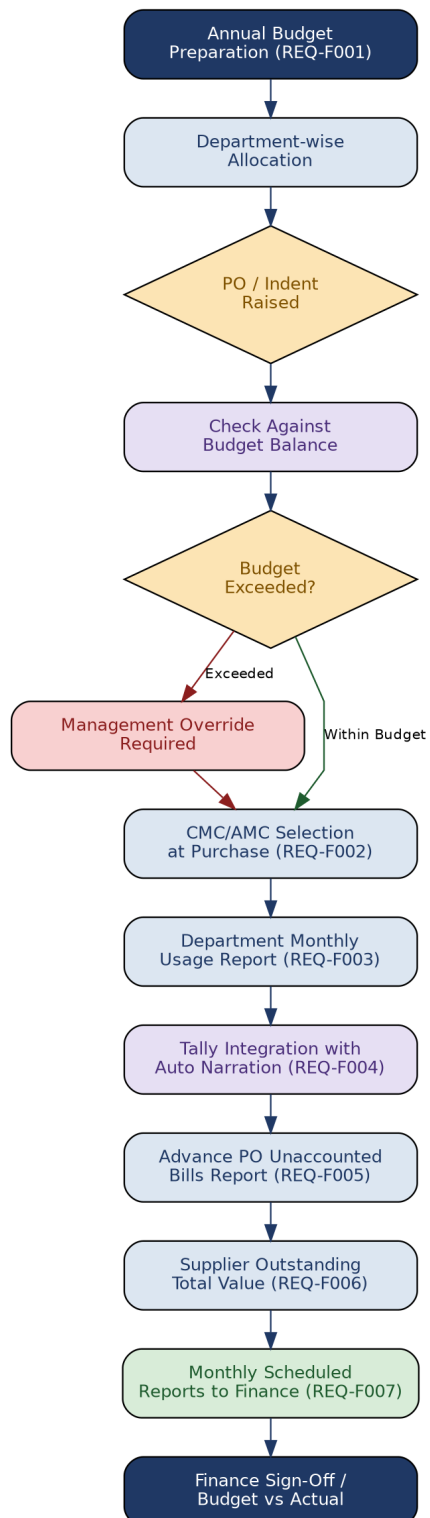


Figure: 3.7 Budget &amp; Finance Integration – Process Flow

## REQ-F001 – Budget Preparation by Accounts

Priority: High | Status: Proposed

Description: Department-wise and category-wise budget. Budget vs actual tracking. Carry-forward config. Revision workflow. Alert at 80% spending. Freeze when exhausted (with override). Utilization report.

Preconditions: The financial year/budget cycle has commenced and department consumption history from prior periods is available.

Inputs: Department, category, financial year, proposed allocation, prior-year actuals, carry-forward eligibility, revision request.

Processing Logic: 1. Accounts prepares department-wise and category-wise budget proposals, referencing prior-year actual spend as a baseline. 2. Draft budgets route through a review and approval workflow before activation for the financial year. 3. Once active, every PR/PO checks against the department's remaining budget balance in real time. 4. System alerts the department and Finance at a configurable utilization threshold (e.g., 80%). 5. Carry-forward of unused budget to the next period, if permitted by policy, is computed automatically at year-end with an audit-logged calculation. 6. Mid-year budget revisions follow the same approval workflow as the original budget and are versioned, not overwritten. 7. Finance can freeze a department's budget entirely (blocking further spend) in exceptional circumstances, with a mandatory reason.

Outputs: Approved departmental/category budget, real-time utilization tracking, 80% threshold alert, carry-forward calculation, revision history, freeze log.

Postconditions: 1. No department spends without an approved, trackable budget ceiling. 2. Budget history – original, revisions, carry-forward, freezes – is fully reconstructable for any past period.

Business Rules: 1. A frozen budget blocks new PRs but does not reverse already-approved, in-progress POs. 2. Carry-forward eligibility (full, partial, or none) is configurable per category, since capital categories are often treated differently from consumable categories.

Exception Handling: 1. If a department's actual spend approaches its ceiling due to a genuinely urgent need, the department can request a documented in-year budget revision rather than being blocked outright, subject to the standard revision approval workflow.



## REQ-F002 – CMC/AMC Selection During Purchase

Priority: High | Status: Proposed

Description: Choose AMC or CMC during PO. Cost comparison display. Auto-populate contract terms. Separate budget tracking. Document attachment. Renewal reminder integration.

Preconditions: A capital equipment purchase requiring ongoing maintenance coverage is being procured or approaching a coverage decision point.

Inputs: Asset/item, AMC cost quote, CMC cost quote, historical repair cost (if available), selected option, contract terms.

Processing Logic: 1. During PO creation for a capital asset, or during a maintenance-coverage renewal decision, the user is presented with an AMC vs. CMC choice. 2. System displays a side-by-side cost comparison: AMC (labor-only coverage, parts billed separately) versus CMC (comprehensive, including parts). 3. Upon selection, contract terms auto-populate from the chosen vendor's standard template, editable within policy limits. 4. The selected contract's cost is tracked against a distinct maintenance budget line, separate from the capital acquisition budget. 5. Supporting contract documents are attached to the asset and contract record. 6. Renewal reminders (per REQ-A007) trigger this same comparison workflow ahead of expiry.

Outputs: AMC/CMC comparison display, selected contract record with auto-populated terms, separate maintenance-budget tracking, attached documents.

Postconditions: 1. Every maintenance-coverage decision is made with visible cost comparison rather than defaulting to whichever option a vendor happens to propose.

Business Rules: 1. For high-value or clinically critical equipment, CMC (comprehensive) is the recommended default unless a documented reason supports AMC. 2. Maintenance contract spend is reported separately from capital spend in budget-vs-actual analysis.

Exception Handling: 1. If neither AMC nor CMC is selected by the time warranty expires, the system flags the asset as having a maintenance coverage gap in the exposure report shared under REQ-A007.

## REQ-F003 – Department Monthly Usage Report for Budget

Priority: High | Status: Proposed

Description: Monthly consumption: product and category breakdown. Trend: current vs previous vs last year. Usage pattern for forecasting. Export to Excel. Input for next budget.

Preconditions: Department consumption transactions exist for the current and prior comparison periods.

Inputs: Department, month, category, product-wise consumption value and quantity, budget allocation.

Processing Logic: 1. System aggregates each department's monthly consumption by product and category. 2. It compares the current month against the department's budget allocation and computes a variance (favorable/unfavorable). 3. A trend comparison against the previous month and against the same month in the prior year is presented to distinguish genuine growth from seasonal variation. 4. Top items driving the department's consumption value are listed to focus attention on the largest cost drivers. 5. The report is exportable to Excel for department-head review meetings.

Outputs: Monthly department usage report with variance analysis, trend comparison, top-consumption-item ranking, Excel export.

Postconditions: 1. Department heads and Finance have a consistent, comparable basis for monthly budget review conversations rather than ad hoc data pulls.

Business Rules: 1. Variance beyond a configurable threshold is highlighted for mandatory discussion at the monthly budget review. 2. Seasonal departments (e.g., higher consumption during specific clinical seasons) can have a seasonality note attached so variance is interpreted correctly.

Exception Handling: 1. If a department has no consumption in a given month (e.g., newly established), the report shows a clear 'no activity' state rather than a misleading zero-variance figure.

## REQ-F004 – Tally Integration with Auto Narration

Priority: High | Status: Proposed

Description: Auto-sync transactions. Narration: 'PO:{PONo} GRN:{GRNNo} Supplier:{Name} Inv:{InvNo} Dt:{Date}'. Pull details as tags. Configurable template. Error log. Reconciliation report.

Preconditions: Tally ERP is configured and reachable for integration, and the transaction (PO/GRN/invoice/payment) is finalized on the IHMS side.

Inputs: PO number, GRN number, supplier name, invoice number, invoice date, amount, account heads, narration template.

Processing Logic: 1. On finalization of a qualifying transaction, IHMS auto-generates the corresponding Tally voucher entry via the configured integration (API or structured export/import). 2. A configurable narration template is applied, e.g., 'PO:{PONo} GRN:{GRNNo} Supplier:{Name} Inv:{InvNo} Dt:{Date}', with tags pulled automatically from transaction data. 3. The synced entry's Tally voucher reference is stored back against the IHMS transaction for two-way traceability. 4. Sync failures (e.g., Tally unreachable) are queued and retried automatically, with a failure alert to Finance if unresolved beyond a configurable window. 5. A daily sync-reconciliation report confirms that every eligible IHMS transaction has a matching Tally entry.

Outputs: Auto-created Tally voucher with structured narration, two-way transaction linkage, sync failure queue and alert, daily reconciliation report.

Postconditions: 1. Financial postings in Tally are consistently narrated and traceable back to their originating stores/procurement transaction without manual re-keying.

Business Rules: 1. The narration template is configurable by Finance but changes are versioned so historical entries remain interpretable against the template in force at the time. 2. Manual Tally entries that were not IHMS-originated are excluded from the reconciliation expectation.

Exception Handling: 1. If Tally rejects an entry (e.g., account head mismatch), the transaction remains queued with the specific rejection reason surfaced to Finance for correction and resubmission, never silently dropped.

## REQ-F005 – Advance PO Unaccounted Bills Report

Priority: High | Status: Proposed

Description: Monthly report of advance POs without invoice. Department-wise. Aging: 30/60/90+ days. Auto-email to HOD. Closure tracking. Audit trail. Month-on-month comparison.

Preconditions: One or more advance-payment POs have been raised and remain without a matching invoice booked.

Inputs: PO ID, department, advance amount, PO date, invoice status.

Processing Logic: 1. System identifies all advance POs where no corresponding invoice has yet been booked. 2. These are aggregated monthly, grouped by department, with an ageing bucket (30/60/90+ days) based on days since the advance PO was raised. 3. Report is auto-emailed to the respective department HOD as well as Finance. 4. Each item is tracked to closure (invoice booked and matched) or to a documented reason for continued delay. 5. All ageing transitions and closures are captured in the audit trail for the monthly report.

Outputs: Monthly advance-POs-without-invoice report (department-wise, aged), auto-emailed notification, closure tracking, audit trail.

Postconditions: 1. Aged, unaccounted advances are never left unmonitored; every one is visible monthly to both the responsible department and Finance until resolved.

Business Rules: 1. Advances aged beyond the 90+ day bucket are escalated to the department head's superior and Finance Head for direct follow-up. 2. An advance PO cancelled or fully invoiced is automatically removed from the report in the following cycle.

Exception Handling: 1. If an advance is tied to an import shipment with an inherently longer lead time, it can be flagged as 'expected-long-cycle' to avoid false escalation, subject to Finance sign-off on the extended timeline.

## REQ-F006 – Supplier Outstanding Total Value

Priority: High | Status: Proposed

Description: Total outstanding: paid + pending. Age-wise breakdown. Reconcile with supplier statements. Payment planning. Credit period utilization. Payment history. Currently missing in IHMS.

Preconditions: One or more suppliers have outstanding (paid and/or pending) balances against invoices booked in the system.

Inputs: Supplier, invoice-wise paid amount, pending amount, invoice age, credit period terms, supplier statement (for reconciliation).

Processing Logic: 1. System computes each supplier's total outstanding as the sum of paid-but-unsettled and fully-pending invoice amounts. 2. An age-wise breakdown (e.g., 0-30/31-60/61-90/90+ days) is presented per supplier. 3. Where a supplier statement has been reconciled (per REQ-V005), the outstanding figure reflects the reconciled position; unreconciled suppliers are flagged separately. 4. Finance uses the age-wise view for payment planning, prioritizing suppliers nearing credit-period expiry to avoid penalty or supply disruption. 5. Credit-period utilization (average days taken to pay versus agreed terms) is tracked per supplier as an input to future negotiation.

Outputs: Supplier-wise total and age-wise outstanding, reconciliation status flag, credit-period utilization metric, payment planning view.

Postconditions: 1. Finance has a single, reliable, age-aware view of every supplier's outstanding position to plan cash outflow and protect vendor relationships.

Business Rules: 1. Disputed invoices (per REQ-V005 mismatch tracking) are shown separately from confirmed outstanding balances, since they should not be included in routine payment planning until resolved. 2. Credit-period breach (payment overdue relative to agreed terms) is highlighted distinctly from within-terms outstanding.

Exception Handling: 1. If a supplier's statement has never been reconciled (no response to reconciliation requests), the outstanding figure is shown with an 'unverified – hospital records only' caveat rather than presented as a confirmed position.

## REQ-F007 – Monthly Scheduled Reports to Finance

Priority: High | Status: Proposed

Description: Pre-defined templates. Schedule: daily/weekly/monthly/quarterly. Auto-email. Types: Purchase, GRN, Outstanding, Stock. Classification: dept/value/product/advance. MIS-ready format.

Preconditions: Pre-defined report templates and a distribution schedule are configured for Finance.

Inputs: Report template, schedule frequency (daily/weekly/monthly/quarterly), recipient list, classification filter (department/category).

Processing Logic: 1. Finance/Administrator configures which report templates (Purchase, GRN, Outstanding, Stock Position, etc.) are auto-generated and on what schedule. 2. At each scheduled interval, the system compiles the report with current data, applying any configured department/category classification filters. 3. Reports are auto-emailed to the configured recipient list in the specified format (PDF/Excel). 4. A delivery log confirms successful transmission; failed deliveries (e.g., invalid email) are flagged for administrator correction. 5. Recipients can additionally pull any scheduled report on demand outside its normal schedule.

Outputs: Auto-generated and auto-emailed reports per schedule, delivery confirmation log, on-demand pull capability.

Postconditions: 1. Finance receives a consistent, predictable cadence of core MIS reports without needing to manually request them each period.

Business Rules: 1. Schedule and recipient-list changes require Finance Head approval, since report distribution can involve sensitive financial data. 2. Reports containing supplier payment/bank details are restricted to roles with the appropriate access level even within the scheduled distribution.

Exception Handling: 1. If the underlying data required for a scheduled report is incomplete at generation time (e.g., month-end close not finalized), the report is generated with a 'provisional – pending close' watermark rather than silently presenting incomplete figures as final.

## REQ-F008 – Non-Cash Expenditures

Priority: Medium | Status: Proposed

Description: Barter, donations, free samples, internal transfers. Separate ledger. Compliance reporting.

Preconditions: A stock movement occurs without an associated cash payment – barter, donation, free sample, or internal zero-value transfer.

Inputs: Transaction type, item, quantity, source/donor, notional value for reporting, approving authority.

Processing Logic: 1. Non-cash transaction types are captured through a dedicated entry screen distinct from standard purchase GRNs (as detailed under REQ-I018). 2. Each entry is posted to a separate non-cash ledger, valued at standard/notional cost for internal reporting without affecting cash-purchase spend figures. 3. Compliance reporting compiles all non-cash

movements for the period, supporting statutory disclosure requirements (e.g., donations received) and internal audit review.

Outputs: Non-cash ledger, compliance report of donations/samples/barter/zero-value transfers for the period.

Postconditions: 1. Non-cash inventory movements are fully visible to Finance and Audit without distorting cash-spend-based budget and procurement metrics.

Business Rules: 1. Non-cash receipts above a configurable value threshold require the same approval discipline as an equivalent cash purchase. 2. Donations of regulated items (e.g., pharmaceuticals) require the additional compliance documentation referenced under REQ-I018 before being reflected as usable stock.

Exception Handling: 1. If a non-cash transaction's notional value cannot be reasonably estimated, it is recorded with quantity only and flagged 'value not determined' for Finance to assign a valuation basis before it is included in the compliance report.

## 3.8 CRM & TICKETING

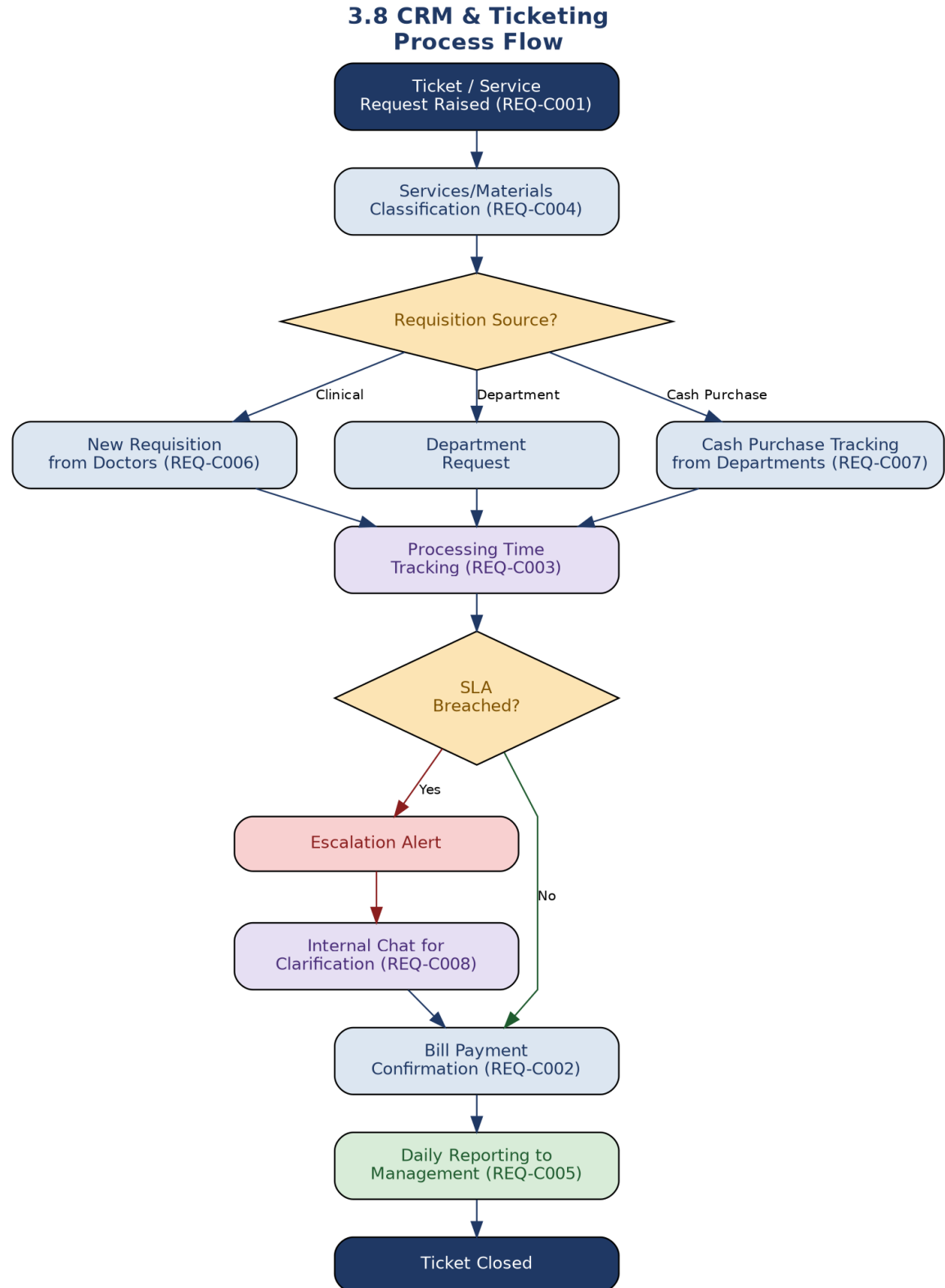


Figure: 3.8 Crm & Ticketing – Process Flow



## REQ-C001 – Ticket/Request Raising

Priority: High | Status: Proposed

Description: Tickets for: new product, new service, bill payment clarification. Classification: Materials/Services. Requester details. Auto ticket number. Status: Open/In-Progress/Resolved/Closed/Escalated. Priority levels.

Preconditions: A department user, doctor, or ward staff member has a need for a new product, service, or clarification not covered by a standard indent.

Inputs: Requester name, department, category (new product/new service/bill clarification), description, priority.

Processing Logic: 1. Requester raises a ticket selecting the classification: Materials or Services. 2. System auto-generates a sequential ticket number and timestamps creation. 3. Requester details (name, department, contact) are captured automatically from the logged-in profile where available. 4. Ticket routes to the appropriate queue (Stores for materials, Biomedical/Engineering for services) based on classification. 5. Requester and assignee both receive status notifications as the ticket progresses.

Outputs: Ticket record with unique number, classification-based routing, requester profile linkage, status notifications.

Postconditions: 1. Every non-standard request has a single, trackable entry point rather than being handled through informal calls or emails that leave no record.

Business Rules: 1. Ticket classification determines the default SLA and routing; misclassified tickets can be reassigned by the receiving team with the reclassification logged. 2. A ticket cannot be closed without a resolution note.

Exception Handling: 1. If the requester's department cannot be auto-detected (e.g., shared login), the system prompts for manual department selection before the ticket can be submitted.

## REQ-C002 – Bill Payment Confirmation/Clarification

Priority: High | Status: Proposed

Description: Request payment confirmation. Clarification on status. Dispute resolution. Attach documents. Escalation if unresolved per SLA. Trend report.

Preconditions: A department or vendor requires confirmation or clarification on the status of a bill/invoice payment.

Inputs: Ticket ID, invoice/PO reference, nature of query (confirmation/clarification/dispute), supporting documents.

Processing Logic: 1. Requester raises a ticket referencing the specific invoice/PO in question. 2. Accounts team reviews the payment status and responds with confirmation, an explanation of the current processing stage, or acknowledgment of a genuine dispute. 3. Disputes are escalated to a dedicated resolution workflow involving Purchase and Finance as needed. 4. Supporting documents (e.g., disputed delivery proof) can be attached to the ticket. 5. If unresolved beyond the configured SLA, the ticket auto-escalates to the Finance Head.

Outputs: Payment status clarification response, dispute resolution record, escalation log, trend report on payment-query volume.

Postconditions: 1. Payment-related queries are resolved through a documented workflow, and disputes are never left informally unresolved.

Business Rules: 1. A dispute ticket cannot be closed by Accounts alone; it requires acknowledgment from the requester or vendor that the matter is resolved. 2. Recurring payment-query volume for a specific vendor or department is monitored as a signal of a possible underlying process issue.

Exception Handling: 1. If the vendor (not an internal department) raises the query, it is routed through the vendor portal's query channel and linked to the same ticket for a unified resolution thread.

### REQ-C003 – Request and Processing Time Tracking

Priority: High | Status: Proposed

Description: Capture request time, processing start, processed by, closure time. Calculate TAT. Receiver feedback rating. SLA breach alert and escalation.

Preconditions: A ticket has been raised and is progressing through the resolution workflow.

Inputs: Ticket ID, creation time, assignment time, processing start time, closure time, assignee, feedback rating.

Processing Logic: 1. System timestamps every stage transition of a ticket automatically: creation, assignment, processing start, and closure. 2. Turnaround time (TAT) is computed as the elapsed time between creation and closure, and stage-wise durations are also captured to

identify where delays occur. 3. On closure, the requester is prompted to rate the resolution (satisfaction feedback). 4. If a ticket approaches its SLA threshold without closure, an automated alert is sent to the assignee, escalating to their supervisor if the SLA is breached. 5. TAT and feedback data feed into the daily/monthly reporting under REQ-C005.

Outputs: Stage-wise timestamp log, computed TAT, requester feedback rating, SLA-breach alert and escalation.

Postconditions: 1. Response and resolution performance is measurable and improvable rather than anecdotal, with SLA breaches surfaced proactively rather than discovered after the fact.

Business Rules: 1. SLA targets are configurable by ticket category/priority; life-critical service requests (e.g., ICU equipment failure) carry a materially shorter SLA than routine requests. 2. A ticket reopened after closure restarts TAT measurement from the reopening point, tracked as a distinct 'reopened' cycle.

Exception Handling: 1. If the assignee is unavailable (leave, shift change) at the point of SLA-breach escalation, the system escalates to the department's on-call or supervisor role instead of a single named individual.

## REQ-C004 – Services/Materials Classification

Priority: High | Status: Proposed

Description: Materials: Consumables, Assets, Equipment. Services: Repairs, Maintenance, Cleaning, Installation. Sub-classification. Different workflows. Category-wise reporting.

Preconditions: A ticket is being classified at creation to determine its handling workflow.

Inputs: Ticket category (Materials/Services), sub-classification (Consumables/Assets/Equipment or Repairs/Maintenance/Cleaning/Installation).

Processing Logic: 1. At ticket creation, the requester first selects Materials or Services as the primary classification. 2. Based on that choice, a relevant sub-classification list is presented (e.g., Materials: Consumables, Assets, Equipment; Services: Repairs, Maintenance, Cleaning, Installation). 3. Sub-classification determines the specific team queue and checklist/workflow applied – e.g., an Equipment repair ticket routes to Biomedical Engineering with an asset-linkage prompt, while a Cleaning ticket routes to Housekeeping with a different checklist. 4. Reporting can roll up by primary classification or drill down by sub-classification for granular analysis.

Outputs: Classified and sub-classified ticket, workflow/queue routing, classification-based reporting rollup.

Postconditions: 1. Every ticket reaches the correctly equipped team on first assignment, reducing misrouting and reassignment delay.

Business Rules: 1. The classification taxonomy (categories and sub-categories) is centrally maintained by System Administration, not editable ad hoc by individual users, to keep reporting consistent over time.

Exception Handling: 1. If a ticket genuinely spans more than one sub-classification (e.g., an installation that also requires a service PO), the primary classification is used for routing while the ticket description captures the secondary need for the receiving team's awareness.

### REQ-C005 – Daily Reporting to Management

Priority: High | Status: Proposed

Description: Daily summary: Open, Resolved, Pending. Department-wise volume. Average resolution time. Escalated tickets. Cash purchases included. Auto-email at configured time. Trend analysis.

Preconditions: Tickets are actively being created, processed, and closed across the reporting period (typically the prior day).

Inputs: Ticket volume by status (Open/Resolved/Pending), department, average resolution time, escalated tickets, associated cash purchases.

Processing Logic: 1. System compiles a daily summary of ticket activity: counts by status, department-wise volume, and average resolution time for the day. 2. Escalated (SLA-breached) tickets are listed explicitly for management attention. 3. Cash purchases logged via tickets (REQ-C007) are included in the summary for a consolidated view of both service requests and informal purchase activity. 4. The summary is auto-emailed to Management each morning covering the prior day's activity.

Outputs: Daily ticket summary report, escalated-ticket list, department-wise volume breakdown, auto-emailed distribution.

Postconditions: 1. Management has daily visibility into service-request volume, backlog, and performance without needing to query the system directly.

Business Rules: 1. The daily report window is a fixed 24-hour period (e.g., midnight to midnight) for consistency; ad hoc custom-range reports are available separately on demand.

Exception Handling: 1. If ticket volume for the day is zero (e.g., a holiday), the system still sends the summary noting 'no activity' rather than skipping the report, to confirm the reporting mechanism itself is functioning.

### REQ-C006 – New Requisition from Doctors

Priority: High | Status: Proposed

Description: Doctors raise new product/service requests. Clinical justification. Route to HOD then stores. Track adoption patterns. Report on new product adoption.

Preconditions: A doctor identifies a need for a new product or service not currently in the approved item/vendor list.

Inputs: Doctor ID, product/service description, clinical justification, requesting department.

Processing Logic: 1. Doctor raises a New Requisition ticket describing the product/service and the clinical justification for its introduction. 2. Ticket routes first to the relevant department HOD for clinical/operational endorsement. 3. On HOD endorsement, it proceeds to Stores/Purchase for feasibility and vendor sourcing, following the standard new-item and vendor evaluation workflow (Section 3.2/3.10). 4. System tracks adoption patterns – how many doctor-originated requisitions proceed to become regularly stocked items versus one-off requests. 5. A periodic report on new-product requisitions supports formulary/product-range review meetings.

Outputs: New requisition ticket, HOD endorsement record, downstream item/vendor onboarding linkage, adoption-pattern report.

Postconditions: 1. Clinically-driven product introduction requests are captured, endorsed, and evaluated through a consistent path rather than informal individual requests to Purchase.

Business Rules: 1. A new requisition cannot bypass HOD endorsement, even when raised by a senior doctor, to preserve departmental oversight of new introductions. 2. Requisitions not pursued must record a reason for the periodic formulary review record.

Exception Handling: 1. If the requested product/service is urgently needed for an individual patient before the full evaluation can complete, it is handled under the Immediate Purchase workflow (REQ-P006) in parallel, with the New Requisition ticket continuing independently for a longer-term stocking decision.

## REQ-C007 – Cash Purchase Tracking from Departments

Priority: Medium | Status: Proposed

Description: Log cash purchases via tickets. Attach receipts. Reimbursement workflow. Track against budget. Report frequency and value by department.

Preconditions: A department has made a purchase using cash/petty cash rather than through the standard PO process, and needs to log it and seek reimbursement.

Inputs: Department, item, amount, receipt/bill image, budget cost center.

Processing Logic: 1. Department logs the cash purchase via a ticket, attaching the receipt image and specifying the cost center to be debited. 2. Ticket routes to Accounts for reimbursement processing. 3. System tracks the cash purchase against the department's budget utilization, exactly as a formal Cash PO would (linking to REQ-P008 where the two overlap). 4. A report aggregates cash-purchase frequency and value by department, surfacing departments that may benefit from a rate contract or pre-approved vendor arrangement to reduce ad hoc cash buying.

Outputs: Logged cash-purchase ticket with attached receipt, reimbursement workflow trigger, budget debit, department frequency/value report.

Postconditions: 1. Informal cash purchases are captured in the same auditable system as formal purchases rather than existing only as paper reimbursement claims.

Business Rules: 1. Cash purchases above the organization's cash-purchase ceiling logged via ticket are flagged as a policy exception requiring Finance review, consistent with REQ-P008. 2. Reimbursement cannot be processed without an attached, legible receipt image.

Exception Handling: 1. If the receipt is lost, the department can submit a signed declaration in lieu, subject to a lower reimbursement ceiling and mandatory Store Manager co-sign, per hospital petty-cash policy.

## REQ-C008 – Internal Chat Option

Priority: Low | Status: Proposed

Description: Department-to-stores and stores-to-vendor chat. History linked to tickets/POs/GRNs. Notifications. File attachments.

Preconditions: Users participating in a ticket, PO, or GRN thread need to communicate contextually without leaving the system.

Inputs: Ticket/PO/GRN reference, sender, message text, file attachment (optional).

Processing Logic: 1. System provides an internal chat panel accessible from any ticket, PO, or GRN record, scoped to that transaction's participants. 2. Department-to-Stores and Stores-to-Vendor (via the vendor portal) chat channels are both supported, each respecting the access boundaries of the participants (a vendor cannot see internal department-only chat). 3. Chat history is permanently linked to and retrievable from the parent transaction, forming part of its complete record. 4. Participants receive notifications for new messages; file attachments (e.g., a photo of a defect) can be shared inline.

Outputs: Transaction-linked chat thread, notification alerts, file attachment sharing, permanent chat history.

Postconditions: 1. Clarifications and informal coordination around a specific transaction are captured in context rather than scattered across email or verbal exchanges, aiding both efficiency and audit completeness.

Business Rules: 1. Vendor-facing chat is restricted to information relevant to that vendor's own transactions; internal department discussion is never exposed to the vendor side. 2. Chat content is retained per the same document-retention policy as its parent transaction.

Exception Handling: 1. If a chat participant's access to the parent transaction is later revoked (e.g., role change), their prior messages remain visible to other participants for continuity, but they lose the ability to post further messages.

## 3.9 REPORTS & ANALYTICS

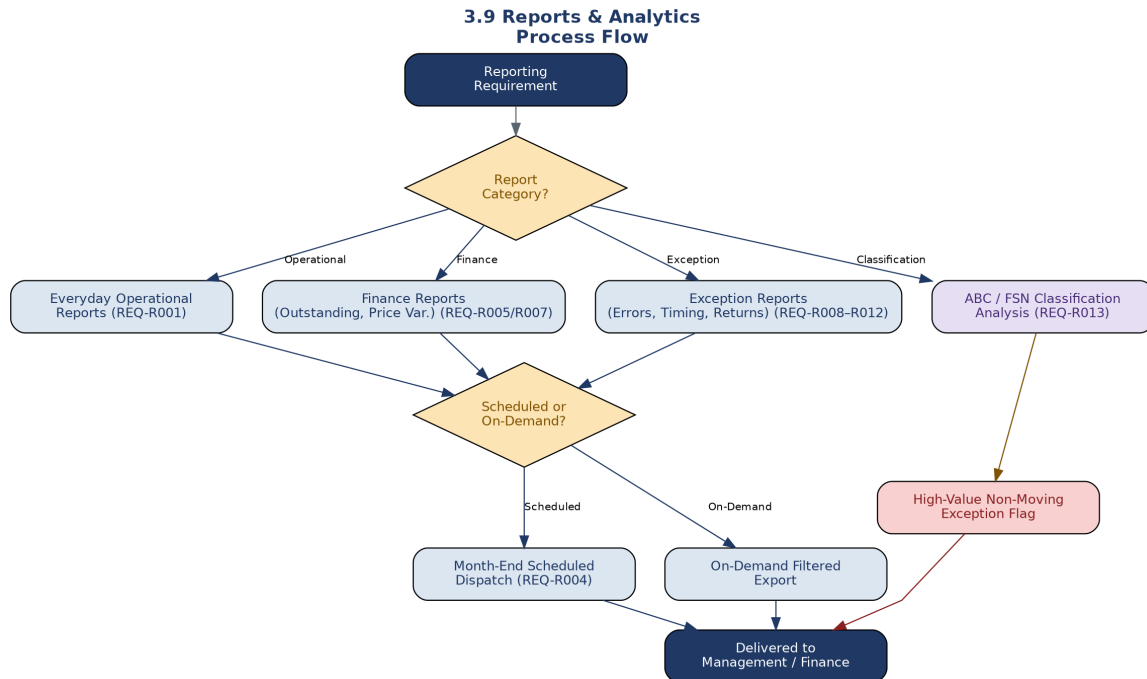


Figure: 3.9 Reports & Analytics – Process Flow

### REQ-R001 – Everyday Report Classification

Priority: High | Status: Proposed

Description: Daily: dept-wise purchase, advance PO status, invoice-wise receipt, value-wise summary (0-10K, 10K-50K, 50K-1L, >1L), product-wise consumption. Auto-generate at 6 PM. Email distribution.

Preconditions: Daily transactional activity (PRs, POs, GRNs, issues) exists for the reporting day.

Inputs: Report category, department, value band, product, date.

Processing Logic: 1. System compiles a fixed set of daily reports each morning covering the prior day: department-wise purchase summary, advance-PO status, invoice-wise receipt (GRN) summary, value-wise purchase summary (banded 0-10K/10K-50K/50K-1L/>1L), and product-wise consumption. 2. Reports are generated from a standard, pre-approved template set to ensure consistency across departments and over time. 3. Each report is auto-distributed to its relevant audience (department heads for their own department report, Finance/Management for the consolidated view). 4. Users can also access any daily report on demand outside the automated schedule.



Outputs: Standardized set of daily operational reports, auto-distributed by audience, on-demand access.

Postconditions: 1. Every stakeholder has a predictable, consistent daily view of purchasing and consumption activity without needing to construct ad hoc queries.

Business Rules: 1. Value bands used for the value-wise summary align with the approval-matrix tiers (Section 5.4) so the report doubles as an approval-workload indicator. 2. New report categories can be added to the daily set only through a change-controlled configuration process.

Exception Handling: 1. If no transactions occurred in a category for the day, the corresponding report section is generated showing zero activity rather than being omitted, so recipients can distinguish 'no activity' from 'report failure'.

## REQ-R002 – Department Usage Monthly Report

Priority: High | Status: Proposed

Description: Monthly consumption value. Category breakdown. Budget comparison. Variance analysis. Trend: current vs previous 3 months. Top 10 items per department.

Preconditions: A full month of department consumption data is available for the reporting period.

Inputs: Department, month, category, consumption value, budget allocation, prior 3 months' data.

Processing Logic: 1. System aggregates each department's monthly consumption value by category. 2. It computes variance against the department's budget allocation for the month. 3. A trend view compares the current month to each of the previous three months to reveal short-term trend direction. 4. The top ten items by consumption value are listed per department to focus review on the largest cost drivers. 5. Report is exportable and schedulable per REQ-F007.

Outputs: Monthly department usage report with variance, three-month trend, top-10-item ranking.

Postconditions: 1. Department-level budget review discussions are grounded in a consistent, comparable monthly report rather than one-off analysis.

Business Rules: 1. This report is the analytics-team equivalent of REQ-F003 and shares the same underlying data model to avoid inconsistent figures between Finance and Analytics views.

Exception Handling: 1. If a department was newly established mid-year, trend comparison against non-existent prior months is shown as 'not applicable' rather than a misleading zero.

### REQ-R003 – Advance PO Unaccounted Bills Monthly

Priority: High | Status: Proposed

Description: List of advance POs without invoice. Department-wise. Aging 30/60/90+ days. Auto-email to HOD. Closure tracking. Audit format. Month-on-month.

Preconditions: Advance-payment POs exist without a matching invoice booked as of the reporting date.

Inputs: PO ID, department, advance amount, PO date, ageing bucket.

Processing Logic: 1. System lists every advance PO with no linked invoice, grouped by department. 2. Ageing buckets (30/60/90+ days) are computed from the PO date. 3. The monthly report is auto-emailed to each department HOD along with Finance. 4. Each entry is tracked to closure and the audit-format version of the report (with full transaction history) is retained for audit review. 5. Month-on-month trend of total unaccounted advance value is tracked to identify whether the overall exposure is improving or worsening.

Outputs: Monthly aged advance-PO report (department-wise), auto-email distribution, closure tracking, audit-format archive, month-on-month trend.

Postconditions: 1. This is the reporting-module counterpart to REQ-F005, ensuring the same underlying exposure is visible both operationally (Finance) and analytically (Reports & Analytics) with consistent figures.

Business Rules: 1. The audit-format version of this report is immutable once generated for a closed period, serving as the point-in-time record for external audit.

Exception Handling: 1. See REQ-F005 for the same exception-handling logic (import/long-cycle advances flagged separately).

**REQ-R004 – Scheduled Month-End Reports to Finance**

Priority: High | Status: Proposed

Description: Purchase summary, stock position (opening/receipts/issues/closing), outstanding summary, depreciation, budget utilization. Auto-email last working day. Excel and PDF.

Preconditions: The financial month is closing or has closed, and all relevant modules (purchase, stock, outstanding, depreciation, budget) have current data.

Inputs: Month, purchase summary data, opening/receipts/issues/closing stock, outstanding summary, depreciation figures, budget utilization.

Processing Logic: 1. On the scheduled month-end date, the system compiles a consolidated report covering: purchase summary, stock position (opening, receipts, issues, closing), outstanding summary, depreciation charge for the month, and budget utilization by department. 2. The report is auto-emailed to Finance on the last working day of the month (or the first working day of the following month, per configuration). 3. Figures are cross-checked for internal consistency (e.g., closing stock = opening + receipts – issues ± adjustments) before distribution, with any inconsistency flagged rather than silently included.

Outputs: Consolidated month-end MIS report, auto-emailed to Finance, internal consistency check flag.

Postconditions: 1. Finance receives a single, reconciled month-end package rather than needing to compile figures manually from multiple modules.

Business Rules: 1. The report cannot be marked 'Final' until the internal consistency check passes; a failed check requires Finance/Stores investigation before the month is formally closed.

Exception Handling: 1. If depreciation for the month has not yet been run (see fixed asset accounting module), the report is issued with depreciation marked 'pending run' rather than omitted or shown as zero.

## REQ-R005 – Supplier Outstanding Reconciliation Report

Priority: High | Status: Proposed

Description: Hospital vs supplier statement. Matched items with reference. Mismatches with reason. Disputed items tracking. Resolution status. Monthly certificate.

Preconditions: Supplier statement reconciliation (REQ-V005) has been performed for the reporting period.

Inputs: Supplier, matched items, mismatched items with reason, disputed items, resolution status.

Processing Logic: 1. System compiles the outcome of the monthly supplier reconciliation process into a formal report: matched items (with hospital payment reference), mismatched items (with categorized reason), and any items still under dispute. 2. Resolution status for each mismatch/dispute is tracked to closure. 3. A reconciliation certificate is generated once all items for a supplier-period combination reach a closed (matched or resolved) state, serving as the audit-ready sign-off document.

Outputs: Supplier reconciliation report, disputed-item tracker, reconciliation certificate.

Postconditions: 1. Every supplier's outstanding position for the period is either fully reconciled or has a clearly tracked, open item – nothing is left ambiguous.

Business Rules: 1. A reconciliation certificate can only be issued once all items are closed; partial certificates are not permitted, preventing premature sign-off with unresolved items outstanding.

Exception Handling: 1. See REQ-V005 for detailed exception handling on non-responding suppliers and format mismatches, which this report surfaces at the aggregate level for management visibility.

## REQ-R006 – Store Purchase Decisions Report

Priority: Medium | Status: Proposed

Description: Daily supplier selection. Price comparison at PO time. Justification for higher price. Better options not selected. Management approval for deviations. Trend analysis.

Preconditions: Purchase decisions involving supplier selection at other than the lowest quoted price have occurred during the day.

Inputs: PO ID, selected supplier, price comparison data, justification for non-lowest selection, approving authority.

Processing Logic: 1. System compiles a daily report of every purchase decision where the selected supplier was not the lowest-priced quotation. 2. Each entry shows the price comparison at the time of decision, the justification provided, and the approving authority who accepted it. 3. Options that were available but not selected are shown alongside, giving Management a complete picture of the trade-off made. 4. Report supports the same Management approval requirement described in REQ-P020 by providing a consolidated daily view rather than requiring Management to review each PO individually in real time.

Outputs: Daily higher-price-selection report with full comparison context and justification trail.

Postconditions: 1. Every deviation from lowest-price selection is visible in aggregate, supporting both real-time oversight and periodic procurement-integrity audit.

Business Rules: 1. This report is generated regardless of whether the individual PO required Management approval under REQ-P020's threshold, giving broader visibility than the approval workflow alone.

Exception Handling: 1. If a PO was raised under a rate contract (fixed price, not subject to comparative quotation), it is excluded from this report since no competitive selection was applicable.

## REQ-R007 – Price Variation Report

Priority: Medium | Status: Proposed

Description: Item-wise and supplier-wise price trend. Day-on-day comparison. Alert on >10% increase. Reason tracking. Negotiation support.

Preconditions: Historical and current pricing data exists for items across multiple purchase transactions and/or suppliers.

Inputs: Item, supplier, date, quoted/invoiced price.

Processing Logic: 1. System plots item-wise and supplier-wise price movement over time using historical purchase/quotation data. 2. A day-on-day comparison view highlights sudden price jumps. 3. Any increase beyond a configurable threshold (e.g., >10%) is flagged for review. 4. The report is designed to support Purchase in renegotiation discussions by presenting a clear, evidence-based pricing history rather than relying on memory or scattered records.

Outputs: Item/supplier price trend visualization, day-on-day comparison, threshold-breach flag, negotiation-support export.

Postconditions: 1. Pricing anomalies and sustained upward trends are identified proactively rather than only being noticed when a specific PO triggers a manual comparison.

Business Rules: 1. This report shares its underlying comparison logic with REQ-P020's real-time alert but is intended for periodic, aggregate review rather than transaction-time decision support.

Exception Handling: 1. Price changes attributable to a documented GST rate change are annotated as such and excluded from the 'anomaly' flag, consistent with the distinction made under REQ-P021.

## REQ-R008 – Timing Report

Priority: Medium | Status: Proposed

Description: Security entry, PO raised, approval sent/approved, GRN time. Calculate delays. Identify bill submission delay reasons. Process efficiency report.

Preconditions: Transactions have progressed through multiple stages (security entry, PO raised, approval, GRN) with timestamps captured at each.

Inputs: Transaction ID, security entry time, PO raised time, approval sent/approved time, GRN time.

Processing Logic: 1. System captures a timestamp at each defined stage of a transaction's lifecycle. 2. Stage-to-stage elapsed time is calculated (e.g., security entry to PO raised, approval sent to approved, PO to GRN). 3. Delays exceeding configurable norms per stage are highlighted. 4. Where delays correlate with bill/invoice submission lag, the report cross-references REQ-G014's department delay data to identify likely root causes. 5. Aggregated process-efficiency metrics are presented by department and transaction type to support continuous process improvement discussions.

Outputs: Stage-wise timing report, delay highlighting, cross-referenced root-cause indicators, process-efficiency metrics.

Postconditions: 1. Bottlenecks in the procurement-to-receipt cycle are identified with evidence rather than assumption, enabling targeted process improvement.

Business Rules: 1. Stage-timing norms are configurable and may differ by transaction type (e.g., emergency purchases are expected to move faster through approval than routine ones).

Exception Handling: 1. If a stage timestamp is missing due to an off-system step (e.g., a manual gate entry not yet digitized), that stage is shown as 'not captured' rather than computed as zero or excluded silently.

### REQ-R009 – Stock Return Report

Priority: Medium | Status: Proposed

Description: Department-wise return frequency. Reason classification. Return value. Transaction timing. Deviation reasons. Trend for quality improvement.

Preconditions: Stock return transactions (REQ-I012) have occurred during the reporting period.

Inputs: Department, item, return reason, return value, transaction date.

Processing Logic: 1. System aggregates all stock returns for the period by department and reason category (expired, excess, not required, damaged). 2. Return value is computed to quantify the financial impact of returns. 3. Transaction timing (how promptly returns were processed relative to the original issue) is analyzed. 4. Deviation patterns – departments or items with disproportionately high return rates – are highlighted as candidates for root-cause review, feeding into quality/process-improvement initiatives (e.g., over-issuing, poor initial estimation).

Outputs: Department-wise return frequency and value report, reason classification breakdown, deviation/trend analysis for quality improvement.

Postconditions: 1. Recurring, avoidable return patterns are surfaced for corrective action rather than being absorbed as routine operational noise.

Business Rules: 1. Returns due to genuine clinical necessity (e.g., treatment plan change) are distinguished in reporting from returns indicating a process gap (e.g., systematic over-ordering).

Exception Handling: 1. If a return's reason is recorded as 'Other' without further detail, it is flagged in the report for the Stores Manager to follow up and reclassify for meaningful trend analysis.

### REQ-R010 – Error and Correction Report

Priority: Medium | Status: Proposed

Description: Corrected entries with original/corrected values. Error type classification. User tracking. Time to correct. Frequency by user/type. Training needs.

Preconditions: One or more data-correction transactions (e.g., invoice error correction per REQ-G010) have occurred in the period.

Inputs: Correction ID, original value, corrected value, error type, user, correction time.

Processing Logic: 1. System compiles every corrected entry for the period, showing the original and corrected values side by side. 2. Errors are classified by type (data entry, rate mismatch, quantity error, tax miscalculation, etc.). 3. Corrections are attributed to the user who made the original entry and the user who performed the correction, with time-to-correct measured. 4. Frequency of errors by user and by error type is aggregated to identify training needs or systemic data-entry issues.

Outputs: Error and correction log, error-type classification, user-level frequency analysis, training-need indicators.

Postconditions: 1. Data-quality issues are tracked as a management metric rather than being invisible once individually corrected, supporting targeted training and process refinement.

Business Rules: 1. This report reads from the correction-entry audit trail (REQ-G010) rather than maintaining a separate log, ensuring the two remain consistent.

Exception Handling: 1. If the same transaction is corrected more than once, all correction iterations are shown in sequence rather than only the latest, to give a complete picture of how the entry evolved.



## REQ-R011 – Internal Transfer Emergency Report

Priority: Medium | Status: Proposed

Description: Requesting dept, source dept, product/qty, reason for stockout, operational factors, recommendations for stock adjustment.

Preconditions: Internal transfers triggered by emergency/unavailable stock (REQ-IO11) have occurred during the period.

Inputs: Requesting department, source department, product, quantity, reason for stockout, operational context.

Processing Logic: 1. System compiles all emergency-driven internal transfers for the period. 2. Each entry captures the requesting and source departments, item and quantity moved, and the stated reason for the originating stockout. 3. Operational context (e.g., delayed PO, higher-than-forecast consumption) is captured where available to support root-cause analysis. 4. The report concludes with system-generated recommendations for stock-level adjustment where a pattern of repeated emergency transfers for the same item/department suggests the reorder level or safety stock should be revised.

Outputs: Emergency internal transfer report, root-cause context, stock-adjustment recommendations.

Postconditions: 1. Recurring emergency transfers – a symptom of misconfigured reorder levels or forecasting gaps – are surfaced with actionable recommendations rather than repeatedly handled as one-off exceptions.

Business Rules: 1. Recommendations generated by this report are advisory; any actual reorder-level change still requires Stores Manager review and approval, consistent with standard master-data governance.

Exception Handling: 1. If an emergency transfer was caused by a one-time event (e.g., a mass-casualty incident) rather than a systemic pattern, it can be tagged 'exceptional – exclude from trend' so it does not skew the reorder-level recommendation.

## REQ-R012 – Indent Modification Instances Report

Priority: Medium | Status: Proposed

Description: Modifications per department. Types: add/delete/change qty. Repeated modifiers. Emergency dept exceptions. Impact on processing time. Process improvement.

Preconditions: Indent modifications (REQ-N009) have occurred during the reporting period.

Inputs: Department, modification type (add/delete/change quantity), indent ID, modifying user, key-department exemption flag.

Processing Logic: 1. System compiles all indent modifications for the period, categorized by type (add item, delete item, quantity change). 2. Modifications are aggregated by department, distinguishing exempt key departments (REQ-N007) from standard departments since different expectations apply. 3. Frequently-modifying departments/users are identified. 4. Impact on Stores processing time (delay introduced by modification) is estimated by comparing processing time for modified versus unmodified indents.

Outputs: Indent modification report by department and type, frequent-modifier identification, processing-time impact estimate.

Postconditions: 1. Departments whose indenting practices introduce avoidable rework for Stores are identified for a targeted process-improvement conversation, supported by measurable impact data.

Business Rules: 1. Exempt key departments' modifications are reported separately and are not included in any 'frequent modifier' flagging intended to prompt corrective coaching, since their operating context differs.

Exception Handling: 1. If a modification was necessitated by a genuine Stores-side error (e.g., wrong item picked initially, later corrected by the department), it is tagged 'Stores-originated correction' and excluded from department-attributed modification frequency.

## REQ-R013 – ABC and FSN Inventory Classification Analysis

Priority: Medium | Status: Proposed

**Description:** The system shall classify inventory items by consumption value, ABC analysis, and by movement frequency, FSN -- Fast, Slow, Non-moving -- analysis, and generate the corresponding reports so that stores and purchase teams can prioritise control effort on the items that matter most.

**Preconditions:** A minimum of one full review period (default 12 months, configurable) of consumption history must exist for an item to be classified with confidence; new items are provisionally classified using estimated consumption.

**Inputs:** Item ID, period consumption value, period consumption quantity, last movement date, classification period (default annual, re-run frequency configurable).

**Processing Logic:** 1. System computes each item's annual consumption value, quantity issued multiplied by unit cost, and ranks all items descending by value. 2. Items are classified into Category A, items whose cumulative value reaches the top 70 percent (configurable) of total consumption value, typically the fewest SKUs; Category B, the next 20 percent; and Category C, the remaining 10 percent by value, typically the largest SKU count. 3. Independently, the system computes FSN classification based on last movement date and issue frequency: Fast-moving, Slow-moving, and Non-moving, using configurable windows (default 30 days for fast-moving activity, 180 days of no issue for non-moving). 4. System combines ABC and FSN into a matrix, for example AF, AN, CN, to flag high-value non-moving stock as a priority review item. 5. Classification is re-run on the configured cycle and prior classification is retained for trend comparison.

**Outputs:** ABC classification report, FSN classification report, combined ABC-FSN matrix report, high-value-non-moving exception list, classification trend history.

**Postconditions:** 1. Classification bands are configurable per organisation policy and changing them re-runs classification for all items. 2. An item's classification history is retained even after re-classification, for trend analysis. 3. Reorder policy can reference ABC class to set tighter review cycles for Category A items.

**Business Rules:** 1. Items with less than the minimum required history are marked Provisional and excluded from committed reorder-policy changes until sufficient history accumulates. 2. A manual override of an item's classification requires a reason and is logged, and is superseded at the next scheduled re-run unless the override is marked permanent.

**Exception Handling:** 1. Sufficient consumption history exists, or item is explicitly flagged Provisional. 2. Classification period is configured. 3. Unit cost is available for value-based ranking.

## 3.10 PRODUCT MASTER MANAGEMENT

### 3.10 Product Master Management Process Flow

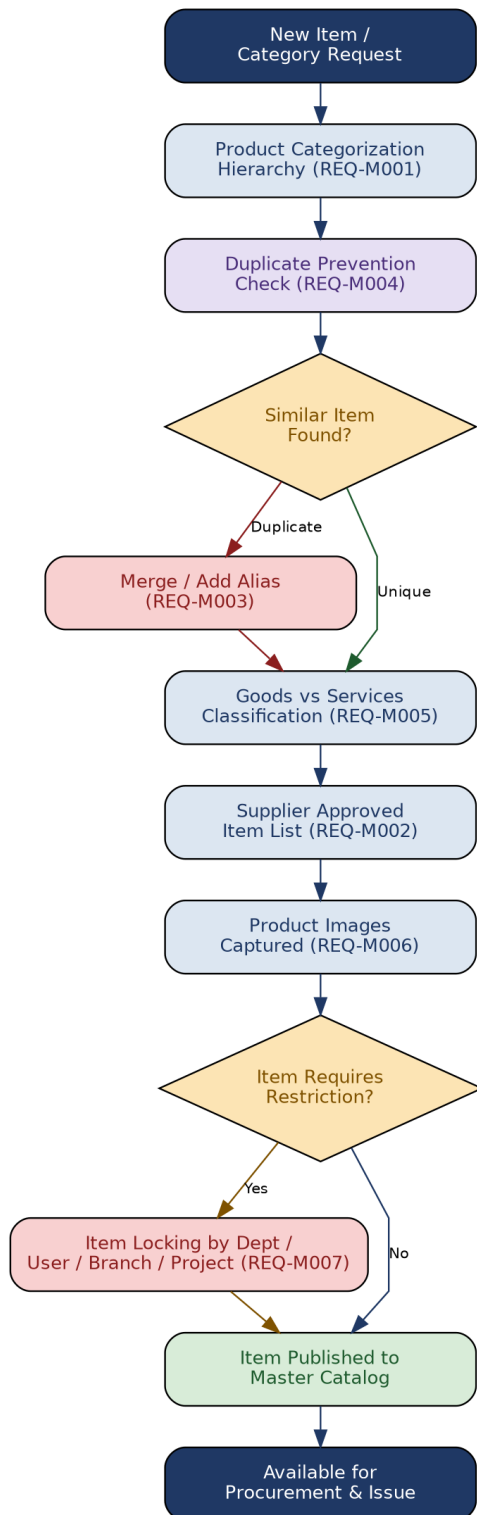


Figure: 3.10 Product Master Management – Process Flow

## REQ-M001 – Product Categorization Hierarchy

Priority: High | Status: Proposed

Description: Five levels: Family → Category → Group → Type → Item. Configurable. Drag-and-drop reordering. Category-wise reports.

Preconditions: A new item needs to be introduced into the system, either proposed by a vendor, department, or Purchase.

Inputs: Item name, generic name, category, sub-category, unit of measure, HSN code, GST rate, proposing user.

Processing Logic: 1. Proposing user (vendor via portal, department via ticket, or Purchase directly) submits the new item's core attributes. 2. System checks for likely duplicates (similar name/generic name) and presents matches for the proposer to confirm this is genuinely new. 3. Product Master Admin reviews and completes any missing classification/hierarchy fields before approving activation. 4. Approved items become available for indenting, quotation, and PO immediately; rejected proposals are returned with a reason.

Outputs: New item master record, duplicate-check results, approval/rejection log.

Postconditions: 1. The item catalog grows in a controlled, de-duplicated manner rather than accumulating near-identical entries from uncoordinated proposals.

Business Rules: 1. Only Product Master Admin can grant final activation; proposers can submit but not self-approve. 2. A rejected proposal can be resubmitted with corrected/additional information, tracked as a revision of the same request.

Exception Handling: 1. If the duplicate check finds a strong match, the system blocks creation and instead offers to route the proposer to request an update/rate addition against the existing item.

## REQ-M002 – Supplier Approved Item List

Priority: High | Status: Proposed

Description: Management-approved item list per vendor. Only approved items in PO screen. New item workflow: Supplier → Store → HOD → Management. Status tracking. Rejection reason. Periodic review.

Preconditions: The organization's category/sub-category structure requires ongoing maintenance as the catalog grows.

Inputs: Category name, parent category, sub-category name, effective hierarchy level.

Processing Logic: 1. Product Master Admin defines and maintains a multi-level category hierarchy (Category > Sub-category > Item Group). 2. New items must be mapped to an existing hierarchy node at creation; ad hoc, unmapped items are not permitted. 3. Hierarchy changes (renaming, merging, splitting categories) are versioned, and historical reports continue to reflect the hierarchy in force at the time of the original transaction where relevant for audit consistency.

Outputs: Maintained category hierarchy, mandatory item-to-hierarchy mapping, versioned hierarchy change log.

Postconditions: 1. Reporting and classification-driven workflows (ABC/VED/FSN/HML, budget categories) remain consistent and meaningful over time.

Business Rules: 1. Merging two categories requires all affected items to be remapped as part of the same change, not left orphaned. 2. Deleting a category with active items mapped to it is blocked until those items are remapped elsewhere.

Exception Handling: 1. If an item genuinely does not fit any existing category, Product Master Admin can create a new node, but must justify why it is not a sub-category of an existing one, to avoid unnecessary hierarchy fragmentation.

### REQ-M003 – Product Alias/Tags

Priority: High | Status: Proposed

Description: Internal name, supplier name, ward colloquial names. Searchable by any alias. Dropdown suggestions. Prevent confusion. Display all aliases in detail view.

Preconditions: An item may be available from multiple approved vendors, potentially at different rates.

Inputs: Item code, vendor ID, vendor-specific rate, effective date, preferred-vendor flag.

Processing Logic: 1. System maintains a many-to-many mapping between items and approved vendors, each with its own current rate and effective date. 2. At quotation/PO time, all mapped vendors for an item are available for selection, with their last-known rate displayed for reference (subject to fresh quotation per the three-quotation policy). 3. A preferred-vendor

flag can be set per item based on historical performance, used as a tie-breaker suggestion, not an automatic override of the competitive process.

Outputs: Item-vendor rate mapping, preferred-vendor indicator, quotation-time vendor shortlist.

Postconditions: 1. Purchase Executives have immediate visibility into which vendors can supply a given item and their recent pricing, speeding up RFQ shortlisting.

Business Rules: 1. A vendor cannot be mapped to an item until both the vendor and item are independently approved and active. 2. Removing a vendor's mapping to an item does not delete historical transaction records referencing that mapping.

Exception Handling: 1. If an item has no mapped vendors at all, the system flags it as 'unsourced' when it appears on an indent, prompting Purchase to initiate vendor sourcing before the indent can proceed to quotation.

## REQ-M004 – Duplicate Prevention

Priority: High | Status: Proposed

Description: Cross-verify against existing names and aliases. Fuzzy matching. Alert on potential duplicate. Admin override with justification. Regular duplicate audit. Merge with history preservation.

Preconditions: An item may be known by different names or codes internally versus in a vendor's or manufacturer's catalog.

Inputs: Internal item code, alias/synonym, vendor catalog code, manufacturer part number.

Processing Logic: 1. Product Master Admin can register one or more aliases, vendor catalog codes, and manufacturer part numbers against a single internal item record. 2. Search functionality across indenting, quotation, and PO screens matches against the internal name as well as any registered alias/code, so users searching by a vendor's terminology still find the correct internal item. 3. This mapping also supports automated matching when importing vendor bulk-rate files (REQ-V011), where the vendor's own code is used.

Outputs: Alias/synonym-enabled item master, cross-referenced vendor/manufacturer codes, improved search matching.

Postconditions: 1. Users are not required to know or guess the exact internal naming convention to find and transact against the correct item.

Business Rules: 1. Aliases are additive and do not change the item's primary/canonical name used in official reports and the statutory item register.

Exception Handling: 1. If two different vendors use the same code to refer to different products, the system requires the code mapping to be vendor-specific (not global) to prevent cross-vendor mismatches.

## REQ-M005 – Goods and Services Categorization

Priority: High | Status: Proposed

Description: Goods: physical with stock tracking. Services: non-physical. Different workflows. Service PO: no GRN, direct payment. Goods PO: full workflow. Separate reporting.

Preconditions: An item is being discontinued, superseded by a newer item, or found to be a duplicate of an existing entry.

Inputs: Item code, deactivation reason, superseding item code (if applicable), effective date.

Processing Logic: 1. Product Master Admin marks an item Inactive rather than deleting it, preserving all historical transaction linkage. 2. Where a superseding item exists, the two are cross-linked so that any user searching the discontinued item is guided to the replacement. 3. Inactive items are excluded from new indent/quotation/PO screens but remain fully visible in historical reports, the asset register, and audit trails. 4. Any open PR/PO referencing an item being deactivated is flagged for resolution before deactivation takes final effect.

Outputs: Deactivated item record with preserved history, supersession cross-link, open-transaction flag.

Postconditions: 1. The active catalog remains clean and current for day-to-day use, while the complete historical record – essential for audit and asset accounting – is never lost.

Business Rules: 1. Items with an active rate contract or open PO cannot be deactivated until those are closed or explicitly reassigned to the superseding item. 2. Deactivation is reversible (reactivation) by Product Master Admin if done in error.

Exception Handling: 1. If an item is deactivated but later found to still be needed (e.g., a discontinued brand becomes available again), it can be reactivated with its full original history intact rather than requiring re-creation as a new item.

## REQ-M006 – Product Images



Priority: Medium | Status: Proposed

Description: Front/back image upload. Thumbnail and full view. Image search. Display in indent/PO/GRN. Mobile capture. Deadline-based collection from suppliers.

Preconditions: Item master data needs to be reviewed or bulk-updated periodically (e.g., annual GST rate revision, category realignment).

Inputs: Item codes (bulk selection or full export), fields to update, update source (manual/bulk file).

Processing Logic: 1. Product Master Admin can export the full item master (or a filtered subset) to Excel for offline review. 2. A bulk-update file, following the same validated-template approach as REQ-V011, can be re-imported to apply changes (e.g., GST rate revisions following a government notification) across many items in a single controlled action. 3. All bulk changes pass through the same validation and are staged for review before being committed, with a before/after comparison available for Product Master Admin sign-off. 4. Every bulk update is logged as a single traceable batch event in the audit trail, in addition to the individual field-level changes.

Outputs: Exported item master data, validated bulk-update staging, before/after comparison, batch-level audit entry.

Postconditions: 1. Large-scale, periodic master-data maintenance (e.g., a GST rate change affecting hundreds of items) can be executed accurately and quickly, with full auditability, rather than requiring error-prone item-by-item manual edits.

Business Rules: 1. Bulk updates affecting GST rates or pricing require Finance sign-off before being committed, even though the change originates from Product Master Admin. 2. A bulk update cannot be partially committed – either the full validated batch is applied, or none of it is, to avoid an inconsistent intermediate state.

Exception Handling: 1. If a bulk-update file contains a mix of valid and invalid rows, the system stages only the valid rows for commit and returns the invalid rows for correction, consistent with the pattern established in REQ-V011.

## REQ-M007 – Item/Product Locking by Department, User, Branch and Project

Priority: High | Status: Proposed

Description: The system shall allow selected items to be restricted so that they can only be requested, issued, or consumed by specific departments, users, branches, or projects,

preventing use of controlled or specialised items outside their intended context, for example restricting an MRI contrast agent to the MRI department only.

Preconditions: Item must exist in the product master. Restriction rules must be defined by an authorized admin.

Inputs: Item ID, restriction type (Department/User/Branch/Project), list of allowed entities, list of explicitly blocked entities (optional), effective date, restriction reason, approving authority.

Processing Logic: 1. Admin marks an item as Restricted in the product master and defines one or more restriction rules: allowed department(s), allowed user(s), allowed branch(es), and/or allowed project(s). 2. Rules can combine, for example allowed only to a specific department, and only for users with a specific role within that department. 3. At indent, issue, or consumption, system checks the requesting department, user, branch, or project against the item's restriction rules before allowing the transaction to proceed. 4. If the requester is not on the allowed list, the transaction is blocked and the requester is shown the reason plus the contact for requesting an exception. 5. An authorized user can grant a one-time exception with a mandatory reason, which is logged against both the item and the requester. 6. Restriction rules and any exceptions granted are visible in the item's audit history.

Outputs: Restricted-item register, blocked-transaction log, exception-grant log, restriction audit trail.

Postconditions: 1. A restricted item cannot be issued to a non-allowed department, user, branch, or project without a logged exception. 2. Removing or loosening a restriction requires the same approval level that originally set it. 3. Exceptions are single-use by default; a recurring need should be handled by widening the rule, not by repeated exceptions.

Business Rules: 1. If an item is restricted at both department and user level, both checks must pass. 2. If a blocked store or department attempts to requisition a restricted item, the request is auto-routed to the item's owning department for review instead of being silently rejected.

Exception Handling: 1. Item is marked Restricted and rules are configured. 2. Requesting entity's department, branch, or project is correctly captured on the transaction. 3. Exception grant, where used, has a valid approver and reason.

## 3.11 PRINTING CONSUMABLES MANAGEMENT

### 3.11 Printing Consumables Management Process Flow

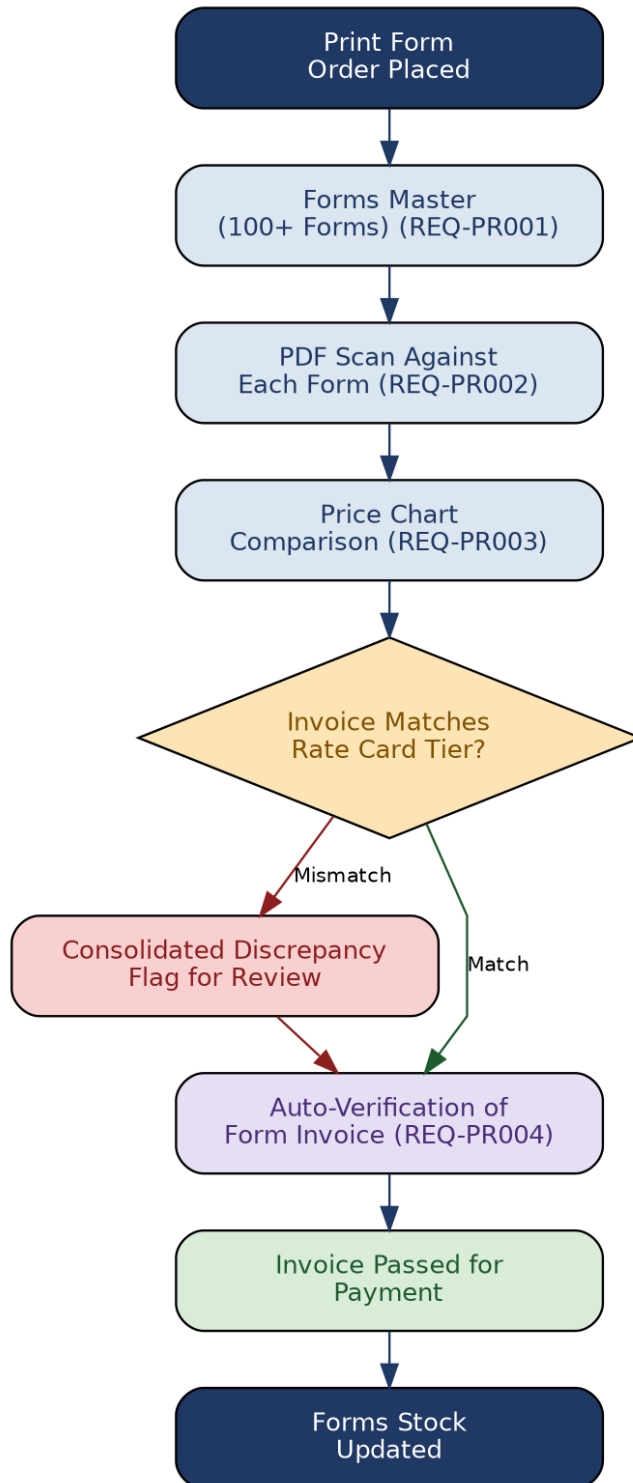


Figure: 3.11 Printing Consumables Management – Process Flow

## REQ-PR001 – Forms Management (100+ Forms)

Priority: Medium | Status: Proposed

Description: Master list of all hospital forms. Form code, name, dept usage, current version. Version control with effective date. Obsolete marking. Dept allocation. Usage tracking.

Preconditions: The hospital uses a large, standardized set of printed forms (registers, consent forms, requisition slips, case sheets) across departments.

Inputs: Form code, form name, department(s) using it, current version, effective date, obsolete flag.

Processing Logic: 1. System maintains a master list of all hospital forms (100+), each with a unique form code, name, and the department(s) that use it. 2. Each form revision is version-controlled with an effective date; the previous version is automatically marked obsolete from that date, but remains retrievable for historical reference. 3. Department allocation determines which departments can indent a given form. 4. Usage tracking records issue frequency and quantity per form per department, feeding into REQ-PR003's price/volume analysis.

Outputs: Versioned form master, department allocation mapping, obsolete-version archive, usage tracking data.

Postconditions: 1. No department can inadvertently order or use an obsolete form version once a new version's effective date has passed. 2. The complete print-form catalog is centrally governed rather than managed informally by individual departments.

Business Rules: 1. A new form version requires the same item-master-style approval as any master data change (Quality/Stores Manager sign-off), given forms often carry regulatory or clinical significance. 2. A form cannot be marked obsolete while unfulfilled indents against it remain open.

Exception Handling: 1. If a department is found still using an obsolete form version (e.g., old physical stock not yet exhausted), the system flags this in the usage-tracking report rather than blocking the transaction outright, allowing controlled stock run-down.

## REQ-PR002 – PDF Scan Against Each Form

Priority: Medium | Status: Proposed

Description: Upload original PDF. Supplier copy storage. Version-wise archive. Preview and download. Link to rate card. Access control: Store + Quality.

Preconditions: A form's approved design/content needs a definitive, retrievable master copy, and each print run needs to be verified against it.

Inputs: Form code, original PDF design file, supplier printed-copy scan, version, uploading user.

Processing Logic: 1. The original approved PDF design is uploaded and stored against the form's master record. 2. When a supplier delivers a print run, a scan of the supplier's printed copy is also uploaded and stored against the same form and version, allowing visual comparison. 3. All versions are retained in a version-wise archive with preview and download available directly from the form record. 4. Each form's document set is also linked to its current rate card entry (REQ-PR003) for a complete design-to-cost view. 5. Access to upload/view is restricted to Stores and Quality roles, given the clinical/regulatory sensitivity of some forms.

Outputs: Stored original PDF and supplier scan per version, version-wise document archive, rate-card linkage, restricted access log.

Postconditions: 1. Any dispute about whether a delivered print run matches the approved design can be resolved instantly by comparing the stored original against the supplier's scanned copy.

Business Rules: 1. Only Store and Quality roles can upload or replace form documents; other roles have view-only access consistent with their form-allocation permissions.

Exception Handling: 1. If a supplier's printed copy is found to deviate from the approved original PDF, the GRN for that print run is held pending Quality review and supplier correction, referencing REQ-PR004's discrepancy flagging.

## REQ-PR003 – Price Chart Comparison

Priority: Medium | Status: Proposed

Description: Form-wise, format-wise, copies-wise pricing. Tiers: 50/100/200/300/500/1000/1000+. Supplier comparison. Historical trend. Best supplier by volume. Export to Excel.

Preconditions: Multiple suppliers quote for form-printing across a range of standard quantity tiers.

Inputs: Form code, format/size, quantity tier (50/100/200/300/500/1000/1000+), supplier, quoted price.

Processing Logic: 1. System captures form-wise and format-wise pricing from each supplier across the standard quantity tiers. 2. A comparison view presents all suppliers' pricing side by side for a given form and tier, along with historical pricing trend over time. 3. Based on the organization's actual order volume for each form, the system highlights the best-value supplier at that specific volume, since per-unit price often varies non-linearly with quantity. 4. The comparison and trend data are exportable to Excel for procurement negotiation.

Outputs: Form/format/tier-wise price comparison, historical trend, best-supplier-by-volume recommendation, Excel export.

Postconditions: 1. Form-printing purchase decisions are based on actual applicable-volume economics rather than a single headline unit price that may not reflect the organization's real order size.

Business Rules: 1. This comparison still operates within the three-competitive-quotation policy (Section 5.2); the tiered price chart informs vendor selection but does not itself substitute for the formal RFQ/PO process.

Exception Handling: 1. If a supplier has not quoted for a specific tier the hospital actually orders at, the system flags the gap rather than extrapolating a potentially inaccurate price.

## REQ-PR004 – Auto Verification of Form Invoice

Priority: Medium | Status: Proposed

Description: On invoice entry: read form number and qty. Cross-verify with approved list. Match qty with rate card. Auto-calculate expected value. Flag discrepancies. One-point check.

Preconditions: A supplier invoice for a form-printing order is being entered against a GRN.

Inputs: Invoice form number, invoice quantity, approved form list, rate card entry, invoiced value.

Processing Logic: 1. At invoice entry, the system reads the form number and quantity stated on the invoice. 2. It cross-verifies the form number against the approved form master (REQ-PR001) to confirm it is a recognized, current form. 3. Invoiced quantity and rate are matched against the applicable rate-card tier (REQ-PR003) for that supplier and form. 4. Expected

value is auto-calculated from the matched rate and quantity, and compared against the invoiced value. 5. Any discrepancy – unrecognized form, quantity or rate mismatch – is flagged as a single consolidated exception for Purchase/Store review before the invoice is passed, rather than requiring separate manual checks.

Outputs: Auto-verified invoice with expected-vs-invoiced value comparison, consolidated discrepancy flag.

Postconditions: 1. Form-printing invoices are checked against the approved catalog and rate card automatically, reducing the manual reconciliation burden and catching pricing errors at entry rather than after payment.

Business Rules: 1. An invoice referencing a form not in the approved master is held entirely, since it may indicate an unauthorized or discontinued form still being printed. 2. Rate mismatches beyond a configurable tolerance require Purchase Manager sign-off before payment.

Exception Handling: 1. If the applicable rate-card tier for the exact invoiced quantity does not exist (e.g., order size falls between configured tiers), the system interpolates conservatively and flags the invoice for manual confirmation rather than auto-passing it.

### 3.12 UNIFORMS & LINEN TRACKING

#### 3.12 Uniforms & Linen Tracking Process Flow

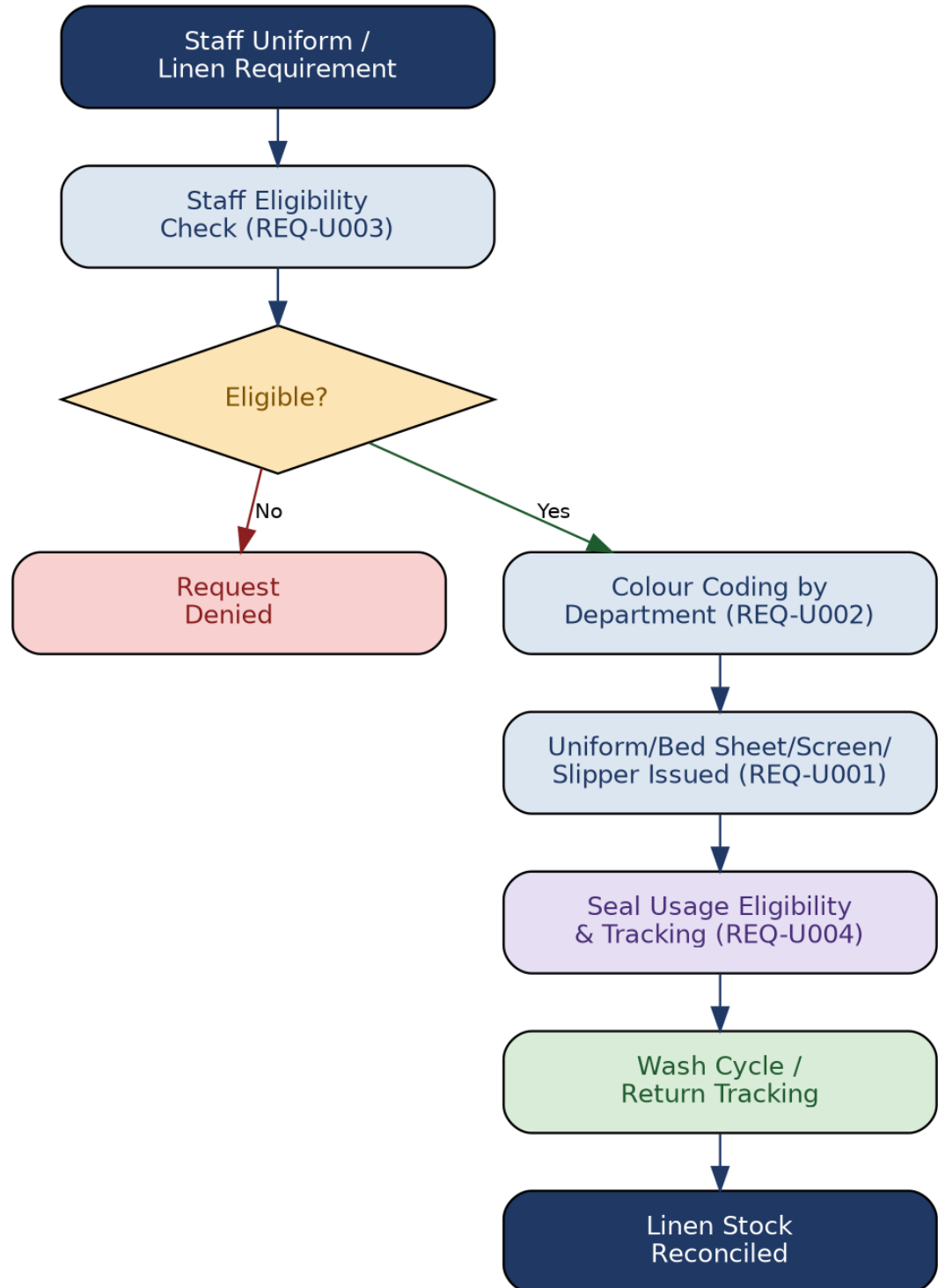


Figure: 3.12 Uniforms & Linen Tracking – Process Flow



## REQ-U001 – Uniform/Bed Sheet/Screen/Slipper Tracking

Priority: Medium | Status: Proposed

Description: Track all linen: uniforms, bed sheets, screens, slippers. Size/spec tracking. Dept allocation. Issue/return tracking. Laundry cycle. Replacement schedule by wear count.

Preconditions: Uniforms, bed sheets, privacy screens, and slippers are in active circulation across departments and require lifecycle tracking distinct from general consumables.

Inputs: Item type (uniform/bed sheet/screen/slipper), size/specification, department, issue quantity, wear count, laundry cycle count.

Processing Logic: 1. Each linen category is tracked with its specific size/specification attributes (e.g., uniform size, bed-sheet dimension). 2. Items are allocated to departments and, for uniforms, to individual staff. 3. Issue and return transactions are logged each time an item moves through use and laundry. 4. A wear-count or laundry-cycle counter increments with each recorded laundry cycle, and the item is flagged for replacement once it reaches the configured threshold for its category, rather than relying on a fixed calendar-based replacement schedule alone.

Outputs: Category-specific linen master, department/staff allocation, issue-return-laundry cycle log, wear-based replacement flag.

Postconditions: 1. Replacement decisions for linen are based on actual usage wear rather than a uniform calendar assumption, extending serviceable life while avoiding overuse of worn items.

Business Rules: 1. The wear-count threshold triggering replacement is configurable per item type and can reflect fabric quality/vendor differences. 2. An item flagged for replacement is not automatically discarded; it follows the standard write-off approval (REQ-UL... general disposal workflow) before removal from stock.

Exception Handling: 1. If an item's laundry-cycle count cannot be tracked precisely (e.g., bulk-washed batches rather than individually tagged items), the system falls back to a calendar-based replacement estimate for that batch, flagged as 'estimated cycle count'.

## REQ-U002 – Colour Coding by Department

Priority: Medium | Status: Proposed

Description: Assign colour codes to departments. Uniform colour mapping. Screen/bed sheet coding. Colour legend. Report by colour/dept. Prevent mix-up.

Preconditions: The hospital assigns distinct colour codes to departments to visually distinguish linen and uniforms and prevent cross-department mix-up.

Inputs: Department, assigned colour code, item category (uniform/screen/bed sheet).

Processing Logic: 1. Administrator defines a colour-code legend mapping each department to a distinct colour, applied consistently across uniforms, screens, and bed sheets issued to that department. 2. New linen/uniform stock received for a department is tagged with its assigned colour at the item level. 3. Reports can be filtered or grouped by colour code as well as by department, supporting both procurement planning and physical sorting (e.g., during laundry return, items are sorted by colour before being routed back to the correct ward). 4. The colour legend is displayed on relevant screens to help staff quickly verify correct allocation.

Outputs: Colour-coded item records, colour legend reference, colour/department-filtered reports.

Postconditions: 1. Physical mix-up of linen and uniforms between departments – a common operational friction point – is reduced through consistent, system-supported colour identification.

Business Rules: 1. Colour-code reassignment for a department requires Stores Manager approval, since it affects a large volume of existing tagged stock and laundry-sorting practice.

Exception Handling: 1. If two departments are inadvertently assigned overlapping or easily confused colours, the system flags the conflict at the point of assignment and requires the administrator to select a distinguishable alternative.

## REQ-U003 – Staff Eligibility Tracking for Uniforms

Priority: Medium | Status: Proposed

Description: Define sets per staff category. Track issued vs eligibility. Alert on excess request. Management approval for excess. Deduction tracking. Report repeated requests.

Preconditions: Each staff category (e.g., nursing, housekeeping, security) has a defined entitlement of uniform sets per period.

Inputs: Staff category, entitlement (sets per period), employee ID, issued-to-date count, excess request justification.

Processing Logic: 1. Administrator defines the standard uniform-set entitlement per staff category and issue period (e.g., 3 sets per year for nursing staff). 2. Each issue against an employee is checked against their remaining entitlement for the period. 3. A request within entitlement is processed through the standard issue workflow. 4. A request exceeding entitlement triggers an alert and requires Management approval, with a mandatory justification (loss, damage, role change) captured before the excess issue is permitted. 5. Deductions against entitlement are tracked per employee, and a report highlights employees with repeated excess requests for HR/Stores Manager review.

Outputs: Entitlement-tracked issue record, excess-request approval workflow, repeated-excess-request report.

Postconditions: 1. Uniform issue remains within policy-defined entitlement by default, with any exception visibly approved and justified rather than issued informally.

Business Rules: 1. Entitlement resets at the start of each configured period; unused entitlement does not carry forward unless explicitly permitted by policy. 2. Role changes that alter a staff member's category (and therefore entitlement) require the entitlement to be recalculated from the effective date of the role change.

Exception Handling: 1. If a staff member's uniform is lost/damaged due to a documented workplace incident (e.g., biohazard exposure), the excess-issue approval can reference the incident record as automatic justification rather than requiring a fresh case-by-case explanation.

## REQ-U004 – Seal Usage Eligibility and Tracking

Priority: Medium | Status: Proposed

Description: Define ward eligibility for seal types. Track issuance by ward/purpose. Current stock of 100+ unused seals. Alert on excessive requests. Usage justification. Utilization report.

Preconditions: Official seals are a controlled, limited-stock item requiring eligibility-based issuance to specific wards for specific purposes.

Inputs: Ward, seal type, eligibility rule, issue quantity, purpose/justification, current stock of unused seals.

Processing Logic: 1. Administrator defines which wards are eligible for which seal types and for what purposes. 2. Seal issuance is logged against the requesting ward with a mandatory

purpose/justification. 3. System maintains a real-time count of unused seal stock (typically 100+ units tracked individually or in small controlled batches). 4. Requests exceeding a configurable normal-usage threshold for a ward trigger an alert requiring higher-level review before issuance. 5. A utilization report shows seal issuance by ward, purpose, and time, supporting periodic control review given the sensitivity of seal usage.

Outputs: Eligibility-checked seal issuance record, real-time unused-seal stock count, excessive-request alert, utilization report.

Postconditions: 1. Issuance of a sensitive, low-volume controlled item like official seals is tightly governed and fully auditable, with abnormal usage patterns surfaced proactively rather than discovered only during a periodic manual stock check.

Business Rules: 1. A ward not on the eligibility list cannot be issued a seal without an explicit, approved exception, distinct from the routine issue workflow. 2. Seal stock counts are subject to more frequent physical verification than general stores items, given their sensitivity.

Exception Handling: 1. If the physical seal count during verification does not match the system record, it is treated as a security-relevant exception requiring escalation to Management and Security, not a routine stock-adjustment entry.

## 3.13 WORKFLOW & APPROVALS

### 3.13 Workflow & Approvals Process Flow

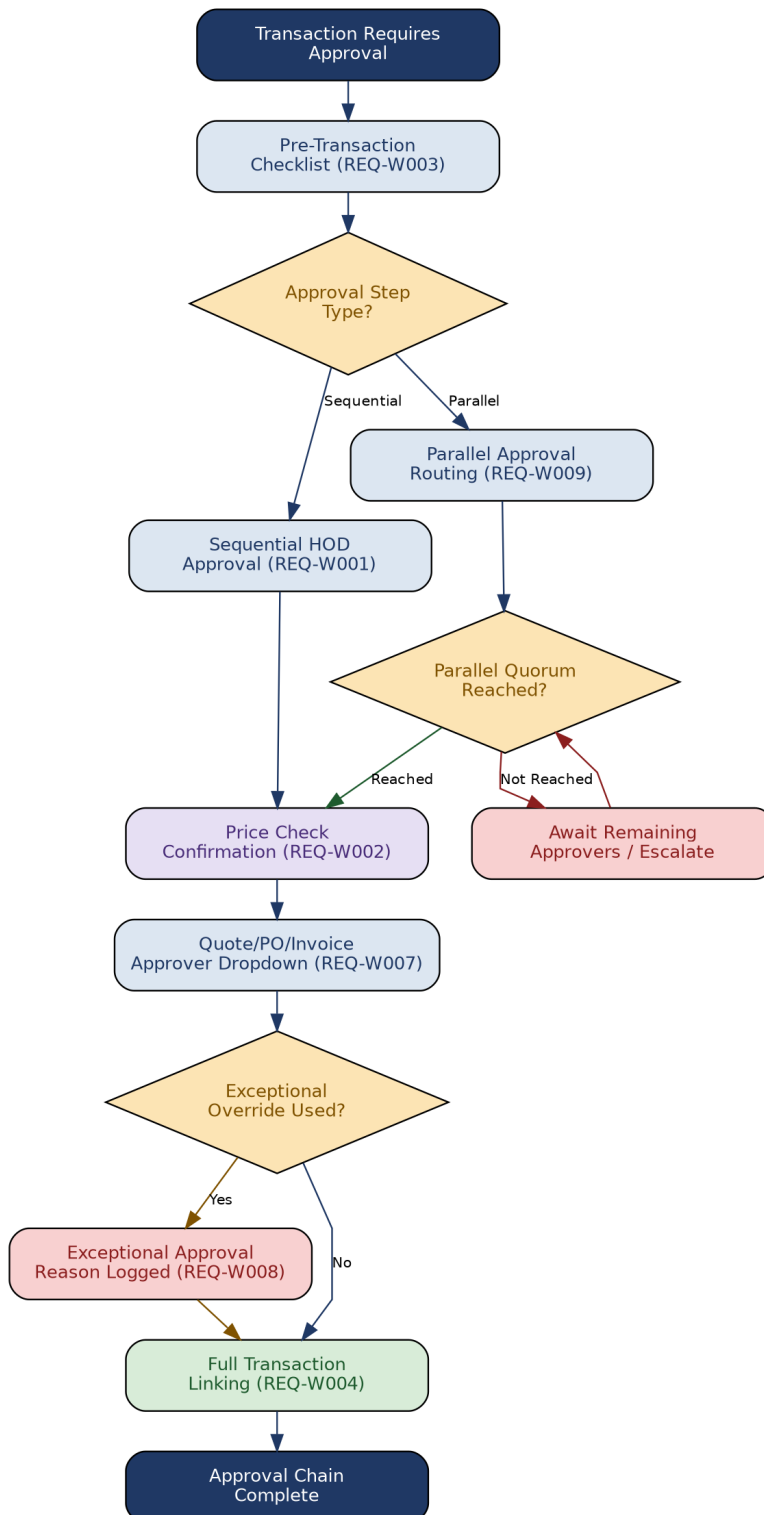


Figure: 3.13 Workflow & Approvals – Process Flow

**REQ-W001 – Department HOD Internal Online Approval**

Priority: High | Status: Proposed

Description: All dept indents route to HOD. Email and dashboard notification. Approve/Reject/Clarify. Mobile-friendly. Delegation during absence. History with timestamps. SLA: 24 hours.

Preconditions: A department user submits an indent that requires HOD sign-off before proceeding.

Inputs: Indent ID, department, HOD ID, delegate ID (if configured), SLA (24 hours).

Processing Logic: 1. Every department indent is automatically routed to the configured HOD's approval queue on submission. 2. The HOD is notified simultaneously by email and an in-portal dashboard alert. 3. The HOD can Approve, Reject, or Request Clarification from either the desktop portal or a mobile-friendly view. 4. If the HOD has configured a delegate for a period of absence, the delegate receives the same routing and their action is labeled 'Approved/Rejected by Delegate' in the history. 5. A 24-hour SLA applies; indents unactioned past this window trigger an automated reminder, then escalation. 6. Every action (approve/reject/clarify) is timestamped and retained in a complete, chronological history against the indent.

Outputs: HOD approval queue entry, multi-channel notification, mobile-accessible approval action, delegate routing, SLA reminder/escalation, timestamped history.

Postconditions: 1. No departmental indent proceeds to Stores/Purchase without a timely, traceable HOD decision, and delays are proactively surfaced rather than silently accumulating.

Business Rules: 1. The 24-hour SLA clock pauses during a 'Clarification Requested' state (since the delay is with the requester, not the HOD) and resumes once the requester responds.

Exception Handling: 1. If neither the HOD nor a configured delegate acts within the SLA plus a configurable grace period, the system escalates to the HOD's reporting manager for interim decision, flagged for the HOD's retrospective awareness.

## REQ-W002 – Price Check Confirmations in Approval

Priority: High | Status: Proposed

Description: Display last purchase price. Display rate contract price. Flag if current > last by threshold. Approver confirms price check. Justification text for higher prices. Historical price chart.

Preconditions: A quotation, PO, or invoice is under HOD/approver review, and historical pricing data exists for the item.

Inputs: Item, current quoted/PO price, last purchase price, active rate-contract price (if any), variance threshold.

Processing Logic: 1. At the approval screen, the system displays the current price alongside the last purchase price and any active rate-contract price for the same item, side by side. 2. If the current price exceeds the last purchase price (or rate-contract price) by more than a configurable threshold, a visible flag is raised on the approval screen. 3. The approver cannot approve a flagged transaction without entering a justification for the higher price. 4. A historical price chart for the item is available in-line for the approver to review trend context without leaving the approval screen.

Outputs: Price-context display at approval, threshold-breach flag, mandatory justification capture, in-line historical price chart.

Postconditions: 1. Price increases are never approved 'blind' – every approver has immediate, contextual pricing history in front of them at the moment of decision.

Business Rules: 1. This price-check display is a hard gate for flagged transactions: the Approve action is disabled until either the price is within tolerance or a justification is entered. 2. The threshold is configurable by item category, consistent with REQ-P020's variation-tolerance design.

Exception Handling: 1. If no prior purchase history exists for the item (first-time purchase), the price-check flag is not applicable and the approval screen indicates 'no historical baseline available' rather than showing a misleading comparison.

### REQ-W003 – Checklist Before Transaction Processing

Priority: High | Status: Proposed

Description: Configurable checklists for PO/GRN/Indent. PO: Vendor approved? Budget available? Rate contract valid? GRN: PO exists? Qty within tolerance? Invoice valid? Mandatory completion. Audit trail. Admin-configurable.

Preconditions: A PO, GRN, or indent is ready to move to the next processing stage.

Inputs: Transaction type, configurable checklist items, completion status per item, processing user.

Processing Logic: 1. Administrator configures a checklist per transaction type – for PO: vendor approved, budget available, rate contract valid (where applicable); for GRN: PO exists, quantity within tolerance, invoice valid. 2. Before the transaction can advance (e.g., PO released to vendor, GRN finalized into stock), every checklist item must be explicitly confirmed by the processing user. 3. Checklist completion is stored as part of the transaction's permanent record, contributing to the audit trail. 4. Checklists can be extended or modified by an administrator as process requirements evolve, without requiring a system change request.

Outputs: Mandatory pre-processing checklist, completion record in audit trail, admin-configurable checklist templates.

Postconditions: 1. Core control checks (vendor approval status, budget availability, rate/PO/quantity validity) are never skipped due to human oversight, since the system will not allow the transaction to proceed until each is explicitly confirmed.

Business Rules: 1. Checklist items cannot be bulk-confirmed without individual review; each item requires its own explicit tick/confirmation action. 2. A checklist item failing its check (e.g., budget not available) blocks progression until resolved, rather than merely warning.

Exception Handling: 1. If a genuine business need requires proceeding despite a failed checklist item (e.g., emergency purchase without prior budget approval), an authorized override is available but requires a higher-tier approval and mandatory reason, fully distinct from routine checklist completion.

### REQ-W004 – Full Transaction Linking

Priority: High | Status: Proposed



Description: Chain: Request → Quote → Comparison → Finalization → Approval → PO → GRN → Payment → Issue → Usage → Billing/Expiry. Single-click navigation. Status indicators. Delay alert per SLA. Cycle time report.

Preconditions: A material request has progressed through multiple stages from initial request to final usage/billing/expiry.

Inputs: Request ID and its linked Quote, Comparison, Finalization, Approval, PO, GRN, Payment, Issue, Usage, and Billing/Expiry references.

Processing Logic: 1. Every stage of a transaction's lifecycle – Request, Quote, Comparison, Finalization, Approval, PO, GRN, Payment, Issue, Usage, and Billing/Expiry – is linked into a single traceable chain sharing a common root reference. 2. Any stage can be opened directly from any other linked stage via single-click navigation, without needing to search separately for related records. 3. Each stage displays a status indicator (Pending/In-Progress/Complete/Delayed) so the full chain's progress is visible at a glance. 4. A delay alert is raised if any stage exceeds its configured SLA, and a cycle-time report aggregates total elapsed time from Request to final Usage/Billing across transactions for process-improvement analysis.

Outputs: Fully linked transaction chain, single-click cross-stage navigation, stage status indicators, delay alerts, cycle-time report.

Postconditions: 1. Any stakeholder can trace a transaction's complete history – from the original request through to final consumption or billing – without needing to consult multiple disconnected screens or ask other departments for status.

Business Rules: 1. The chain linkage is established automatically as each stage is created from its predecessor (e.g., a PO generated from an approved requisition inherits the link); it cannot be manually broken or reassigned without an audit-logged exception.

Exception Handling: 1. If a stage is skipped by design (e.g., a rate-contract PO with no separate Quote/Comparison stage), the chain displays that stage as 'Not Applicable' rather than 'Missing', so process-completeness reporting is not distorted.

**REQ-W005 – Purchase Requester with HOD Approval**

Priority: High | Status: Proposed

Description: Capture requester name and dept. HOD approval in their login. Requester cannot self-approve. Track request-to-approval time. Report by user.

Preconditions: A purchase requisition or indent is being raised by a department user.

Inputs: Requester name, department, HOD ID, request timestamp, approval timestamp.

Processing Logic: 1. System captures the requester's name and department automatically from their logged-in profile at the point of request creation. 2. The request routes to the HOD associated with that department for approval, using the HOD's own login (not the requester's), enforcing that the requester cannot approve their own request under any circumstance, including if they hold a dual role. 3. Request-to-approval elapsed time is measured and available in a per-user report, supporting both HOD responsiveness review and requester pattern analysis (e.g., a requester whose requests are frequently rejected or heavily modified).

Outputs: Requester-attributed request record, HOD-login-gated approval, request-to-approval time metric, per-user report.

Postconditions: 1. Segregation of duties between requester and approver is structurally enforced by the system, not dependent on individual discipline.

Business Rules: 1. Even where a single individual holds both a requesting role and an HOD role for a different department, the system only allows them to approve requests originating from departments other than their own requesting context for that specific transaction.

Exception Handling: 1. If the requester and configured HOD are unavoidably the same person (e.g., a very small department with only one senior staff member), the request is automatically escalated one level up the reporting hierarchy for approval instead.

## REQ-W006 – Quote Provider Details with HOD Approval

Priority: High | Status: Proposed

Description: Record quote provider: name, contact, validity. Upload quote document. HOD approval in login. Auto-generated comparison table. Link to PO. Expiry alert.

Preconditions: A quotation has been obtained from a vendor and needs to be captured for HOD/Purchase review.

Inputs: Quote provider name, contact details, quote validity date, quote document, HOD ID.

Processing Logic: 1. Purchase Executive records the quote provider's name, contact details, and quote validity date, and uploads the quote document itself. 2. The quotation is presented for HOD approval within the HOD's own login, consistent with the segregation-of-duties principle in REQ-W005. 3. Once all required quotations for the RFQ are captured, the system auto-generates the comparison table (per Section 5.2/5.3) directly from the recorded quote data, without manual re-entry. 4. The approved, winning quotation is linked directly to the resulting PO for full traceability. 5. System monitors quote validity dates and alerts if a PO is being raised against an expired quotation.

Outputs: Structured quote-provider record with document, HOD-approved quote, auto-generated comparison table, PO linkage, expiry alert.

Postconditions: 1. Every PO can be traced back to its specific, dated, HOD-reviewed quotation, eliminating ambiguity about which quote a purchase decision was actually based on.

Business Rules: 1. A PO cannot be finalized against a quotation whose validity date has passed without an explicit re-confirmation from the vendor, re-captured as an updated quote record.

Exception Handling: 1. If a vendor provides a quote informally (e.g., verbally or via a phone call, common for very low-value items), the Purchase Executive can record the quote details with a 'no document available' flag, subject to Purchase Manager acknowledgment of the informal source.

## REQ-W007 – Quote/PO/Invoice Approver Dropdown

Priority: Medium | Status: Proposed

Description: Dropdown of authorized approvers. Filter by dept and value limit. Display workload. Auto-suggest based on rules. Track delegation. Approver performance report.

Preconditions: A quotation, PO, or invoice requires an approver to be selected or auto-suggested for routing.

Inputs: Transaction value, department, category, list of authorized approvers with value limits, approver current workload.

Processing Logic: 1. System presents a dropdown of approvers who are authorized for the transaction's department and whose configured value limit covers the transaction, filtering out anyone not eligible. 2. Each listed approver's current pending-approval workload is displayed, allowing a submitter (where more than one eligible approver exists at the same tier) to route toward a less-loaded approver where policy permits discretion. 3. Where the routing is not discretionary (fixed hierarchy), the system auto-suggests the correct approver(s) based on the configured rules rather than presenting an open choice. 4. Delegation arrangements are reflected in the dropdown automatically. 5. An approver-performance report tracks average turnaround time per approver, supporting workload-balancing and process-improvement decisions.

Outputs: Filtered, eligibility-based approver dropdown, workload visibility, auto-suggested routing, delegation-aware listing, approver-performance report.

Postconditions: 1. Transactions are always routed to a genuinely eligible approver, and bottlenecks caused by uneven approver workload are visible and addressable.

Business Rules: 1. The dropdown never presents an approver who lacks the value/department authority for the specific transaction, regardless of workload considerations, since eligibility is a hard constraint and workload balancing is only a secondary, discretionary factor.

Exception Handling: 1. If no eligible approver is currently active (e.g., all on leave without delegates configured), the system flags the transaction as 'unroutable' and alerts System Administration rather than leaving it silently unassigned.

## REQ-W008 – Exceptional Approval Reasons

Priority: Medium | Status: Proposed

Description: Mandatory reason for exceptions: urgent purchase, single vendor, budget exceed. Dropdown + free text. Management review. Trend analysis. Process improvement.

Preconditions: A transaction is being processed outside the standard rule set – urgent purchase, single-vendor sourcing, or a budget-exceeding request.

Inputs: Exception type, reason code (dropdown), free-text justification, approving authority.

Processing Logic: 1. Whenever a transaction is flagged as an exception (urgent purchase bypassing the standard cycle, single-vendor purchase bypassing the three-quotation rule, or a request exceeding approved budget), the system requires a reason to be captured via a structured dropdown plus supplementary free text. 2. The exception routes to Management for review, distinct from the transaction's normal approval routing. 3. All exceptions across the organization are compiled into a periodic trend-analysis report, broken down by exception type, department, and frequency, to identify whether exceptions represent genuine one-off circumstances or a recurring workaround of standard process. 4. Findings feed into process-improvement discussions and, where warranted, policy or master-data adjustments (e.g., adding a genuinely sole-source vendor to a formal single-source-approved list rather than repeatedly flagging it as an exception).

Outputs: Structured exception reason capture, Management review routing, periodic exception trend-analysis report, process-improvement input.

Postconditions: 1. Deviations from standard procurement discipline are never unexplained, and patterns indicating a need for policy or master-data change are surfaced systematically rather than remaining scattered across individual transactions.

Business Rules: 1. The dropdown reason list is centrally maintained and periodically reviewed by Purchase Management to ensure it reflects genuine, distinct exception categories rather than becoming a catch-all.

Exception Handling: 1. If free-text justification is provided without selecting a matching dropdown category (reason doesn't fit existing options), the transaction is still processed but flagged 'uncategorized exception' for the trend report, and the reason list is reviewed periodically for potential expansion.

## REQ-W009 – Parallel Approval Routing

Priority: Medium | Status: Proposed

**Description:** In addition to the existing sequential approval chain, the system shall support parallel approval routing, where a request is sent simultaneously to two or more approvers and proceeds once the configured quorum of approvals is received, for cases where waiting on a strict sequence would slow down time-sensitive requests.

**Preconditions:** Approval matrix must define a step as Parallel type with its member approvers and required quorum, for example 2 of 3, or all.

**Inputs:** Request ID, approval step type (Sequential/Parallel), list of parallel approvers, quorum rule, individual approver decisions and timestamps.

**Processing Logic:** 1. Admin configures a given approval step as Parallel and specifies the approver group and quorum, for example any 2 of 3 department heads, or unanimous. 2. When a request reaches that step, notifications are sent to all approvers in the group at the same time rather than one after another. 3. Each approver's decision, Approve, Reject, or Delegate, is recorded independently as it arrives. 4. Once the quorum of approvals is reached, the request proceeds to the next step automatically; remaining pending approvers are notified the step has closed. 5. A single Reject from any approver in a unanimous-quorum group stops the request and returns it to the requester with all remarks; for a partial quorum such as 2 of 3, a Reject only blocks progress if it makes quorum unreachable. 6. Parallel and sequential steps can be combined within the same overall approval chain, for example a parallel department-head review followed by sequential finance and management sign-off.

**Outputs:** Parallel approval log with individual timestamps, quorum-reached notification, step-closure record, combined approval-chain audit trail.

**Postconditions:** 1. A request cannot proceed past a parallel step until quorum is met or the step is explicitly overridden by an authorized escalation. 2. All individual approver decisions are retained even after quorum is reached, for audit. 3. Delegation within a parallel step still counts toward that approver's vote once the delegate decides.

**Business Rules:** 1. If quorum is not reached within the configured SLA, the request is escalated to the next-level authority defined for that step. 2. If an approver is unavailable and has not set a delegate, system logs the gap and adjusts the effective quorum denominator only if configured to allow it; otherwise the SLA escalation rule applies.

**Exception Handling:** 1. Approval step is correctly configured as Parallel with a valid quorum rule. 2. All named approvers are active users with approval rights. 3. Quorum threshold does not exceed the number of configured approvers.

### 3.14 TRACKING & MONITORING

#### 3.14 Tracking & Monitoring Process Flow

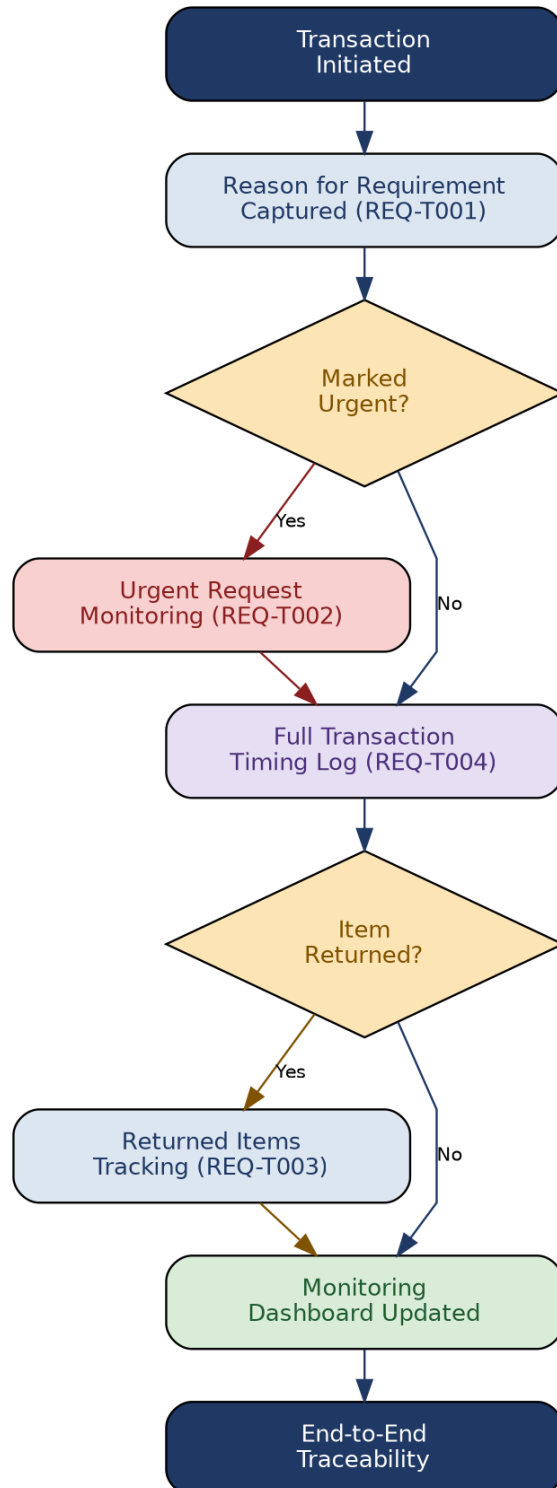


Figure: 3.14 Tracking & Monitoring – Process Flow

## REQ-T001 – Reason for Requirement Field

Priority: High | Status: Proposed

Description: Mandatory 'Reason for Requirement' for all ward requests. Sub-fields: Requested By, Reason, Total Qty Needed, Reason for Total Qty. Free text with minimum limit. Dropdown suggestions. Report on patterns.

Preconditions: A ward or department is raising a stock/material request.

Inputs: Requested by, reason for requirement, total quantity needed, reason for total quantity, item, department.

Processing Logic: 1. Every ward/department request mandates a 'Reason for Requirement' field before submission is permitted. 2. The field is structured into sub-fields: Requested By, Reason, Total Quantity Needed, and Reason for Total Quantity, each with a minimum free-text character limit to discourage low-effort entries such as a single word. 3. A dropdown of common, pre-defined reason suggestions is offered to speed entry while still allowing free text for anything not covered. 4. Accumulated reason data is analyzed periodically for recurring patterns – common justifications, unusually high-quantity requests, or specific items associated with vague reasoning – to inform Stores policy and department training.

Outputs: Mandatory structured reason capture, minimum-length enforcement, dropdown-assisted entry, pattern-analysis report.

Postconditions: 1. Every material request carries a genuine, reviewable justification rather than being submitted without explanation, supporting both approval quality and later audit review.

Business Rules: 1. The minimum character limit is configurable but is enforced strictly – a request cannot be submitted with a reason field below the threshold, regardless of urgency.

Exception Handling: 1. For the exempt key departments (REQ-N007: ICU, OT, Emergency, NICU), the reason field remains mandatory but is not permitted to delay or block submission under time pressure – the request proceeds immediately with the reason logged, and any incompleteness is followed up retrospectively rather than blocking real-time care.



## REQ-T002 – Urgent Request Monitoring

Priority: High | Status: Proposed

Description: Track frequently urgent items. Post-usage verification. Report items returned after urgent purchase. Flag high urgent ratio departments. Management review. Deterrent through visibility.

Preconditions: Requests are being flagged and processed as 'urgent' across departments.

Inputs: Department, item, urgent-flag frequency, post-usage verification outcome.

Processing Logic: 1. System tracks how frequently each item and department combination is flagged urgent over a rolling period. 2. For items procured urgently, the system prompts a post-usage verification – confirming the item was in fact used for the stated urgent purpose rather than, for example, being returned unused shortly after. 3. Items found returned or unused after being sourced urgently are cross-referenced against REQ-T003's returned-items tracking and highlighted specifically in this report. 4. Departments with a high urgent-to-total-request ratio are flagged for Management review, with the visibility itself intended as a deterrent against habitual, unwarranted use of the urgent pathway.

Outputs: Urgent-flag frequency tracking, post-usage verification outcome, cross-referenced unused-after-urgent report, high-urgent-ratio department flag.

Postconditions: 1. The urgent-purchase pathway remains available for genuine need while being protected from becoming a routine workaround for poor planning, through visible, ongoing monitoring rather than after-the-fact investigation only.

Business Rules: 1. Being flagged in this report is not itself a disciplinary action; it is a visibility and trend-monitoring mechanism feeding into Management's periodic process review, consistent with the same principle applied under REQ-P017.

Exception Handling: 1. If a department's high urgent-ratio is attributable to a documented, exceptional operational period (e.g., a disease outbreak surge), it can be annotated as such so the trend data is interpreted in context rather than triggering an unwarranted review.

## REQ-T003 – Returned Items Tracking

Priority: High | Status: Proposed

Description: Track 'Not of Use' returns. Link to original request and requester. Calculate wastage value. Identify patterns: product, dept, requester. Feedback to requester. Process improvement.

Preconditions: An issued item is returned to Stores marked 'Not of Use' rather than for a standard reason such as excess or expiry.

Inputs: Return ID, original request ID, requester, item, quantity, return reason ('Not of Use'), return value.

Processing Logic: 1. Returns specifically reasoned 'Not of Use' are tracked as a distinct sub-category within the general stock-return workflow (REQ-IO12/REQ-R009). 2. Each such return is linked back to its original request and requester, and its value is computed to quantify wastage. 3. System aggregates patterns by product, department, and requester to identify recurring sources of unnecessary issue (e.g., a specific item consistently requested but not used, or a specific requester whose requests are disproportionately later returned unused). 4. Feedback is routed back to the requester/department, and the aggregate pattern data supports process-improvement discussions on indenting accuracy and forecasting.

Outputs: 'Not of Use' return sub-classification, wastage value calculation, product/department/requester pattern analysis, requester feedback loop.

Postconditions: 1. Avoidable wastage from over-requesting or inaccurate indenting is quantified and attributed, supporting targeted coaching rather than being absorbed as unexplained return volume.

Business Rules: 1. This tracking operates as a refinement layer over the general return workflow (Section 3.3), sharing the same underlying transaction data rather than maintaining a separate return process.

Exception Handling: 1. If a 'Not of Use' return is due to a genuine, unavoidable change in clinical plan (e.g., patient discharged before scheduled procedure), it can be annotated as 'unavoidable – clinical' so pattern analysis distinguishes it from avoidable over-requesting.

## REQ-T004 – Full Transaction Timing

Priority: Medium | Status: Proposed

Description: Timestamp at every stage. Stage-to-stage duration. Bottleneck identification. SLA per transaction type. Escalation on breach. Continuous improvement data.

Preconditions: A transaction has progressed through multiple defined stages from initiation to closure.

Inputs: Transaction ID, transaction type, stage, timestamp at entry into each stage, configured SLA per transaction type.

Processing Logic: 1. System timestamps every stage transition for every transaction type (indent, PO, GRN, ticket, asset workflow, etc.) automatically, without requiring manual logging. 2. Stage-to-stage duration is calculated and compared against the SLA configured for that transaction type. 3. Transactions/stages that exceed their SLA are highlighted for escalation to the relevant supervisor. 4. Aggregated timing data across all transaction types is used to identify systemic bottlenecks (a specific stage that is consistently the slowest across many transactions) rather than only flagging individual late cases. 5. This timing data feeds the various cycle-time and turnaround reports referenced throughout this specification (REQ-C003, REQ-R008, REQ-W004), providing the single underlying timestamp source they all draw from.

Outputs: Automatic stage timestamps for all transaction types, SLA-comparison and breach escalation, systemic bottleneck identification, shared timing data source for downstream reports.

Postconditions: 1. Process efficiency across the entire stores, procurement, and asset lifecycle is measurable on a consistent, system-wide basis, supporting continuous improvement rather than isolated, transaction-type-specific timing efforts.

Business Rules: 1. SLA configuration is maintained centrally so that all reports referencing timing data (turnaround, cycle-time, bottleneck) remain consistent with one another rather than each defining its own standard.

Exception Handling: 1. If a transaction type is newly introduced without a configured SLA, its timing is still captured and reported, but escalation is suppressed until an SLA is explicitly set, to avoid inappropriate default escalation.

## 3.15 GATE PASS MANAGEMENT

### 3.15 Gate Pass Management Process Flow

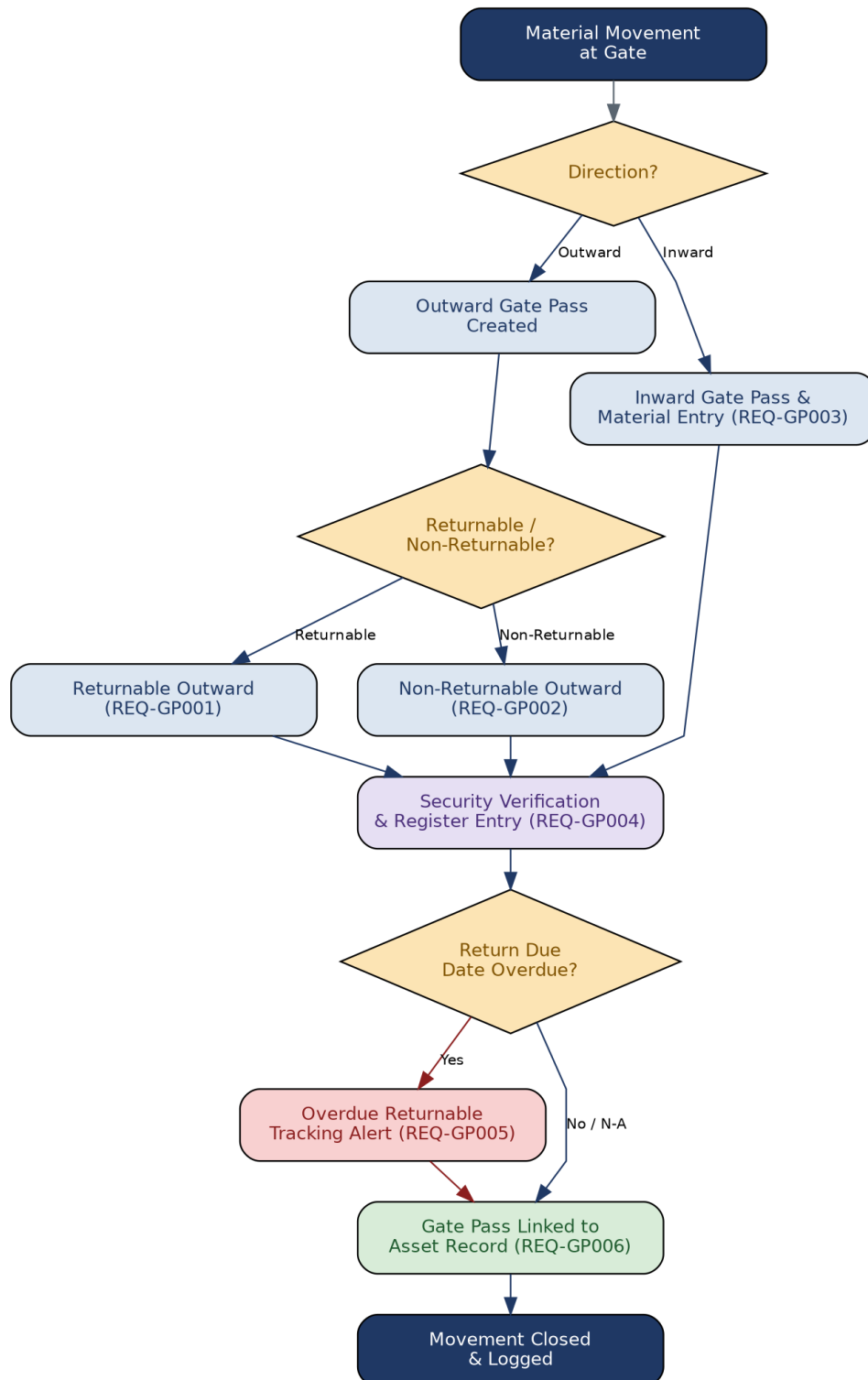


Figure: 3.15 GATE PASS MANAGEMENT – Process Flow

**REQ-GP001 – Outward Gate Pass (Returnable)**

Priority: High | Status: Proposed

Description: The system shall generate a returnable outward gate pass whenever an asset, equipment, or material leaves the premises with an expectation of return, such as items sent for repair, calibration, demo, loan, or exhibition.

Preconditions: A valid business reason exists to move an item outside the premises. The item is registered in the Asset Register or Product Master.

Inputs: Item/asset ID, reason (repair/calibration/demo/loan/exhibition), destination, carrier/vendor name, expected return date, requesting department, approving authority.

Processing Logic: 1. Requesting department or Biomedical Engineering initiates an outward gate pass, selecting 'Returnable' and the specific reason. 2. System auto-populates item details from the Asset Register or Product Master, including serial number and current custodian. 3. Expected return date is mandatory and drives the overdue-tracking workflow (REQ-GP005). 4. Gate pass routes for approval per the value/criticality matrix (REQ-GP003) before it becomes valid for exit. 5. On approval, a uniquely numbered gate pass is generated, printable and available for Security to verify at the exit point (REQ-GP004). 6. System links the gate pass to the underlying Service PO (for repairs) or transfer/loan record, where applicable. 7. On physical return, Security or Stores confirms receipt, verifies condition, and closes the gate pass.

Outputs: Gate pass document with unique number, linkage to asset/Service PO record, exit authorization for Security, return-confirmation record.

Postconditions: 1. No returnable item leaves the premises without an approved, uniquely numbered gate pass. 2. An asset with an open (unreturned) gate pass cannot be marked Active/In-Use in the Asset Register until the gate pass is closed. 3. Gate pass closure requires physical verification, not just a status update.

Business Rules: 1. Items above a configurable value/criticality threshold require Management approval in addition to department HOD. 2. A gate pass cannot be raised for an asset that already has an open outward gate pass. 3. Repair-related gate passes must reference the associated Service PO (REQ-A008).

Exception Handling: 1. If the item is returned in damaged or incomplete condition, the discrepancy is logged against the gate pass and routed to the requesting department and, where relevant, the vendor for resolution. 2. If Security identifies a mismatch between the physical item and the gate pass description at exit, the item is held and the discrepancy escalated before release.

## REQ-GP002 – Outward Gate Pass (Non-Returnable)

Priority: High | Status: Proposed

Description: The system shall generate a non-returnable outward gate pass for items permanently leaving the premises, such as scrapped/disposed assets, items sold, donations, or courier dispatch of documents and samples.

Preconditions: An item has an approved disposal, sale, or dispatch decision (e.g., via REQ-A005 Scrap/Sold/Red-Tag Marking) or is being sent out with no expectation of return.

Inputs: Item/asset ID, reason (scrap/sold/donation/courier/sample dispatch), buyer/recipient details, sale value (if sold), approving authority, dispatch date.

Processing Logic: 1. Requester selects 'Non-Returnable' and the specific reason for the gate pass. 2. For scrap/sold items, the system cross-checks that a disposal or sale approval already exists (REQ-A005); a gate pass cannot be raised for disposal-reason items without it. 3. For courier/sample dispatch, item description, quantity, and recipient are captured without requiring an asset-disposal linkage. 4. Gate pass routes for approval per REQ-GP003. 5. On approval, the gate pass is generated and presented to Security at exit for verification and logging (REQ-GP004). 6. System updates the Asset Register or stock ledger to reflect permanent removal once the gate pass is confirmed exited.

Outputs: Non-returnable gate pass document, updated Asset Register/stock status (removed), buyer/recipient record, dispatch confirmation.

Postconditions: 1. Asset or stock status is only updated to 'Disposed/Sold/Dispatched' after Security confirms physical exit against the gate pass, not at approval stage. 2. Every permanent removal from premises has a documented, approved gate pass on record.

Business Rules: 1. A non-returnable gate pass for a scrap/sold asset must reference the REQ-A005 disposal record; the reference cannot be bypassed. 2. Sale-reason gate passes below the recorded book value trigger the same escalation as REQ-A005's disposal approval, applied at the gate-pass stage as a final check.

Exception Handling: 1. If Security finds no matching disposal/sale approval for a scrap/sold gate pass at the point of exit, the item is held and the discrepancy is escalated to Stores Manager and Biomedical Engineering. 2. If the recipient refuses or fails to accept a dispatched item, the gate pass is reopened and the item's return is tracked through the returnable workflow.

## REQ-GP003 – Inward Gate Pass and Material Entry

Priority: High | Status: Proposed

Description: The system shall record inward gate passes for material entering the premises outside the standard GRN/PO flow, such as vendor demo equipment on trial, visitor-brought material, returned loan items, and courier receipt of documents.

Preconditions: Material is arriving at the premises that does not correspond to a GRN against an existing PO (which is handled under Section 3.4).

Inputs: Item description, quantity, bringing party (vendor/visitor/courier), purpose (demo/trial/return/document delivery), expected duration on-site (for trials/demos), Security guard ID.

Processing Logic: 1. Security records the inward material at the gate: item description, quantity, bringing party, and stated purpose. 2. For vendor demo/trial equipment, the system flags a required trial-end date and links the entry to a pending Purchase or Ticket record if one exists. 3. For returned loan items (matching an open outward returnable gate pass), the system automatically matches and closes the corresponding outward gate pass (REQ-GP001). 4. For courier/document receipt, the entry is logged for record purposes without inventory impact. 5. Concerned department is notified of the material's arrival for acknowledgment. 6. Demo/trial equipment nearing its trial-end date triggers a reminder to decide: purchase, return, or extend trial.

Outputs: Inward gate pass/entry record, auto-matched closure of corresponding outward pass (loan returns), trial-end reminder, department notification.

Postconditions: 1. All material entering the premises outside the GRN/PO flow has a Security-logged entry point. 2. Demo/trial equipment is never left on indefinite, untracked trial status; every trial has a defined end date and decision trail.

Business Rules: 1. Demo/trial equipment on-site beyond its trial-end date without a purchase or return decision is escalated to the requesting department HOD and Stores Manager. 2. Visitor-brought material of significant value may require the visitor's photo ID and a Security supervisor countersignature.

Exception Handling: 1. If a returned loan item does not match any open outward gate pass, it is logged as an unreconciled inward entry pending investigation rather than auto-matched. 2. If the bringing party cannot be identified or authorized, Security holds the material at the gate pending department confirmation.

## REQ-GP004 – Gate Pass Security Verification and Register

Priority: High | Status: Proposed

Description: The system shall provide Security staff with a verification screen and consolidated gate register to confirm every material movement against its approved gate pass, transfer note, or Service PO before allowing exit or entry.

Preconditions: A gate pass, transfer note, or Service PO exists for the material movement being verified.

Inputs: Gate pass/transfer note number (scanned or entered), vehicle/carrier details, Security guard ID, timestamp.

Processing Logic: 1. Security scans or enters the gate pass number at the point of movement. 2. System displays the approved item list, quantity, and authorization status for visual cross-check against the physical material. 3. Security confirms match and records vehicle/carrier details and exact time of movement. 4. System logs the entry in a consolidated, time-stamped Gate Register covering all outward, inward, and inter-facility movements. 5. Any gate pass presented without a valid approved status is rejected at this screen, with the attempted use logged. 6. Daily Gate Register summary is available to Stores Manager and Security Head for reconciliation against approved passes.

Outputs: Verified exit/entry confirmation, time-stamped Gate Register entry, rejected-attempt log, daily reconciliation summary.

Postconditions: 1. No material movement is recorded as complete without a Security-verified match against an approved gate pass. 2. The Gate Register is the single source of truth for all physical material movement, reconciled daily against system-approved passes.

Business Rules: 1. A gate pass can be used for exactly one exit and one corresponding return entry (returnable) or one exit only (non-returnable); reuse is blocked. 2. Security cannot approve a movement for a gate pass in 'Pending Approval' status, regardless of urgency claimed verbally.

Exception Handling: 1. If Security detects a quantity or item mismatch against the approved gate pass, the movement is held and escalated to the Stores Manager for resolution before release. 2. If the gate pass system is unavailable at the point of movement, a manual register entry is permitted with mandatory retrospective system entry within the same shift.

## REQ-GP005 – Overdue Returnable Gate Pass Tracking

Priority: High | Status: Proposed



Description: The system shall track all open returnable gate passes against their expected return date and escalate overdue items to prevent assets and loaned equipment from being lost track of outside the premises.

Preconditions: A returnable outward gate pass (REQ-GP001) is open and has not yet been closed by a matching inward entry.

Inputs: Gate pass ID, expected return date, current date, custodian, department, escalation thresholds.

Processing Logic: 1. System runs a daily check comparing expected return date against the current date for all open returnable gate passes. 2. At the expected return date, a reminder is sent to the custodian and requesting department. 3. If unreturned 7 days past the expected date (configurable), an escalation is sent to the department HOD. 4. If unreturned 15 days past the expected date, escalation goes to the Stores Manager and Biomedical Engineering Head with a consolidated overdue list. 5. Overdue gate passes are shown on a dedicated dashboard widget, categorized by age and value/criticality of the item. 6. Department HOD or vendor can request a return-date extension, which requires the original approver's re-approval and updates the expected return date.

Outputs: Daily overdue check results, tiered reminder and escalation notifications, overdue dashboard widget, extension approval record.

Postconditions: 1. No returnable gate pass remains open indefinitely without visibility to management once past its expected return date. 2. Extensions are always explicitly approved and logged, never silently rolled forward.

Business Rules: 1. Critical/high-value assets on an overdue returnable gate pass are escalated on a shorter timeline than routine items, per the same criticality classification used in Asset Management. 2. Three or more overdue gate passes for the same vendor (e.g., repair turnaround) are flagged in the vendor's quality rating (REQ-V010).

Exception Handling: 1. If an item is confirmed lost or the vendor is unable to return it, the gate pass is converted to a disposal/write-off workflow (REQ-A005) rather than left open indefinitely. 2. If the custodian has left the department/organization, the overdue escalation is redirected to their reporting HOD.

## REQ-GP006 – Gate Pass Reporting and Asset Linkage

Priority: Medium | Status: Proposed

Description: The system shall maintain full linkage between every gate pass and the underlying asset, Service PO, or transfer record, and provide consolidated reporting on material movement for audit and operational review.

Preconditions: One or more gate passes have been created, approved, and processed through Security verification.

Inputs: Date range, department, reason category, gate pass status filters.

Processing Logic: 1. Every gate pass, once linked to an Asset Register entry, Service PO, or transfer note, appears in that record's full history for lifecycle traceability. 2. Consolidated reports are available by department, reason (repair/demo/disposal/courier/sample), status (open/closed/overdue), and value. 3. A monthly gate-pass audit report highlights items pending closure beyond policy, frequent-repair assets (candidates for AMC/CMC review), and any Security-flagged mismatches from the period. 4. Reports export to Excel for statutory audit (e.g., NABH material-movement traceability) and support drill-down to the individual gate pass and its Security Register entry.

Outputs: Gate-pass-linked asset/PO history, consolidated movement reports by multiple filters, monthly audit report, Excel export.

Postconditions: 1. Every material movement in or out of the premises is traceable end-to-end from the originating request through approval, Security verification, and closure, supporting both operational and statutory audit needs.

Business Rules: 1. Report access to sale-value and vendor-financial fields is restricted to Stores Manager, Finance, and Management roles, consistent with the sensitive-data masking policy (REQ-NF014).

Exception Handling: 1. If a gate pass has no linked asset/PO/transfer record (e.g., ad-hoc visitor material), it is still retained in the Gate Register and reporting, tagged as 'unlinked' rather than excluded.

## 3.16 CONSIGNMENT STOCK MANAGEMENT

### 3.16 Consignment Stock Management Process Flow

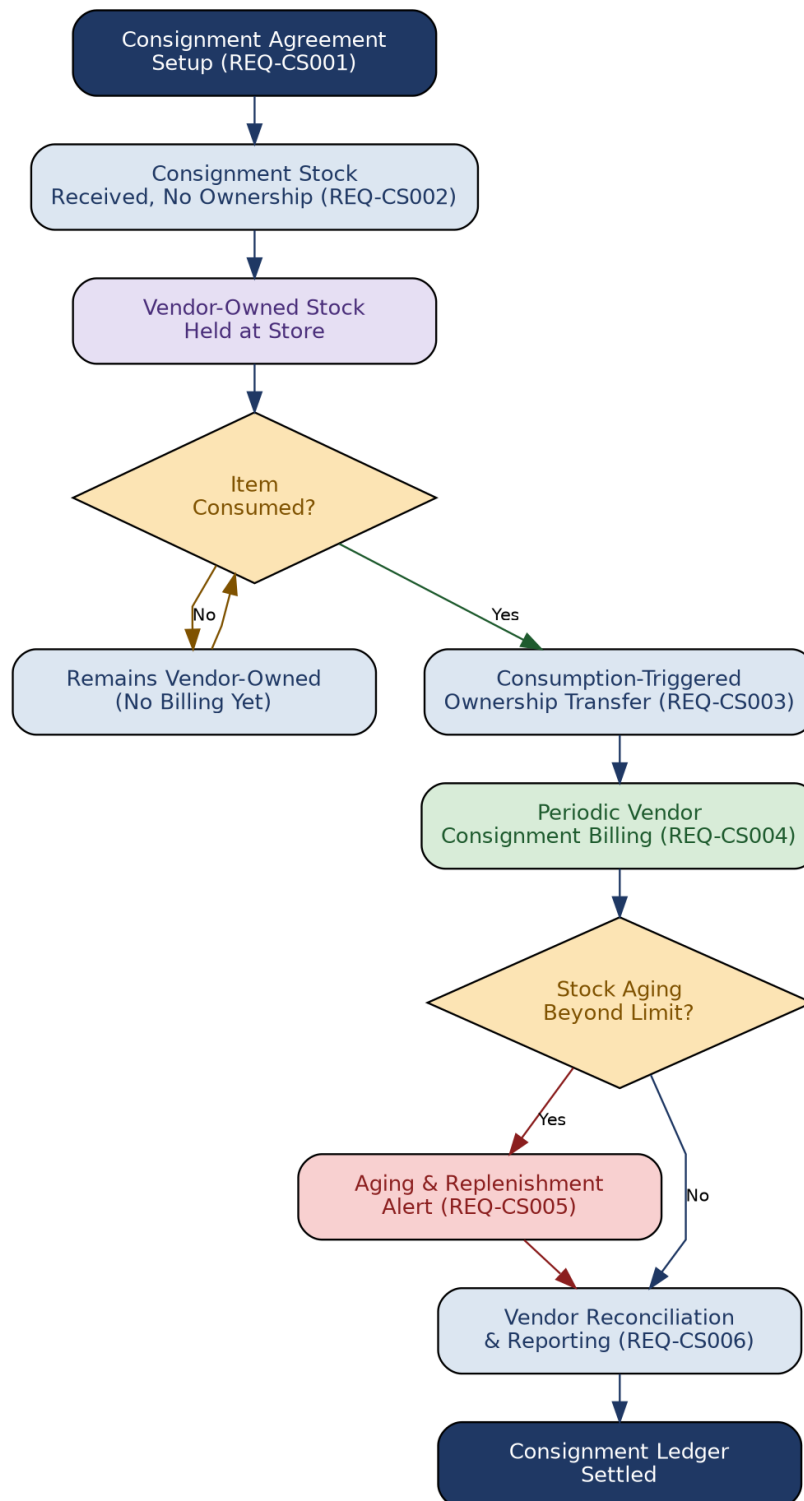


Figure: 3.16 CONSIGNMENT STOCK MANAGEMENT – Process Flow

## REQ-CS001 – Consignment Agreement Setup

Priority: High | Status: Proposed

Description: The system shall capture and maintain vendor consignment agreements, defining the terms under which vendor-owned stock is held on-site without transfer of ownership until consumption.

Preconditions: A vendor has been approved (Section 3.2) and a consignment arrangement has been commercially agreed for specific products.

Inputs: Vendor ID, product(s) covered, agreed consignment quantity/value cap, replenishment trigger level, billing cycle, agreement start/end date, pricing terms, scanned agreement document.

Processing Logic: 1. Purchase or Stores Manager creates a Consignment Agreement record against the vendor, specifying the covered products and commercial terms. 2. Agreed consignment cap (maximum quantity/value the vendor may hold on-site) and replenishment trigger level are configured per product. 3. Products under a consignment agreement are flagged in the Product Master as 'Consignment-Eligible' for that vendor, distinguishing them from standard purchase-based stock. 4. The scanned agreement is stored against the vendor record, with renewal reminders following the same tiered cadence as REQ-A007 (90/60/30/15 days). 5. Agreement terms (pricing, billing cycle) feed the vendor invoicing workflow (REQ-CS004) automatically.

Outputs: Consignment Agreement record, consignment-eligible product flags, stored agreement document, renewal reminders.

Postconditions: 1. No product can be received as consignment stock (REQ-CS002) without an active, approved Consignment Agreement covering it. 2. Consignment cap and replenishment terms are enforced at receipt, not left to manual tracking.

Business Rules: 1. Consignment agreements above a configurable value require Management approval before activation. 2. A product cannot simultaneously be under both a rate contract (standard purchase) and a consignment agreement with conflicting terms from the same vendor without explicit resolution.

Exception Handling: 1. If the agreement expires while consignment stock is still on-site, the system blocks further consumption postings against it and escalates for renewal or stock return decision before any further consumption is recorded.

## REQ-CS002 – Consignment Stock Receipt

Priority: High | Status: Proposed

Description: The system shall receive vendor-owned consignment stock into a distinct sub-inventory that is tracked for quantity and location but excluded from owned-stock valuation and capitalization.

Preconditions: An active Consignment Agreement exists for the vendor and product being received.

Inputs: Vendor ID, product ID, quantity received, batch/serial/expiry (as applicable), consignment agreement reference, receiving location.

Processing Logic: 1. Store keeper receives consignment stock using a distinct 'Consignment GRN' type, referencing the active agreement. 2. System validates the received quantity against the agreement's consignment cap; quantity exceeding the cap is blocked pending vendor/Stores Manager confirmation. 3. Stock is added to a separate consignment sub-ledger, tracked by quantity, batch, and location identically to owned stock for operational purposes (FEFO/FIFO, expiry alerts), but excluded from the owned-stock valuation and Asset/Inventory Register capitalization. 4. Batch and expiry tracking (REQ-IO13 Short-Expiry Highlight) applies equally to consignment stock. 5. Receipt is confirmed to the vendor portal for their own record reconciliation.

Outputs: Consignment GRN, consignment sub-ledger entry, cap-validation result, vendor portal receipt confirmation.

Postconditions: 1. Consignment stock is never included in owned-inventory valuation reports or Tally stock postings until consumed. 2. Every unit of consignment stock is traceable to its governing agreement at all times.

Business Rules: 1. Consignment stock follows the same FEFO/FIFO and expiry-alert discipline as owned stock (Section 3.3). 2. Consignment stock cannot be transferred to another department/location as owned stock; it remains vendor-owned wherever it physically moves within the premises.

Exception Handling: 1. If received quantity would exceed the agreed cap, the excess is held in a 'Pending Vendor Confirmation' status rather than rejected outright, pending the vendor's agreement to a cap revision or return of the excess.

**REQ-CS003 – Consumption-Triggered Ownership Transfer**

Priority: High | Status: Proposed

Description: The system shall transfer ownership of consignment stock from vendor to hospital only at the point of actual consumption (issue to a department/patient), never at receipt, and generate the corresponding billing event.

Preconditions: Consignment stock is available in the consignment sub-ledger and is being issued against a valid indent or patient/procedure usage.

Inputs: Product ID, quantity consumed, issuing department, patient/procedure reference (where applicable), consumption date.

Processing Logic: 1. When a department issues or a patient/procedure consumes consignment stock, the system records the consumption exactly as it would for owned stock (Section 3.5 Indent & Issue). 2. At the moment of consumption, the system posts an ownership-transfer entry: the consumed quantity moves from the vendor-owned consignment sub-ledger to owned-stock valuation, at the agreed consignment price. 3. This ownership-transfer entry simultaneously creates a billable consumption line against the vendor's consignment account. 4. Department-wise and patient-wise consumption value is attributed correctly for costing and, where applicable, patient billing, exactly as with owned stock. 5. Unconsumed consignment stock at any point in time remains vendor-owned and is excluded from the hospital's owned-inventory valuation.

Outputs: Ownership-transfer ledger entry, billable consumption line, department/patient cost attribution, updated consignment sub-ledger balance.

Postconditions: 1. Ownership of any given unit of consignment stock changes exactly once, at the moment of consumption, and is never backdated or bulk-transferred. 2. Owned-stock valuation always reflects only consumed consignment quantities, never the full on-site consignment holding.

Business Rules: 1. Consumption price used for ownership transfer is fixed by the governing Consignment Agreement and cannot be manually overridden at the point of issue. 2. Returned/unused issued consignment stock reverses the ownership transfer and the associated billing line, consistent with the standard Stock Return Module (REQ-IO12).

Exception Handling: 1. If the governing Consignment Agreement has expired at the point of intended consumption, the system blocks the transaction and requires either agreement renewal or reclassification of the stock as an emergency purchase before issue can proceed.

## REQ-CS004 – Periodic Vendor Consignment Billing

Priority: High | Status: Proposed

Description: The system shall compile consumption-triggered ownership transfers into periodic vendor billing statements, replacing the standard PO/invoice flow for consignment stock with a consumption-based reconciliation.

Preconditions: One or more consumption-triggered ownership transfers (REQ-CS003) have occurred since the last billing cycle for a given vendor agreement.

Inputs: Vendor ID, agreement reference, billing cycle (weekly/monthly, per agreement), consumption transactions since last cycle.

Processing Logic: 1. At the configured billing cycle interval, the system compiles all consumption-triggered ownership transfers for the vendor's consignment agreement into a consolidated consumption statement. 2. The statement itemizes product, quantity consumed, agreed unit price, and total payable, mirroring a standard GRN/invoice but sourced entirely from consumption events rather than a delivery. 3. The consumption statement is published to the vendor portal for the vendor to raise their formal invoice against. 4. On receipt of the vendor's invoice, the system cross-validates it against the system-generated consumption statement (quantity and value), following the same variance-handling discipline as REQ-PO21 (Price Variation Report). 5. Validated consignment invoices route through standard Accounts payment processing and Tally posting.

Outputs: Periodic consumption statement, vendor portal publication, invoice cross-validation result, Tally-postable payment entry.

Postconditions: 1. Vendor payment for consignment stock is always based on validated consumption, never on quantity delivered/held on-site. 2. Every consignment billing cycle is fully reconciled and auditable back to individual consumption transactions.

Business Rules: 1. A vendor invoice that does not match the system consumption statement beyond a configurable tolerance is held for joint Purchase/Finance/vendor resolution before payment. 2. Billing cycle frequency is fixed by the governing agreement and cannot be changed unilaterally by either party without an agreement amendment.

Exception Handling: 1. If a vendor disputes the system-recorded consumption quantity, the dispute is logged against the billing cycle and resolved by reference to the department/patient-level consumption trail (REQ-CS003) before the statement is finalized.

## REQ-CS005 – Consignment Stock Aging and Replenishment

Priority: Medium | Status: Proposed

Description: The system shall monitor consignment stock aging on-site and trigger replenishment or return-to-vendor workflows to prevent long-dormant, vendor-owned stock from accumulating without action.

Preconditions: Consignment stock has been on-site for a duration approaching or exceeding the agreement's defined aging threshold.

Inputs: Product ID, quantity on-site, receipt date, agreement-defined aging threshold, replenishment trigger level, consumption trend.

Processing Logic: 1. System runs a daily check on consignment stock age (days on-site since receipt) and remaining balance against the replenishment trigger level. 2. When on-site consignment stock for a product falls below the replenishment trigger, the system notifies the vendor (via portal) to replenish, consistent with a vendor-managed-inventory model. 3. When consignment stock for a slow-moving item exceeds the agreement's maximum aging threshold without consumption, the system flags it for a return-to-vendor decision rather than allowing indefinite on-site holding. 4. Expiry-sensitive consignment items follow the same short-expiry highlighting as owned stock (REQ-I013), with FEFO issue enforced regardless of ownership. 5. A consolidated consignment aging report is available to Stores Manager and Purchase, by vendor, product, and age band.

Outputs: Replenishment notification to vendor, return-to-vendor workflow trigger, expiry alerts on consignment batches, consignment aging report.

Postconditions: 1. Consignment stock is never allowed to sit on-site indefinitely without either consumption, replenishment cycling, or an explicit return-to-vendor decision. 2. Aging and expiry monitoring applies uniformly to consignment and owned stock, so patient safety and FEFO discipline are never compromised by ownership status.

Business Rules: 1. Return-to-vendor of aged, unconsumed consignment stock does not require the standard disposal/write-off approval used for owned stock, since ownership was never transferred to the hospital. 2. Repeated replenishment-trigger breaches for the same product are factored into a review of whether the agreed cap and trigger levels remain appropriate.

Exception Handling: 1. If a consignment item expires on-site before being returned or consumed, it is logged as a vendor-owned expired-stock event (not a hospital write-off) and routed to the vendor per the agreement's expiry-liability terms.



## REQ-CS006 – Consignment Vendor Reconciliation and Reporting

Priority: Medium | Status: Proposed

Description: The system shall provide periodic reconciliation between the hospital's consignment records and vendor-confirmed stock/consumption figures, closing out each billing cycle with a balanced ledger.

Preconditions: One or more billing cycles (REQ-CS004) have been completed for a vendor's consignment agreement.

Inputs: Vendor ID, agreement reference, system consignment sub-ledger balance, vendor-confirmed on-site balance, reconciliation period.

Processing Logic: 1. At each reconciliation point (aligned to the billing cycle or a longer audit interval), the system compares its consignment sub-ledger balance (quantity/value still vendor-owned on-site) against the vendor's own confirmed figure, obtained via the vendor portal. 2. Matches are closed automatically; mismatches are logged as reconciliation exceptions with the variance quantity and value. 3. A consolidated consignment reconciliation report is produced per vendor, per agreement, showing opening balance, receipts, consumption, returns, and closing balance for the period. 4. The report feeds into the same monthly MIS distribution as other supplier outstanding reconciliations (Section 3.9), giving Management a unified view of both owned and consignment vendor exposure.

Outputs: Reconciliation match/exception log, consolidated per-vendor consignment reconciliation report, monthly MIS inclusion.

Postconditions: 1. Every consignment agreement has a demonstrable, periodic reconciliation trail, preventing vendor-owned stock on hospital premises from becoming an unmanaged, unreconciled liability. 2. Unresolved reconciliation exceptions beyond a configurable period are escalated to Purchase, Finance, and the vendor jointly.

Business Rules: 1. Reconciliation exceptions above a configurable value threshold require Finance sign-off before the cycle can be marked closed. 2. A vendor's consignment agreement cannot be renewed (REQ-CS001) while unresolved reconciliation exceptions remain open beyond policy.

Exception Handling: 1. If the vendor is unresponsive to a reconciliation request beyond a configurable period, the system escalates to Management and flags the vendor's quality/reliability rating (REQ-V010).

## 4. EXTERNAL INTERFACE REQUIREMENTS

### 4.1 User Interfaces

The system shall provide a responsive web-based user interface accessible through standard web browsers. The interface shall be designed for both desktop and tablet use, with mobile-responsive views for approval workflows.

| Requirement ID                   | Description                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------------------|
| REQ-UI001<br>Web-based Dashboard | Role-based dashboard with widgets for pending approvals, stock alerts, expiry items, and key metrics |
| REQ-UI002<br>Responsive Design   | Adaptable layout for screens from 1366×768 to 1920×1080 and tablets                                  |
| REQ-UI003<br>Mobile Approval     | Simplified mobile view for HOD and management approvals with one-click actions                       |
| REQ-UI004<br>Search & Filter     | Global search with autocomplete, advanced filters on all list screens                                |
| REQ-UI005<br>Bulk Operations     | Bulk approve, bulk print, bulk export capabilities                                                   |
| REQ-UI006<br>Colour Coding       | Red/Amber/Green status indicators for quick visual assessment                                        |
| REQ-UI007<br>Drag & Drop         | Drag-and-drop for file uploads, reordering categories, and workflow design                           |
| REQ-UI008<br>Contextual Help     | Tooltip help and guided tours for new users                                                          |

### 4.2 Hardware Interfaces

- Barcode Printers: Zebra ZT230, TSC TTP-244 Pro, or equivalent (USB/Network)
- Barcode Scanners: Honeywell, Datalogic 1D/2D scanners (USB/Bluetooth)
- ID Card Scanners: Magnetic stripe and RFID readers for employee/patient ID
- Document Scanners: Flatbed and ADF scanners for invoice and document digitization
- Mobile Devices: Android/iOS tablets for warehouse operations and approvals
- Network Infrastructure: Hospital LAN with minimum 100 Mbps backbone

### 4.3 Software Interfaces

| External System                   | Integration Method                                                              |
|-----------------------------------|---------------------------------------------------------------------------------|
| Tally ERP                         | XML/ODBC/API integration for accounting entries, narration sync, GST data       |
| Hospital Information System (HIS) | REST API or database-level integration for ward indent creation and status sync |
| Email Server (SMTP)               | SMTP/IMAP for automated notifications,                                          |

|                         |                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------|
|                         | approval emails, and report distribution                                             |
| Active Directory / LDAP | Single Sign-On (SSO) and user authentication integration                             |
| Online Marketplaces     | API integration with Amazon Business, Flipkart Wholesale, IndiaMART for order import |
| GST Portal              | Excel-based data exchange for GSTR reconciliation (until API availability)           |
| SMS Gateway             | HTTP API for urgent approval SMS notifications                                       |
| Cloud Storage           | Optional AWS S3/Azure Blob for document and image storage                            |

## 4.4 Communications Interfaces

- HTTPS (TLS 1.2+) for all web traffic
- VPN or IP-whitelisting for vendor portal access
- SFTP for bulk file transfers (rate cards, statements)
- REST API with JSON for real-time integrations
- Webhook support for event-driven notifications
- Email protocols: SMTP for outgoing, IMAP/POP3 for incoming (vendor replies)

## 5. NON-FUNCTIONAL REQUIREMENTS

### 5.1 Performance Requirements

| Req ID    | Parameter          | Requirement                                                                              |
|-----------|--------------------|------------------------------------------------------------------------------------------|
| REQ-NF001 | Response Time      | Any screen load or transaction save shall complete within 3 seconds under normal load    |
| REQ-NF002 | Report Generation  | Standard reports shall generate within 10 seconds; complex MIS reports within 60 seconds |
| REQ-NF003 | Concurrent Users   | System shall support minimum 50 concurrent users without performance degradation         |
| REQ-NF004 | Data Volume        | System shall handle 1 million+ transactions per year with no performance impact          |
| REQ-NF005 | Search Performance | Search results shall display within 2 seconds for datasets up to 100,000 records         |
| REQ-NF006 | File Upload        | Document uploads up to 5MB shall complete within 10 seconds on standard network          |

### 5.2 Security Requirements

REQ-NF007 – Authentication: Multi-factor authentication (MFA) for admin and finance users; password + OTP for critical approvals

REQ-NF008 – Authorization: Role-based access control (RBAC) with minimum 8 predefined roles; custom role creation capability

REQ-NF009 – Data Encryption: AES-256 encryption for data at rest; TLS 1.2+ for data in transit

REQ-NF010 – Audit Trail: Complete audit log for all financial transactions: who, what, when, before/after values

REQ-NF011 – Session Management: Auto-logout after 30 minutes of inactivity; concurrent session limit per user

REQ-NF012 – Password Policy: Minimum 8 characters, complexity requirements, 90-day expiry, history prevention

REQ-NF013 – Vendor Portal Security: IP whitelisting option, CAPTCHA, failed login lockout after 5 attempts

REQ-NF014 – Data Masking: Mask sensitive data (bank accounts, PAN) for non-finance users

### 5.3 Availability & Reliability

REQ-NF015 – Uptime: 99.5% uptime during business hours (8 AM – 8 PM); 95% uptime overall

REQ-NF016 – Disaster Recovery: RPO (Recovery Point Objective) ≤ 1 hour; RTO (Recovery Time Objective) ≤ 4 hours

REQ-NF017 – Backup: Automated daily full backup, hourly incremental backup, 30-day retention

REQ-NF018 – Failover: Automatic failover to secondary server within 5 minutes of primary failure

### 5.4 Scalability

The system shall be horizontally scalable to support:

- 500+ active products in master
- 200+ suppliers in vendor portal
- 50+ hospital departments/wards
- 500+ concurrent users (future expansion)
- 10,000+ transactions per month
- Multi-location deployment capability

### 5.5 Data Integrity & Backup

REQ-NF019 – Data Validation: Field-level validation, referential integrity, business rule validation at entry point

REQ-NF020 – Duplicate Prevention: Database-level unique constraints + application-level duplicate detection

REQ-NF021 – Transaction Atomicity: All financial transactions shall be atomic (all-or-nothing commit)

REQ-NF022 – Data Archival: Auto-archive transactions older than 3 years to separate storage; searchable archive

REQ-NF023 – Backup Verification: Monthly backup restoration testing with documented results

## 6. DATA REQUIREMENTS

### 6.1 Data Models

The following core entities form the data model for IHMS. Each entity includes standard audit fields: CreatedBy, CreatedDate, ModifiedBy, ModifiedDate, IsActive.

| Entity Name            | Key Attributes                                                                                                                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Master         | ProductID, ProductCode, ProductName, AliasNames, CategoryID, UOM, ReorderLevel, ReorderQty, IsAsset, IsConsumable, IsService, DepreciationRate, WarrantyPeriod, Images, IsActive |
| Supplier Master        | SupplierID, SupplierCode, SupplierName, PAN, GSTIN, TAN, Address, BankDetails, ContactPerson, Email, Phone, IsApproved, ApprovalDate, PortalAccessEnabled                        |
| Vendor Product Mapping | MappingID, SupplierID, ProductID, VendorProductCode, VendorProductName, Rate, GSTPercent, HSNCode, EffectiveDate, ExpiryDate, IsApproved                                         |
| Purchase Order         | POID, PONumber, PODate, SupplierID, DepartmentID, RequestedBy, TotalValue, Currency, ExchangeRate, Status, ApprovalLevel, RateContractID, IsCashPO, IsUrgent                     |
| PO Line Items          | LineItemID, POID, ProductID, Qty, UnitRate, GSTPercent, Discount, LineTotal, BatchNo, ExpiryDate, DeliveryDate                                                                   |
| PO Addendum            | AddendumID, POID, AddendumType, Description, Amount, ApprovedBy, ApprovedDate                                                                                                    |
| GRN                    | GRNID, GRNNumber, GRNDate, POID, SupplierID, InvoiceNo, InvoiceDate, DCNo, DCDate, TotalGross, TotalTax, GrandTotal, Status, ReceivedBy                                          |
| GRN Line Items         | GRNLineID, GRNID, ProductID, QtyReceived, UnitRate, BatchNo, ManufactureDate, ExpiryDate, MRP, LocationID                                                                        |
| Indent                 | IndentID, IndentNumber, IndentDate, DepartmentID, RequestedBy, HODApprovedBy, Status, IsAutoIndent,                                                                              |

|                     |                                                                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | IsEmergency, ReasonForRequirement                                                                                                                                                    |
| Indent Line Items   | IndentLineID, IndentID, ProductID,<br>RequestedQty, IssuedQty, IssueDate,<br>IssuedBy, Status                                                                                        |
| Asset Register      | AssetID, AssetCode, ProductID, SerialNo,<br>ManufactureDate, PurchaseDate,<br>WarrantyStart, WarrantyEnd,<br>DepartmentID, LocationID, Status,<br>BookValue, AccumulatedDepreciation |
| Asset Maintenance   | MaintenanceID, AssetID,<br>MaintenanceType, ScheduleDate,<br>CompletionDate, PerformedBy, Cost,<br>NextDueDate, Status                                                               |
| Rate Contract       | ContractID, SupplierID,<br>ContractStartDate, ContractEndDate,<br>ContractValue, TermsConditions,<br>IsActive, RenewalReminderDate                                                   |
| Budget              | BudgetID, DepartmentID, FinancialYear,<br>CategoryID, AllocatedAmount,<br>SpentAmount, CarryForward, IsFrozen                                                                        |
| Ticket/CRM          | TicketID, TicketNumber, TicketDate,<br>DepartmentID, RequestedBy, Category,<br>Priority, Status, AssignedTo,<br>ResolutionDate, FeedbackRating                                       |
| User Master         | UserID, UserName, Email, Phone, RoleID,<br>DepartmentID, ApprovalLimit, IsActive,<br>LastLogin, PasswordExpiryDate                                                                   |
| Approval Workflow   | WorkflowID, TransactionType,<br>TransactionID, ApproverLevel,<br>ApproverID, Action, ActionDate,<br>Comments, DelegatedFrom                                                          |
| Audit Log           | LogID, TableName, RecordID, Action,<br>OldValues, NewValues, PerformedBy,<br>ActionDate                                                                                              |
| Location Master     | LocationID, LocationName,<br>LocationType (Store/Ward/Facility),<br>ParentLocationID, IsActive                                                                                       |
| Department Master   | DepartmentID, DepartmentName,<br>HODUserID, CostCenterCode,<br>IsKeyDepartment, IsActive                                                                                             |
| Transfer Note (ITN) | TransferID, TransferNumber,<br>SourceLocationID,<br>DestinationLocationID, ProductID, Qty,<br>Status (Requested/In-<br>Transit/Received), RequestedBy,<br>ApprovedBy                 |

|                   |                                                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gate Pass         | GatePassID, GatePassNumber, Direction (Outward/Inward), ReturnableFlag, ItemID, Qty, Reason, LinkedAssetID, LinkedServicePOID, ExpectedReturnDate, Status, ApprovedBy, SecurityVerifiedBy |
| Consignment Stock | ConsignmentID, VendorID, AgreementID, ProductID, QtyOnHand, ConsignmentPrice, ReceiptDate, ConsumedQty, OwnershipTransferDate, BillingCycleStatus                                         |

## 6.2 Data Retention

- Active transactions: Indefinite retention in primary database
- Completed transactions (>3 years): Archive to secondary storage with full searchability
- Audit logs: 7 years retention (regulatory compliance)
- Deleted records: Soft delete only; retained for 1 year before hard delete
- Vendor portal documents: Retained for duration of relationship + 3 years
- Email notifications: 1 year retention
- System logs: 2 years retention



## 7. APPENDICES

### Appendix A: Implementation Phases

The implementation is recommended in six phases to ensure smooth transition and early value realization:

#### Phase 1: Core Foundation (Months 1-3)

- Product Master with categorization, aliases, and images
- Supplier Master with basic registration
- PO creation, approval workflow, and GRN processing
- Basic inventory tracking and stock levels
- Tally integration for accounting entries
- User management and role-based access
- Basic reports: Purchase, GRN, Stock position

#### Phase 2: Workflow & Integration (Months 4-5)

- HIS integration for ward indent creation
- Auto-indent based on reorder levels
- Department HOD approval workflows
- Cash PO and spot PO approval
- Barcode generation and printing
- Stock transfer between locations
- Enhanced reporting with filters and exports

#### Phase 3: Vendor Portal (Months 6-7)

- Vendor self-registration with document upload
- Vendor product catalog and rate card upload
- Payment status visibility
- Supplier outstanding reconciliation
- GST cross-validation module
- Invoice scan upload by vendors
- Sample tracking and quality rating

## Phase 4: Advanced Features (Months 8-9)

- Asset lifecycle management with depreciation
- Warranty/AMC/CMC reminder system
- Preventive maintenance scheduling
- Rate contract management
- PO addendums and landed cost calculation
- Multi-currency PO support
- Online marketplace integration (Amazon, Flipkart, Indiamart)

## Phase 5: Analytics & MIS (Months 10-11)

- Management dashboards with KPIs
- Automated scheduled reports
- Price comparison and variation analysis
- Budget vs. actual tracking
- Department usage analytics
- Process timing and deviation reports
- Mobile approval app

## Phase 6: Specialized Modules (Months 12-13)

- Printing consumables management (forms, price charts)
- Uniforms and linen tracking with colour coding
- CRM ticketing system
- Internal chat module
- Advanced audit and compliance reports
- System optimization and performance tuning
- User training and documentation completion

## Appendix B: Glossary of Hospital-Specific Terms

| Term                     | Definition                                                               |
|--------------------------|--------------------------------------------------------------------------|
| Indent                   | A formal request from a ward/department to stores for materials/supplies |
| Issue                    | The act of transferring stock from stores to the requesting department   |
| GRN (Goods Receipt Note) | Document acknowledging physical                                          |

|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
|                     | receipt of goods from a supplier                                                     |
| PO (Purchase Order) | Formal document issued to a vendor authorizing supply of goods/services              |
| Rate Contract       | Agreement with vendor fixing prices for specified items for a defined period         |
| AMC                 | Annual Maintenance Contract – yearly service agreement for equipment                 |
| CMC                 | Comprehensive Maintenance Contract – includes spare parts replacement                |
| HOD                 | Head of Department – responsible for departmental approvals and oversight            |
| HIS                 | Hospital Information System – manages patient records, admissions, billing           |
| Tally               | Popular Indian accounting software used for GST compliance and financial reporting   |
| Red-Tag             | Status marking items as out-of-use, pending disposal decision                        |
| FEFO                | First Expired First Out – inventory issue method prioritizing near-expiry items      |
| FIFO                | First In First Out – inventory issue method prioritizing oldest stock                |
| Landed Cost         | Total cost including purchase price, shipping, insurance, customs, and other charges |
| MRP                 | Maximum Retail Price – printed price on packaged goods                               |
| HSN Code            | Harmonized System of Nomenclature – GST classification code for products             |
| DC                  | Delivery Challan – document accompanying goods when invoice is not yet available     |

## Appendix C: Compliance & Regulatory Requirements

- GST Compliance: Accurate HSN code mapping, multi-slab GST calculation (0%, 5%, 12%, 18%, 28%), GSTR reconciliation
- TDS Compliance: Section 194C/194J TDS calculation and deduction tracking
- Drug & Cosmetics Act: Batch tracking, expiry management, and recall capability for pharmaceuticals
- Medical Device Rules: Unique device identification (UDI) where applicable
- Income Tax Act: Depreciation calculation as per applicable rates (15%, 30%, 40%)
- Companies Act: Asset register maintenance and audit trail requirements
- Data Protection: Personal data of vendors and staff protected; no sharing without consent
- ISO 13485 (if applicable): Medical device quality management traceability

## Appendix D: Risk Mitigation

| Risk                     | Mitigation Strategy                                                                 |
|--------------------------|-------------------------------------------------------------------------------------|
| Data Migration Risk      | Parallel run for 2 months; manual reconciliation; rollback plan                     |
| User Adoption Risk       | Phased rollout by department; intensive training; super-user program                |
| Integration Failure Risk | API fallback to manual import; daily sync monitoring; alert on sync failure         |
| Vendor Portal Adoption   | Mandate portal usage for new vendors; incentive for early adoption; manual fallback |
| Performance Degradation  | Load testing at 2x expected capacity; database indexing; caching strategy           |
| Security Breach          | Penetration testing; regular security audits; incident response plan                |

### DOCUMENT CONTROL

Document Version: 1.0

Date: July 24, 2026

Status: Final

Classification: Internal – Confidential

## Appendix E: Future Enhancement Roadmap

The following capabilities are recommended as post-implementation roadmap items. They are not part of the core scope in this SRS but are noted here for future planning.

- RFID-based inventory and asset tracking, as a future upgrade path from the barcode/QR foundation built in this system.
- IoT integration for real-time condition monitoring of high-value or environment-sensitive assets, such as cold-chain storage.
- AI-based demand forecasting to refine reorder levels and safety stock beyond the rule-based reorder logic already defined for auto-indent.
- A Customer Portal, in addition to the Vendor Portal in Section 3.2, for external parties who consume billed services or products, where applicable to the business model.
- E-invoicing and GST-compliant electronic invoice generation, and OCR-based extraction from scanned vendor invoices to reduce manual GRN/invoice data entry.
- Offline mobile stock counting, allowing physical verification to continue in locations with poor network connectivity and to sync once reconnected.
- A Business Intelligence dashboard layer, such as Power BI or Tableau integration, on top of the reports in Section 3.9.